L4 Safety Island for Autoware

Software Design, Part 1 — node allocation

REVIEW EDITION

for download, reading and annotation

Autoware Foundation • Reference Design Working Group
NEWSLab, National Taiwan University

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How to annotate this document

Every line is numbered, and the right-hand margin is deliberately wide. A comment can therefore be pinned precisely — *"p.14, line 302"* — whether you write on paper or in a PDF reader.

On screen. Any PDF reader's comment tools work: sticky notes, highlights, strike-through. Preview (macOS), Acrobat, Okular and Firefox all round-trip standard PDF annotations, so a marked-up copy can be sent back and opened by anyone.

On paper. Print single-sided; the margin is sized for handwriting.

Either way, the comment log on the last pages collects the points you want discussed in the session, so they do not have to be extracted from the margins by someone else.

What is most useful to comment on

- Anything asserted that you believe is *wrong* — especially a number, a deadline, or a claim about what Autoware does today.
- Anything allocated to the island that you think belongs on the HPC, or the reverse.
- Any place where the safety argument depends on a property that is asserted rather than demonstrated.
- Gaps: a failure mode the design does not name.

The twenty-three items already known to be open are collected in §11, and

> argued with options and a recommendation in the companion discussion document. No need to re-raise them here unless you disagree with how they are framed.
>
> **Version 20261023 is a major revision** of version 20260918. The island sensors are now **ranked** (D8) — forward camera mandatory, 2D lidar optional — so every behaviour above B4 is written against the camera, and **B4 is the loss of the camera**. **B2 first slows to the safe speed of the current ODD** (about 60 km/h on a highway, 30 km/h on a city road), holds for **8–15 s**, then hands over to B3 if the camera still sees the lane and to B4 if it does not; **B3 finds its pull-over space with the camera**. An island communication module to the remote command centre and a local control port (D9) carry two new human-driven behaviours, **B5 tele-operation** and **B6 manual joystick control**, whose detailed design is pending (§6.5). The design is now carried in **two branches** (§9.5): **Least-Cost**, with no radar on a microcontroller, which makes it a city-road, fair-weather branch, and **Maximum-Safety**, with four radars on an R-Car X5H; every behaviour is adjusted to each. **B5 and B6 have a stage-1 detailed design** (§6.6, §8.6): B6 entered from standstill; B5 from standstill or, on the ADS's request when the HPC behaves abnormally, from a moving vehicle in B1–B3; B5's speed bound by the latency the island measures, and both stopped by the island when their link goes quiet. Open decisions 19 to 23 are new. The revision history at the head of the design lists the earlier revisions. **Only what is new since 20260918 — the WG session of 2026-10-17 onward — is marked in yellow**, so a reviewer of 20260918 can read the marking alone and see the whole change; §3 is the place to start.

This edition is generated from `L4SafetyIsland_SoftwareDesign.md`, version 20261023. The Markdown original, where every identifier is a live cross-reference, remains the editable source of truth.

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Part 1 — Node Allocation between the Safety Island and the High-Performance Computer

Reference Design WG · **Release V1.0** · **Version 20261023**

> **Document version.** The version is a date-code, `YYYYMMDD`. The four artefacts of this deliverable — this Markdown source, the slide deck, the review edition and the discussion document — carry the *same* code on their title page. A set that does not share one code is not a coherent revision and should not be reviewed as one. **Release V1.0.** With the WG discussion of the design closed, version 20261023 is released as **V1.0** (2026-11-05). The release changes no content: it is version 20261023 exactly. V1.0 comprises this Markdown source, the slide deck, the reading and review editions, and the diagrams; the discussion document served the WG sessions and is not part of the release.

> **Another way in — the slide deck.** The same design is presented as a discussion deck, `L4SafetyIsland-SoftwareDesign.pdf`: the six behaviours, the ladder drawn as a state diagram, the criticality tiers and the five-phase node allocation, one figure per slide. Read it first for the shape of the design; read this document for the arguments, the interface tables and the open decisions. **The whole document as PDF.** The `reading edition` — normal margins, for reading — and the `review edition` — numbered lines, a wide margin and a comment log, for sending comments back. Both are generated from this document and carry its version.

Revision history — major revisions only.

Version	Issued	Status and scope
V1.0 = 20261023	2026-11-05	**Release.** Version 20261023, released as V1.0 after the WG discussion of the design closed. No content change.

Version	Issued	Status and scope
20261023	2026-10-23	**Major revision**, superseding version 20260918; content from the WG session of 2026-10-17. *(1)* **Ranks the island sensors** (D8): forward camera **mandatory**, radar second, ultrasonic third, IMU / GNSS fourth, 2D lidar **optional**. A behaviour may require only the camera; a lower-ranked sensor buys speed or ODD width, never the behaviour itself. Every behaviour above B4 is rewritten against the camera, the §9.4 sensor rungs are re-cut to the ranking (B2 and B4 survive at rung A and rung 0 respectively; B3 drops to rung A at or below ~30 km/h), and the AEB trigger moves from the 2D lidar to radar. *(2)* **B2 first slows to the safe speed of the current ODD**, `v_odd` — of the order of 60 km/h on a highway and 30 km/h on a city road, the class latched from `H4`'s last `active_odd` — and its hold window `T_hold` is set at **8–15 s** per road class, long enough to contain the descent to `v_odd` and still leave the HPC time to return. The lane must be observable **by the camera**: the buffered map corridor refines it but cannot stand in for it, and the heading / yaw-rate tube is withdrawn as a corridor source. When `T_hold` expires, B2 hands over to exactly one successor — B3 if the camera still sees the lane, B4 if it does not. *(3)* **B3 searches with the camera**: `N14` holds the lane lines, finds the pull-over space between the outer lane line and the road edge, and bounds the lateral move; radar confirms what is coming on the pull-over side and may disqualify a space, never qualify one. *(4)* **B4 is redefined as the loss of the camera** (or of the lane lines beyond the bridging budget); a surviving radar keeps AEB armed and may steepen the ramp, but does not lift the blackout. *(5)* **Declares two human-driven behaviours** on a new island communication module to the remote command and control centre and a local control port (D9): B5 **tele-operation** from the RCC on streamed camera, radar and ultrasonic, and B6 **manual joystick control** over a wired or short-range radio link with no stream. Adds §6.5 and the nodes `N21` `si_rcc_link`, `N22` `si_teleop_streamer` and `N23` `si_local_control`; their stage-1 detailed design is *(7)* below. *(6)* **Carries the design in two branches** (D10, §9.5): **Least-Cost** — camera, ultrasonic, 2D lidar and IMU / GNSS on an S32Z2, AURIX TC4x or Stellar microcontroller, with no radar, which makes it a city-road, fair-weather branch — and **Maximum-Safety** — camera, four radars, ultrasonic and IMU / GNSS on an R-Car X5H; every behaviour is adjusted to each branch's sensors. Also restores the missing ultrasonic row of §6.2. *(7)* **Gives B5 and B6 a stage-1 detailed design** (§6.6, §8.6): B6 entered from standstill, B5 from standstill or — on the ADS's remote take-over request when the HPC behaves abnormally — from a moving vehicle in B1–B3; B5 drives on bounded setpoints at a `v_teleop` set by the latency measured through the frame-stamp echo, is stopped by the island after `T_link` = 300 ms of silence, and may re-engage the ADS from `SAFE_STOP`; B6 runs a wired joystick pendant at 5 km/h and stops after `T_local` = 200 ms. The RCC communication — module, session, security, video transport — and a radio link for B6 are deferred to the next design. Adds the SI-E2E/R profile for the open RCC link, and the P3s partition. Adds open decisions 19 to 23, and reshapes 3, 4, 8, 13, 14, 15 and 18. **Marker yellow now marks only what is new since 20260918** — the material from the WG session of 2026-10-17 onward; the highlights of earlier revisions are removed.

Version	Issued	Status and scope
20260918	2026-09-18	**Major revision**, superseding version 20260912. *(1)* Recasts B1 from a constraint ladder into an **ODD switch negotiation**. The island declares the current operating domain unsatisfiable and **names no target**; the HPC selects a domain it can meet and declares it through `H4`; the island verifies that answer against its own measured envelope *and* against the vehicle's measured behaviour. B1 gains a **Demand** row, an **Acceptance test** row and both exits, and becomes the only behaviour that returns the vehicle to normal L4 operation — in a narrower domain than it started in. Adds `T_ack` and the safety timeout `T_odd`, the latter bounded from above by `t_margin − t_pullover`. *(2)* Makes the **takeover explicit**: when `T_odd` expires the successor is chosen by what the island still *has*, not by what the HPC did wrong — blackout → B4, HPC lost → B2, otherwise → B3, tested in that order (§3.2). *(3)* Defines **`v_hold` as a ramp** rather than a constant. It is the running minimum of five bounds, monotone non-increasing within an episode, decaying to `v_entry` so that escalation from B2 to B3 costs no further deceleration time: the hold window becomes preparation instead of dead time. *(4)* Gives B3 a **second refuge source**. When `H8` offers none, `N18` detects a lateral clearance from island sensing alone — always `EMERGENCY_ONLY`, never displacing a nominated refuge, bounded by `T_refuge` — and the three outcomes are held apart: *found*, *positively none* (→ in-lane stop, the corridor still in hand) and *inconclusive* (→ B4, because an inconclusive search is a capability statement). *(5)* Adds the §6.4 **Unsatisfiable** rung and its six-step verification, §5.3 rung 1b, and new material on `H4`, `H8`, `N18`, `N19` and the §8 interface tables. Adds open decisions 17 and 18. **Marker yellow continues to mark everything new since 20260827**, so this revision's material is highlighted alongside 20260903's rather than in place of it.
20260912	2026-09-12	**Editorial revision**, superseding version 20260903. Adds to §4 a five-phase figure series elaborating the node allocation — Autoware as it stands, the same graph abstracted, the same boxes after the island, what changed at node level, and the island alone. No normative change: no behaviour, node, interface row or open decision is added, removed or altered
20260903	2026-09-03	**Major revision**, superseding version 20260827. *(1)* Adds **B4** as the fourth declared behaviour: what the island does when it has lost the sensing that B2 and B3 both depend on. Emergency stop was previously only step 3b of the §5.3 graded reaction and the bottom rung of the §6.3 abort ladder; it is now *declared*, with its own trigger, deadline, sensor rung and traceability row. *(2)* Admits the **first optional HPC-produced input** — the object list of §3.5 — which all four behaviours may use under the new **monotone-conservatism** rule, through the optional node `N20`. *(3)* **Corrects a misconception about termination:** B3 was written as *the* terminal behaviour. Both B3 and B4 end the drive cycle and neither gives it back; they differ in *where* the vehicle stops and on what evidence, not in whether the drive continues (§3.2). *(4)* **Both terminal behaviours now raise the hazard lamps at entry**, not at standstill, and the indicator conflict this creates for B3 is recorded for the WG. *(5)* Corrects the claim that B4 has no edge out: its one successor is `SAFE_STOP`, and what it has no edge to is any behaviour above it. *(6)* Adds the §6.3 **abort-ladder illustration** — each rung down removes a requirement and never adds one. Adds open decisions 15 and 16. Material new since 20260827 is marked in **marker yellow** throughout this document and the three artefacts built from it.
20260827	distributed 2026-08-28	First WG review draft. Three declared behaviours (B1–B3), node allocation, interface contract, fourteen open decisions.

0 Scope of this part

The project deliverable is a design document covering computing devices, communication, and software architecture for an L4 Safety Island for Autoware, with the software design as the focus. This part answers the first question:

> Which existing Autoware ROS nodes execute on the Safety Island, which remain on the HPC, and which **new** nodes must be created on each side?

It first **declares the six behaviours the island must exhibit** (§3), because the allocation is only defensible as a means of delivering them. It then defines the interface contract between the two domains, because the allocation cannot be justified without that either: a node's placement is only defensible once the messages it must still receive after the other domain has failed are enumerated.

Out of scope for Part 1 (planned as Parts 2–4): FTTI budget derivation and timing analysis, HARA / ASIL allocation evidence, fault-injection test plan, and the ISO 26262 work-product mapping.

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1 Inputs already fixed by the WG

These are taken as given from `SafetyIsland_RefDesignWG.pdf` (p.18–20) and `SaftyIsland_ReferenceDesign.md`, and constrain the allocation:

#	Decision	Source
D1	The L4 Safety Island is a **standalone, safety-certified ECU**, not an on-SoC partition	Deck p.18; RFC §"Hardware devices"
D2	HPC hosts perception / planning / VLM; the island hosts **status monitoring, the control module, and MRM**	Deck p.18
D3	HPC ↔ island link is **TSN / Vehicle Ethernet**; island → actuators is CAN FD / FlexRay	Deck p.20
D4	Island middleware is **nano-ros** (zenoh-pico core, ROS-message compatible, RTOS-portable)	CLAUDE.md; deck p.12
D5	Island OS and nodes must be **real-time compliant**: predictable execution time, memory safe	Deck p.18
D6	Every safety-relevant payload carries **end-to-end protection** — CRC, sequence counter, data ID, age; gPTP provides the time base. **AUTOSAR is out of scope: this is an Autoware in-house design**	RFC §"Communication protocol stack"; WG, 2026-08-27
D7	**Low-resolution, high-robustness sensors are connected directly to the island** — 2D lidar, forward camera, ultrasonic, radar — so that the MRM nodes execute without the HPC	WG, 2026-08-27
D8	**Not every island carries every D7 sensor, and they are ranked.** (1) **Forward camera — mandatory.** (2) **Radar** — forward, and rear-corner on the pull-over side. (3) **Ultrasonic array.** (4) **Island IMU / GNSS.** (5) **2D lidar — optional.** A behaviour may *require* only what is mandatory; what a lower-ranked sensor adds is a higher speed or a wider ODD, never the behaviour itself. The proprioceptive chassis signals — wheel speed, steering report, the ESC yaw rate and acceleration — arrive over the vehicle bus and are not island sensors in this ranking	WG, 2026-10-17
D9	**The island carries its own communication module to the remote command and control centre (RCC)** — a cellular (LTE / 5G) modem on the island's harness and power rail, not the HPC's — **and a local control port** for a joystick. Two human-driven behaviours are declared on them, B5 tele-operation and B6 manual joystick control; their stage-1 detailed design is in §6.6. **In this design the RCC communication is specified only by what B5 requires of it; the communication module, the RCC session and its security, and the video transport are deferred to the next design. The local control port is wired; a short-range radio link is deferred to the next design**	WG, 2026-10-17; scope, 2026-10-24

#	Decision	Source
D10	**The design is carried in two branches** that share the six behaviours, the node set and the interface contract, and differ in the hardware fitted to the island. **Least-Cost:** forward camera, five ultrasonic sensors, a 2D lidar and an IMU / GNSS on an NXP S32Z2, Infineon AURIX TC4x or ST Stellar microcontroller, with no radar. **Maximum-Safety:** forward camera, four radars (two front, two rear), five ultrasonic sensors and an IMU / GNSS on a Renesas R-Car X5H. How each behaviour is adjusted to each branch's sensors is in §9.5	Design lead, 2026-10-23

Two further facts come from work already completed at NEWSLab and are treated as the engineering baseline rather than as assumptions:

- **The Autoware trajectory follower already runs on an island.** `autoware-safety-island` (branch `nano-ros`) runs the vendored Autoware MPC lateral + PID longitudinal controllers as standalone firmware on Zephyr/FreeRTOS, subscribing to trajectory, odometry, steering report, acceleration and operation-mode state, and publishing `control_cmd` — wire-compatible with unmodified ROS 2 Humble Autoware, closed loop at $\geq$ 19 Hz against a 10 Hz trajectory, with a 250-point trajectory bound and `KEEP_LAST` depth 1 on every input.
- **Failover cadence has been measured.** On the RT-evaluation island (six nodes drawn from the Autoware vehicle-interface subsystem, FreeRTOS/zenoh-pico), control cadence held through a Linux outage at a mean period of 9.96 ms, the 500 ms watchdog tripped in 0.49–0.67 s, and MRM ramped the vehicle to standstill.

The allocation below is designed so that this existing island grows into the target island by addition, not by rewrite.

---

2 Allocation method

2.1 Partitioning rules

Each rule is a test applied to every candidate node. R1 is the decisive one.

Rule	Statement	Consequence
R1 — Survivability	If the function must still work *after the HPC is gone*, it executes on the island.	Forces the whole final-command path onto the island.
R2 — No orphan inputs	A node on the island must not depend on a topic that only the HPC can produce, unless it has a defined safe behaviour when that topic goes silent.	Forces new island-local state estimation and fault sources.
R3 — Monitor independence	A monitor must not share a failure mode with what it monitors.	HPC liveness/deadline supervision is island-resident, not HPC-resident.
R4 — Bounded resources	Island nodes must have statically bounded memory, bounded worst-case execution time, and no dynamic discovery.	Excludes map-dependent, learning-based, and solver-unbounded nodes from the ASIL tiers.

Rule	Statement	Consequence
R5 — Narrow interface	The HPC→island interface is typed, fixed-rate, rate-limited and E2E-protected. Anything that would widen it materially stays on the HPC.	Keeps HD map, point clouds and images off the link. The one object list admitted (§3.5) crosses only as a **bounded digest** — fixed object count, fixed horizon, fixed fields — and only as an option.
R6 — Single actuation authority	Exactly one component owns the final command to the actuator gateway, and it is on the island.	The gate is the arbitration point and the ASIL-D anchor.
R7 — Diversity over duplication	Where the island needs an independent view, prefer a *different* sensing/algorithmic channel over a copy of the HPC's.	Per D7: a physically separate low-resolution sensor set on the island, not a second copy of the HPC's high-resolution pipeline — ranked by D8, with the camera the one sensor every behaviour above B4 may assume.
R8 — Certifiability of the critical set	The ASIL-D set must be small enough to argue about. Complexity is permitted only below a monitoring layer.	Two-tier controller (§4.2).

2.2 Criticality tiers on the island

The island is not uniformly ASIL-D. It is layered, and the safety argument is ASIL decomposition by monitoring: a lower-integrity nominal controller is admissible because a higher-integrity arbiter bounds its output.

Tier	Target integrity	Contents	Cadence
T0 — Safety core	ASIL-D	Command gate, safety state machine, watchdog, emergency-stop MRM, actuator TX, E2E gateway, island vehicle-state estimator	100 Hz
T1 — Supervision	ASIL-B(D)	Control validator, fallback controller, safety-envelope monitor, corridor monitor, actuator-feedback supervisor	50 Hz
T1s — Safety sensing	ASIL-B(D)	Island sensor ingest, safety world model, AEB, near-field collision detection	20–50 Hz
T2 — Nominal control	QM, monitored by T0/T1	MPC lateral + PID longitudinal trajectory follower, raw vehicle command converter, shift decider	33–50 Hz
T3 — Non-real-time	QM	Event recorder, UDS/DTC diagnostics, telemetry egress	1–10 Hz

This is what makes the design tractable: **the MPC and its OSQP solver never need to be ASIL-D**, because T0 bounds every command it produces and can replace it within one control period.

The staleness rule for sensing. T1s runs slower than T0 and has a heavier, more variable execution profile, so T0 must never block on it. The island sensing chain publishes a small, bounded **safety world state** — nearest in-path obstacle, its range and range-rate, time-to-collision, free-space distance ahead, perceived corridor bounds, per-sensor validity — which T0 consumes with an explicit age check. **If that state is stale or invalid, T0 does not wait and does not guess: it substitutes the most conservative assumption and executes a blind deceleration MRM.** Island perception therefore *improves* the fallback; it is never a precondition for it. This is the property that keeps the ASIL-D argument independent of the perception chain.

3 Expected behaviours and functional allocation

This section is **normative**. Everything after it — the node allocation, the interface contract, the partitioning — exists to deliver these six behaviours, and each allocation decision must be traceable to one of them.

3.1 The six declared behaviours

ID	Behaviour	One-line statement
B1	**Degrade the ADS when the ODD cannot be held**	When the operating envelope is no longer satisfied, require the ADS to move to an operating domain it *can* satisfy — progressively constraining it, and demanding that it constrain itself, until it does — before any authority is taken.
B2	**Keep the vehicle in its lane after loss of the HPC**	When the HPC is unavailable and B1 therefore cannot be used, take authority and hold the vehicle in its current lane — for a bounded window of typically 8–15 s, at a decaying speed the island can itself justify, *first reduced to the safe speed of the current ODD* (e.g. 60 km/h on a highway, 30 km/h on a city road), while staging the exit manoeuvre — while the lane is still observable by the island's camera, pending the HPC's return or escalation to B3 or B4.
B3	**Pull over if the ODD continues to fail**	When degradation does not restore the envelope within a declared deadline, bring the vehicle out of the running lane to a standstill in a refuge — nominated by the HPC where one is offered, detected by the island's own sensors where one is not — with the **camera** tracking the lane lines and finding the space to pull over into.
B4	**Emergency stop when the island is blind**	When the island has lost the sensing that B2 and B3 both depend on — **the camera**, the one sensor both may assume (D8) — decelerate to a standstill where the vehicle already is — no lane keeping, no refuge, no lateral manoeuvre.
B5	**Tele-operation from the remote command centre**	A remote driver at the RCC drives the vehicle over the island's communication module, on sensor data the island streams to them — camera, radar and ultrasonic — while the island keeps its own guards armed and stops the vehicle itself if the link fails. *From standstill, or taken over from B1–B3 on the ADS's request; at most 20 km/h: §6.6.*
B6	**Manual joystick control without streamed sensing**	An operator with direct sight of the vehicle drives it with a joystick pendant on a cable, with no sensor stream; the island holds the speed to a creep and stops the vehicle on release or on loss of the link. *Stage 1, from standstill, 5 km/h: §6.6.*

B1–B4 are automated; B5 and B6 are human-driven. In the first four the island decides. In the last two a person decides — a remote driver at the RCC, or an operator standing at the vehicle — and the island executes, bounds and guards what that person commands. Both are declared by D9, so that the allocation reserves their nodes, their command sources and their place in the state machine; their sections below give entry, authority, speed, the guards that stay armed, the response to a lost link and the minimum sensor rung, and §6.6 holds the stage-1 detailed design.

Each behaviour is written below against the full D8 ranking. The two design branches of D10 each carry only part of that set, and §9.5 states how every behaviour is adjusted on each: the speed, the ODD and the evidence it can claim.

Each behaviour below closes with the island's node graph lit for it — phase 5 of the §10 series, one page per behaviour, drawn by the convention of `layers/L4SafetyIsland-Layers.md`. The six pages are collected in `layers/L4SafetyIsland-Behaviours.pdf`. The pages for B5 and B6 follow their §3.3 rows and the stage-1 detailed design of §6.6.

B1 — Degrade the ADS when the ODD cannot be held

Read as a constraint ladder, B1 is a demand that the ADS *slow down*. That is only its enforcement. B1's actual object is an **operating domain**: the island measures the world, declares that the domain the vehicle is currently being driven in is no longer satisfiable, and requires the HPC to move to one that is. The speed cap is what holds the vehicle safe *while* that happens — never what resolves it.

The island does not choose the new domain. It does not hold the HPC's catalogue of operating domains, does not know which L4 behaviours each of them admits, and choosing between them is planning — which is not on the island by R4 and R5. So the exchange is asymmetric, and deliberately so:

> **The island declares what the world will support. The HPC declares which domain it will operate in. The island verifies that answer against its own measurements, and accepts it or does not.**

That asymmetry is what makes B1 the only *exitable* behaviour. B3 and B4 end the drive cycle; B2 holds one open. B1 is the only behaviour that returns the vehicle to normal L4 operation — in a narrower domain than the one it started in, driven by the HPC, with the island back to supervising. It exits **upward** when the HPC is driving normally inside a domain the island has accepted, and **downward** — to B2, B3 or B4 — when the safety timeout `T_odd` expires without one.

Trigger	`N19` ODD margin falls below threshold (visibility, friction, precipitation, ambient light); or `N17` reports sensor degradation; or `H4` reports the HPC's own ODD exit; or `N18` reports `pull_over_available = false` while margin is already marginal
Precondition	**The HPC is alive and responsive.** B1 is cooperative by construction: every rung of it is executed by the HPC. If the HPC is unavailable there is nothing to constrain, and the island enters **B2** instead
Authority	**The island constrains; the HPC executes.** The island does not take control under B1.
Demand	`N19` publishes `odd_satisfiable = false` on `/si/out/odd_verdict`, with the measured envelope (`d_visible`, `a_brake`, derived `v_max`), the ODD dimension that failed, and the remaining `T_odd` budget. **The island names no target domain**: the demand is *leave this one*, not *enter that one*
Response	Graded, in order: *notify* → *demand an ODD switch* → *restrict* (hard speed cap from the speed law, §6.4) → *request MRM* from the HPC. The rungs are cumulative, not alternative — the cap stays on while the switch is negotiated, and is lifted only by an accepted switch or by the envelope's own recovery
Acceptance test	Two tests, both measurable on the island, neither needing the HPC's cooperation. **(1) Declaration** — the domain `H4` now reports as active must lie inside the island's measured envelope: speed ceiling at or below `v_max`, required sensing range at or below `d_visible`, assumed friction at or below the estimate. **(2) Behaviour** — measured speed, acceleration and curvature converge on that domain's bounds within `T_odd`. A declaration that passes (1) but fails (2) is a *false declaration*, and counts as no declaration at all

Deadlines	Declare a new active domain — or declare exhaustion — within `T_ack` = **2 s** of the demand; satisfy it within the safety timeout `T_odd` = **10 s** (open decision 17). Comply with a restriction within **3 s**; begin a requested MRM within **10 s**; complete it within **60 s**
Observable output	`/si/out/odd_verdict` (carrying `odd_satisfiable`, the failed dimension, the accepted domain once there is one, and the remaining `T_odd` budget), `/si/out/constraints` (mode `ALLOW` → `CONSTRAIN` → `MRM`). `MrmState` remains `NORMAL` throughout: **B1 is not an MRM**
Success criterion	The HPC declares a domain that passes both acceptance tests and holds it over the recovery window — *or*, where the envelope recovers on its own, measured vehicle speed converges to `v_max` within the deadline, and the ODD margin is restored without a switch having been needed
Exit — upward: the ADS operates in an accepted domain	Both acceptance tests pass and hold for the §6.4 recovery window — ~30 s of clear evidence, with `H4`'s own verdict agreeing. `N3` returns `/si/out/constraints` to `ALLOW` and the island to supervision. **The vehicle is driving normally again, in a narrower domain than the one it started in** — that narrowing is B1's *outcome*, not a residual fault, and it persists until the envelope itself recovers
Exit — downward: `T_odd` expires	Escalate. *Which* behaviour takes over is decided by what the island still **has**, not by what the HPC did wrong — sensing blackout (`N17`) → **B4**; HPC unavailable (`N2`) → **B2**; otherwise, the HPC being alive and either non-compliant or out of satisfiable domains → **B3**. The tests are applied in that order
Failure	Non-compliance within any deadline is raised to `N8` as a fault and escalates by the rule in the row above. A domain that passes acceptance test (1) but fails (2) is additionally recorded by `N11` as a SOTIF event: an HPC reporting a domain it is not in is §6.4's confidently-wrong case, arriving through a second channel
Hysteresis	Degrade on ~2 s of consistent evidence; recover only after ~30 s of clear evidence **and** agreement from `H4`
Optional enrichment (§3.5)	The HPC object list, while it is fresh. Feeds the **self-consistency veto** and adds a conservative term to `N10`'s envelope. Never relaxes a bound and never satisfies a deadline
Minimum sensor rung (§9.4)	**B** — visibility estimation needs both the radar-versus-camera divergence test and the camera contrast test. The friction dimension needs the island IMU, rung **D**; below it, friction falls back to the chassis ESC signals and is claimed with correspondingly less margin

Honest exhaustion is not non-compliance. The HPC may answer the demand with *none of my domains is satisfiable.* That is a **compliant** answer and must not be run out as a timeout. It is the rung the §6.4 ladder already calls **Request MRM** — ask the HPC to pull over while it still can, with the map, the traffic law and the full planner available to it. An HPC-executed pull-over under B1 is strictly better than an island-executed one under B3: the island can verify a refuge, but it cannot choose one (§6.3). Exhaustion therefore routes to the MRM rung immediately, and only the MRM deadlines — 10 s to begin, 60 s to complete — then apply. **`T_odd` bounds silence and non-compliance, not honesty.**

`T_odd` is not a free parameter. The safety timeout is bounded from above by physics the island is already measuring. While the switch is negotiated the envelope keeps shrinking, and B3 has a precondition — a refuge that can still be reached and verified. `T_odd` must be shorter than the time in which the ODD margin, falling at its measured rate, would invalidate that precondition:

text — source form; renders as a figure in the Markdown original

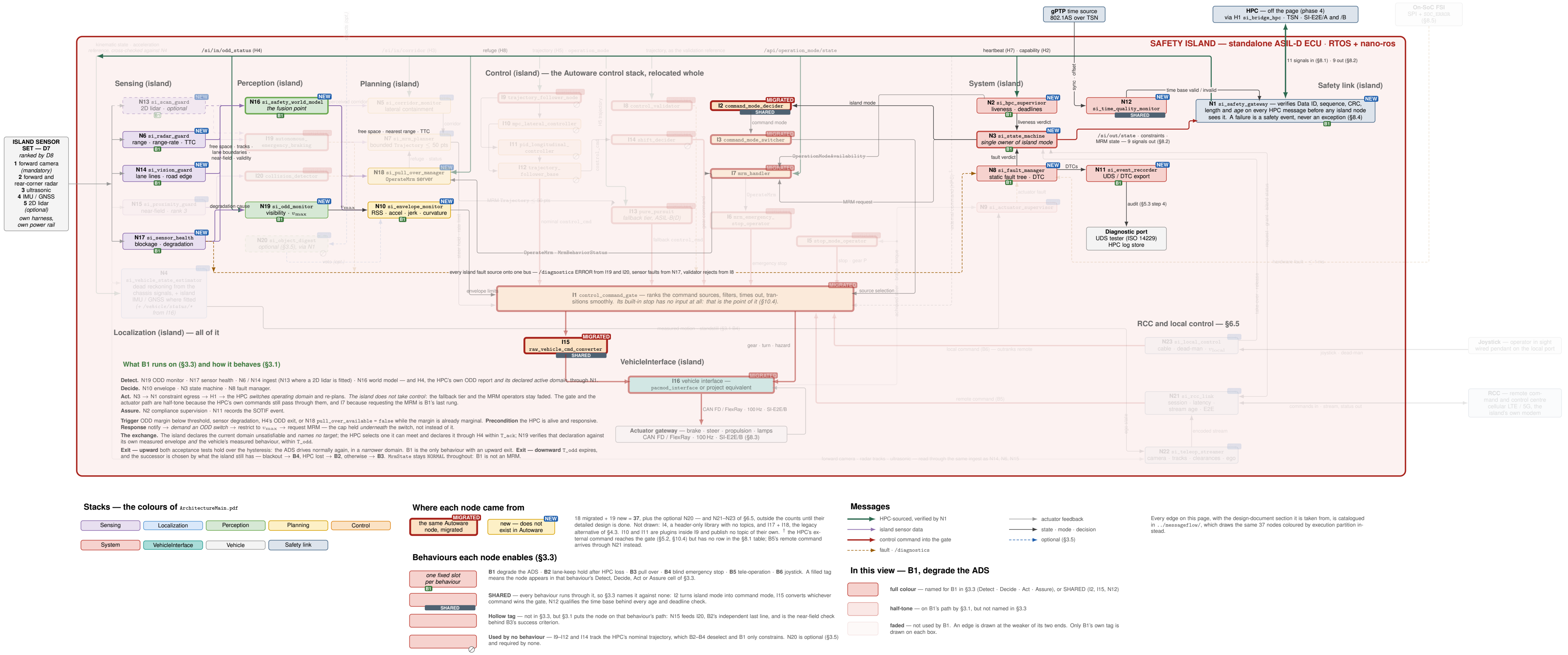
```
T_odd  <  t_margin  -  t_pullover

  t_margin    time until the measured ODD margin reaches zero, at its current rate
  ↪  (N19)
```

```
t_pullover  time to execute a pull-over from the current speed to the refuge
↪  (N18)
```

An island that negotiates until the moment the envelope fails has spent the budget the fallback needed. This is §6.3's refuge coverage guarantee, restated on the time axis instead of the distance axis, and it is why `T_odd` is declared per-situation by `N19` rather than fixed as a constant.

Phase 5 · B1 — Degrade the ADS when the ODD cannot be held

When the operating envelope is no longer satisfied, require the ADS to move to an operating domain it *can* satisfy — progressively constraining it, and demanding that it constrain itself, until it does — before any authority is taken.
The phase-5 drawing, unchanged: what B1 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.


SAFETY ISLAND — standalone ASIL-D ECU · RTOS + nano-ros
ISLAND SENSOR SET — D7
ranked by D8
1 forward camera (mandatory)
2 forward and rear-corner radar
3 ultrasonic
4 IMU / GNSS
5 2D lidar (optional)
own harness, own power rail
kinematic state · acceleration
reference, cross-checked against N4
/si/in/odd_status (H4)
/si/in/corridor (H3)
objects (opt.)
refuge (H8)
trajectory (H5) · operation_mode
trajectory, as the validation reference
/api/operation_mode/state
heartbeat (H7) · capability (H2)
Sensing (island)
N13 si_scan_guard
2D lidar · optional
N6 si_radar_guard
range · range-rate · TTC
N14 si_vision_guard
lane lines · road edge
N15 si_proximity_guard
near-field · rank 3
N17 si_sensor_health
blockage · degradation
NEW
B1
Perception (island)
N16 si_safety_world_model
the fusion point
I19 autonomous_emergency_braking
I20 collision_detector
N19 si_odd_monitor
visibility · $v_{\max}$
N20 si_object_digest
optional (§3.5), via N1
free space · tracks · lane boundaries · near-field · validity
degradation cause
$v_{\max}$
veto (opt.)
perceived corridor
Planning (island)
N5 si_corridor_monitor
lateral containment
corridor
free space · nearest range · TTC
N7 si_mrm_planner
bounded Trajectory ≤ 50 pts
refuge · status
N18 si_pull_over_manager
OperateMrm server
N10 si_envelope_monitor
RSS · accel · jerk · curvature
MRM Trajectory ≤ 50 pts
steer hold · rate limit
N4
si_vehicle_state_estimator
dead reckoning from the chassis signals, + island IMU / GNSS where fitted
(+ /vehicle/status/* from I16)
Localization (island) — all of it
Control (island) — the Autoware control stack, relocated whole
I9 trajectory_follower_node
I10 mpc_lateral_controller
I11 pid_longitudinal_controller
I12 trajectory_follower_base
H5 trajectory
control_cmd
I8 control_validator
I14 shift_decider
I13 pure_pursuit
fallback tier, ASIL-B(D)
nominal control_cmd
fallback control_cmd
gear command
I2 command_mode_decider
MIGRATED
SHARED
command mode
I3 command_mode_switcher
island mode
OperationModeAvailability
I7 mrm_handler
OperateMrm
MRM request
I6 mrm_emergency_stop_operator
emergency stop
I5 stop_mode_operator
stop · gear P
OperateMrm · MrmBehaviorStatus
every island fault source onto one bus — /diagnostics ERROR from I19 and I20, sensor faults from N17, validator rejects from I8
envelope limits
source selection
I1 control_command_gate — ranks the command sources, filters, times out, transitions smoothly. Its built-in stop has no input at all: that is the point of it (§10.4).
measured motion · standstill (§3.1 B4)
I15
raw_vehicle_cmd_converter
gear · turn · hazard
VehicleInterface (island)
I16 vehicle interface — pacmod_interface or project equivalent
CAN FD / FlexRay · 100 Hz · SI-E2E/B
Actuator gateway — brake · steer · propulsion · lamps
CAN FD / FlexRay · 100 Hz · SI-E2E/B (§8.3)
external command (HPC)
achieved steer · brake · torque
gPTP time source
802.1AS over TSN
sync · offset
HPC — off the page (phase 4)
via H1 si_bridge_hpc · TSN · SI-E2E/A and /B
On-SoC FSI
SPI + SOC_ERROR
(§8.5)
11 signals in (§8.1) · 9 out (§8.2)
System (island)
N2 si_hpc_supervisor
liveness · deadlines
liveness verdict
N3 si_state_machine
single owner of island mode
fault verdict
N8 si_fault_manager
static fault tree · DTC
actuator fault
N9 si_actuator_supervisor
N12
si_time_quality_monitor
time base valid / invalid
/si/out/state · constraints · MRM state — 9 signals out (§8.2)
DTCs
N11 si_event_recorder
UDS / DTC export
audit (§5.3 step 4)
Diagnostic port
UDS tester (ISO 14229)
HPC log store
Safety link (island)
N1 si_safety_gateway — verifies Data ID, sequence, CRC, length and age on every HPC message before any island node sees it. A failure is a safety event, never an exception (§8.4)
request · grant · island status
take-over · release
hardware fault · ≤ 1 ms
RCC and local control — §6.5
N23 si_local_control
cable · dead-man · v_{local}
local command (B6) — outranks remote
Joystick — operator in sight
wired pendant on the local port
joystick · dead-man
N21 si_rcc_link
session · latency · stream age · E2E
remote command (B5)
RCC — remote command and control centre
cellular LTE / 5G, the island's own modem
commands in · stream, status out
encoded stream
ego state
N22 si_teleop_streamer
camera · tracks · clearances · ego
forward camera · radar tracks · ultrasonic — read through the same ingest as N14, N6, N15

What B1 runs on (§3.3) and how it behaves (§3.1)

Detect. N19 ODD monitor · N17 sensor health · N6 / N14 ingest (N13 where a 2D lidar is fitted) · N16 world model — and H4, the HPC's own ODD report *and its declared active domain*, through N1.

Decide. N10 envelope · N3 state machine · N8 fault manager.

Act. N3 → N1 constraint egress → H1 → the HPC *switches operating domain* and re-plans. *The island does not take control*: the fallback tier and the MRM operators stay faded. The gate and the actuator path are half-tone because the HPC's own commands still pass through them, and I7 because requesting the MRM is B1's last rung.

Assure. N2 compliance supervision · N11 records the SOTIF event.

Trigger ODD margin below threshold, sensor degradation, H4's ODD exit, or N18 `pull_over_available = false` while the margin is already marginal. **Precondition** the HPC is alive and responsive.
Response notify → *demand an ODD switch* → restrict to $v_{\max}$ → request MRM — the cap held *underneath* the switch, not instead of it.

The exchange. The island declares the current domain unsatisfiable and *names no target*; the HPC selects one it can meet and declares it through H4 within `T_ack`; N19 verifies that declaration against its own measured envelope *and* the vehicle's measured behaviour, within `T_odd`.

Exit — upward both acceptance tests hold over the hysteresis: the ADS drives normally again, in a *narrower* domain. B1 is the only behaviour with an upward exit. **Exit — downward** `T_odd` expires, and the successor is chosen by what the island still has — blackout → **B4**, HPC lost → **B2**, otherwise → **B3**. `MrmState` stays `NORMAL` throughout: B1 is not an MRM.

Stacks — the colours of `ArchitectureMain.pdf`

Sensing
Localization
Perception
Planning
Control
System
VehicleInterface
Vehicle
Safety link

Where each node came from

MIGRATED
the same Autoware node, migrated
NEW
new — does not exist in Autoware

18 migrated + 19 new = **37**, plus the optional N20 — and N21–N23 of §6.5, outside the counts until their detailed design is done. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. † the HPC's external command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table; B5's remote command arrives through N21 instead.

Behaviours each node enables (§3.3)

one fixed slot per behaviour
B1
SHARED

B1 degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

Used by no behaviour — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

HPC-sourced, verified by N1
island sensor data
control command into the gate
fault · /diagnostics
actuator feedback
state · mode · decision
optional (§3.5)

Every edge on this page, with the design-document section it is taken from, is catalogued in `../messageflow/`, which draws the same 37 nodes coloured by execution partition instead.

In this view — B1, degrade the ADS

full colour — named for B1 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)

half-tone — on B1's path by §3.1, but not named in §3.3

faded — not used by B1. An edge is drawn at the weaker of its two ends. Only B1's own tag is drawn on each box.

Figure 1. *B1 on the island's node graph: what B1 runs on in full colour, what it only passes through in half-tone, the rest of the island faded*

B1, on the island's node graph. The nodes of the B1 row of §3.3 are in full colour (the optional N13 half-tone, where fitted); the gate and the actuator path are half-tone because the HPC's own commands still pass through them, and so is I7, since requesting the MRM is B1's last rung. The fallback tier and the MRM operators stay faded: the island does not take control. Full size: `layers/L4SafetyIsland-Layer6-B1-DegradeADS.pdf`; all six behaviours on one handout: `layers/L4SafetyIsland-Behaviours.pdf`.

B2 — Keep the vehicle in its lane after loss of the HPC

B2 is the island's **autonomous holding behaviour**, and it exists for exactly one situation: the HPC is unavailable, so B1 cannot be used — every rung of B1 is executed by the HPC — while B3 is not yet warranted, because the loss may still be transient and a pull-over is the more dangerous manoeuvre. Rather than commit immediately to a terminal manoeuvre, the island takes lateral and longitudinal authority, holds the vehicle in its current lane at a speed it continuously reduces, and waits, for a bounded window, to see whether the HPC comes back.

Two things about that window are now fixed rather than open. **Its first act is to bring the vehicle down to the safe speed of the ODD it is in** — the speed at which a fallback controller on the island's reduced sensor set may hold a lane of that road class: of the order of **60 km/h on a highway and 30 km/h on a city road**. The HPC was entitled to drive faster because it had the full sensor suite, the map and the planner; the island has none of them, and the first thing it does on taking authority is to stop driving at a speed only the HPC could justify. **And its length is of the order of 8–15 s**, not a few seconds: long enough to ride out a restart of the HPC's planning stack, short enough that the vehicle does not loiter under a fallback controller. When it closes with the HPC still lost, B2 hands over to exactly one of two behaviours — B3 if the island can still see the lane, B4 if it cannot.

The hold is not a plateau. Throughout the window the island is doing two things at once: waiting for the HPC, and **preparing to stop without it** — shedding speed onto the ramp defined below and keeping `N18`'s refuge evaluation live, so that when the window closes the exit manoeuvre is already staged. A hold that merely maintains a constant speed and then discovers, at `T_hold`, that it must begin decelerating from that speed has spent the whole window buying nothing.

The behaviour is only enterable while the **lane is still observable** — by the island's own camera, which is the one sensor D8 makes mandatory. The buffered map corridor refines what the camera sees while it is fresh, but it does not substitute for it: a map corridor with no live camera behind it is a remembered lane, not an observed one. That precondition is what distinguishes B2 from a blind stop, and it is checked continuously, not only on entry.

Trigger	`N2` declares the HPC unavailable — heartbeat timeout, per-topic rate violation, or latency beyond the declared budget — and `N8` raises `HPC_LIVENESS_LOST` (or `HPC_COMMAND_INVALID`) while the ADS is engaged. Stated behaviourally: **B1 is unavailable because its executor is gone**
Precondition	A **valid corridor** exists — `N17` holds the camera valid and `N14` reports lane boundaries with sufficient confidence. `H3`'s buffered map corridor, while within its age limit, is fused with them but cannot stand in for them. Without one, B2 is not entered: the island cannot see the lane, and goes directly to B4
Corridor source, in priority order	1. `N14` camera lane boundaries — live, and required · 2. `H3` map-derived corridor while fresh — fused with 1 to refine its geometry, and to bridge a camera dropout no longer than `N17`'s debounce (open decision 15). There is no third source: the heading / yaw-rate tube of earlier revisions is withdrawn, because a lane the island cannot see is one it hands to B4
Authority	**The island steers and brakes.** `I9`–`I11` are deselected; no HPC command is in the loop
Response	Take authority → `I13 pure_pursuit` tracks the corridor centreline → **decelerate to the safe speed of the current ODD, `v_odd`**, then continue down the `v_hold` ramp defined in the next row — not to a constant hold speed → maintain lane, obstacle-aware against `N16` → hazard lamps on → keep `N18`'s refuge evaluation running throughout — the camera searching the road edge for pull-over space while it holds the lane lines — so B3 inherits a staged manoeuvre rather than starting one → hold for at most `T_hold`
Speed — `v_odd`	**The safe speed of the ODD the vehicle is in**, by road class: of the order of **60 km/h on a highway** and **30 km/h on a city road** (the table of classes and values is open decision 13). The class is the one carried by the `active_odd` `H4` last declared, buffered by `N3` at B2 entry. A declaration that is stale or missing selects the **most restrictive class**; rung-C GNSS may corroborate the class against a coarse road-class geofence, and may only lower it. `v_odd` is a ceiling, never a target to accelerate to: a vehicle already below it stays below it
Speed — `v_hold`	**Not a constant, and not `v_max`.** `v_hold` is the running minimum of five bounds, re-evaluated every cycle, and it *decays across the hold window* — the full definition is below the table. `v_odd` is the term that dominates first; the others take over as the window runs. The island is driving on its own reduced sensor set, under a fallback controller, inside the island's own narrower ODD (open decision 13); every one of those is a reason `v_hold` sits below the speed the HPC was holding
Deadlines	Authority transfer complete within the §9.2 fault-reaction budget ($\le$ 70 ms from first heartbeat miss); `v_odd` reached at the declared fallback deceleration (≤ 2 m/s^2, open decision 14), `v_hold` tracked continuously thereafter as it decays; `T_hold` is declared and finite — **8–15 s**, set per ODD class (open decision 14) — the hold is never open-ended — and at `T_hold` the vehicle must already be at `v_entry`, so that escalation to B3 costs no further deceleration time
Observable output	`MrmState` = `MRM_OPERATING`; `N3` mode = `LANE_KEEP_HOLD`; `/si/out/state` carries the authority holder, the corridor source in use, lateral deviation and the remaining hold budget
Bound	Lateral deviation from corridor centre within the declared threshold; corridor validity age-checked per source; `N7` bounds lateral acceleration, steering rate and jerk ; `v_hold` is monotone non-increasing for the whole episode
Success criterion	Lateral deviation stays inside the threshold for the whole episode, measured speed stays at or below `v_hold` throughout, **and** the episode ends in one of the two authorised exits below — never by `T_hold` simply lapsing into an undefined state
Exit — upward: the HPC resumes	`N2` observes the heartbeat restored **and** valid, in-envelope trajectories for a declared confirmation window. `N3` then hands authority back through `I3`'s smooth-transition path and the island returns to supervision under **B1** — entering at `CONSTRAIN`, not `ALLOW`, and releasing only under the §6.4 hysteresis. **This is the only reversible *authority transfer* in the ladder** — the other edge back up, B1 → `ADS_ACTIVE`, moves no authority, because the HPC drives throughout it
Exit — downward: the HPC stays lost	`T_hold` expires with no confirmed recovery → **exactly one of two successors**, decided by what the island can still see, tested in this order: **B4** if `N17` no longer holds the camera valid, or `N14` no longer sees the lane lines; otherwise **B3**, entered at `v_entry` with a refuge already under evaluation rather than at the hold speed with nothing staged. If `N18` then concludes that no refuge exists, B3 degenerates to an in-lane stop, executed under B2's own authority and control law — an outcome of B3's search, not a third exit from B2
Failure	Camera lost, or lane lines lost beyond the bridging budget, mid-episode → **B4 at once**, no waiting for `T_hold`: it is the downward exit's own first test, applied early. `I19` AEB fires, or `N9` reports the actuators not tracking → emergency stop (B4)
Optional enrichment (§3.5)	The last HPC object digest inside its carry-over budget, to lower `v_hold` or shorten `N7`'s ramp. Never extends `T_hold` and never substitutes for the corridor — and the same holds for every sensor below the camera: an ultrasonic or 2D lidar return may add a hazard, never a corridor

Minimum sensor rung (§9.4)	**A** — the forward camera plus the chassis signals. B2 is the behaviour that must survive on the smallest sensor set. At rung A `v_max` is computed on the camera's range alone and will usually sit below the highway class's 60 km/h, so the highway `v_odd` is in practice reached only at rung **B**, where forward radar supplies the range

The `v_hold` ramp

`v_hold` is the running minimum of five bounds, re-evaluated every cycle. Four of them are static ceilings; the fifth is what makes the hold a ramp:

text — source form; renders as a figure in the Markdown original

```text
v_hold(t) = min( v_max,         S6.4 speed law, on island sensing alone -- stop
↪  within what you see    (N19)
                 v_odd,         safe speed of the current ODD class -- ~60 km/h
                 ↪  highway, ~30 km/h city
                                road; class latched from H4's last active_odd (N3),
                                ↪  open decision 13
                 v_corridor,    the ceiling the corridor source currently in use
                 ↪  can support         (N5)
                 v_track,       speed at which I13 holds the corridor curvature
                 ↪  inside N7's bounds
                 v_exit(t) )    speed from which the staged exit manoeuvre is still
                 ↪  executable       (N18)

v_exit(t)  =  the profile that arrives at v_entry when t reaches T_hold
v_entry    =  the speed at which B3 enters the pull-over, or at which the in-lane
↪  stop begins
```

Two rules govern it, and both are normative.

`v_hold` is monotone non-increasing within an episode. It never rises. A corridor source that degrades mid-episode — the map corridor ageing out, the markings fading, the camera's lane confidence falling — lowers `v_corridor` and therefore `v_hold`, and nothing raises it again. The only release is the authority hand-back, and that leaves B2 altogether. An island that recovers speed on its own evidence, while the HPC is still gone, has quietly promoted itself from *holding* to *driving*.

The ramp lands on `v_entry`, not on the hold speed. `v_exit(t)` is what converts the hold window from dead time into preparation. At `t = T_hold` the vehicle is already at the speed B3 needs, so the downward exit is a *continuation* of a deceleration already underway rather than the start of a new one. This is the same bounding argument B1's `T_odd` makes: the window a behaviour spends waiting must not consume the budget the behaviour underneath it needs.

`v_odd` is where the window's budget goes first, and it is why the window is 8–15 s. The deceleration to the ODD's safe speed is not free, and it is the largest single item in the hold:

Road class	**Typical entry speed**	**v_odd**	**Time to v_odd at 2 m/s^2**	**Distance**	**Left of an 8 s / 15 s window**
Highway	100 km/h	60 km/h	5.6 s	124 m	2.4 s / 9.4 s
City road	50 km/h	30 km/h	2.8 s	31 m	5.2 s / 12.2 s

Road class	**Typical entry speed**	**v_odd**	**Time to v_odd at 2 m/s^2**	**Distance**	**Left of an 8 s / 15 s window**

A window of 2–5 s, which earlier revisions proposed, would spend itself entirely on the first row and leave nothing for the HPC to come back in — every transient loss on a highway would become a pull-over. The window therefore has to be long enough to *contain* the descent to `v_odd` and still leave the HPC a real chance to recover, and that is what sets its floor near 8 s. Its ceiling near 15 s is set by the other side: the time a fallback controller on a camera-held corridor should be trusted to keep a lane at all. Within those bounds `T_hold` is declared per road class (open decision 14).

Note what this rules out. `v_hold` is **not** the HPC's last commanded speed, **not** the posted limit, and **not** `v_max` alone — `v_max` answers only *can I stop within what I can see*, which is necessary and nowhere near sufficient for a fallback controller tracking a corridor it may be inferring from camera boundaries. Every term above is measured or declared on the island, so `v_hold` remains defensible after the HPC is gone — which is the only condition under which it is ever used.

The two corridor sources degrade differently, and B2's ceiling degrades with them: the buffered map corridor is exact but ages out within seconds of the HPC's loss, while the camera's lane boundaries hold for as long as the markings are visible. For most of an 8–15 s window the camera is therefore the corridor, and `v_corridor` is set by its lane-detection confidence and range. When the camera can no longer hold the lane there is nothing below it to fall to — the heading tube is not a lane-keeping source — and the episode leaves for B4.

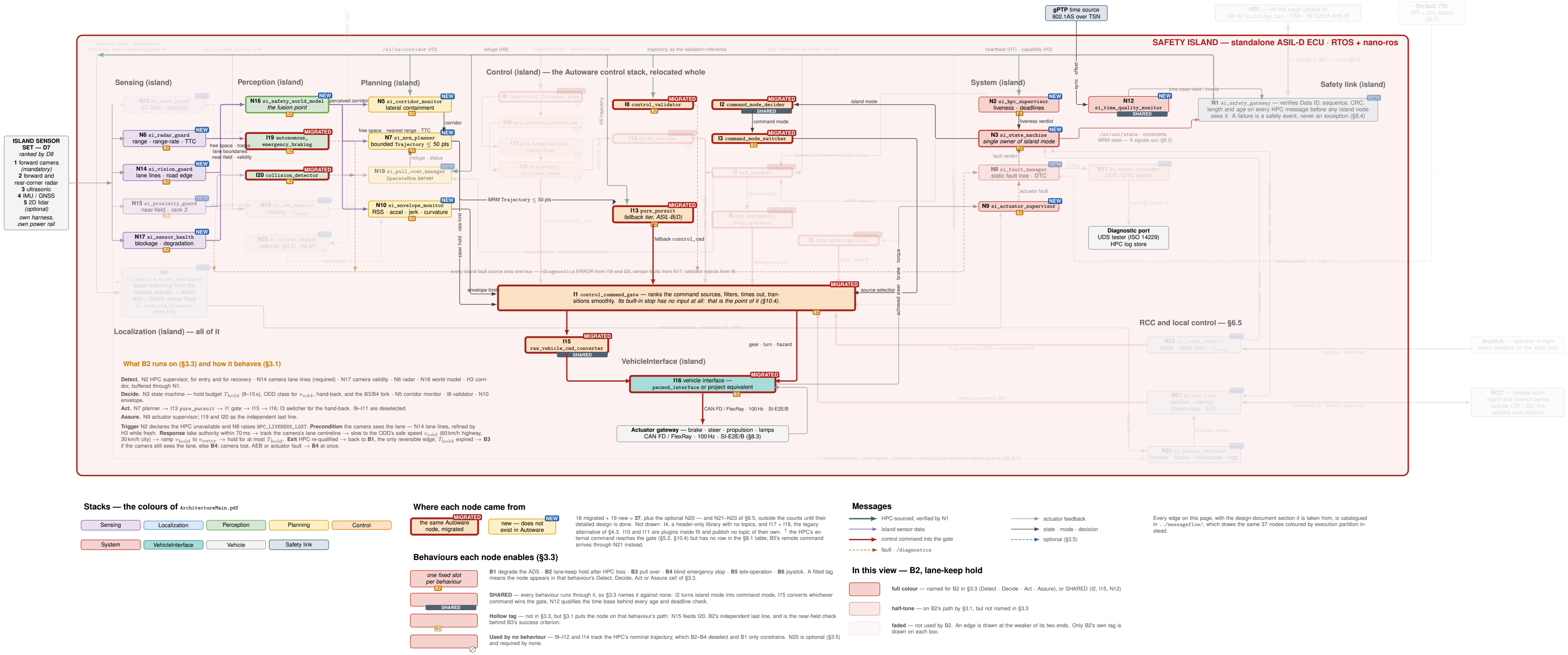
Lateral containment supervision is not B2. The continuous check — `N5` against the corridor, with `I8` and `N10` bounding commanded curvature and steering rate *while the HPC steers* — runs whenever the ADS is engaged, including throughout B1 and B3. It is a **supervisory function**, the *Validate* block of §3.4, and its output is a veto and a fault, not a behaviour. B2 is what the island does when it must produce the steering itself. The distinction is worth holding because the two differ in every property that matters for the safety argument:

	Containment supervision (always on)	**B2 (episode)**
When	Whenever the ADS is engaged, HPC steering	HPC lost, corridor still valid
Authority	Island **vetoes**; the HPC steers	Island **steers**
Mechanism	`N5` corridor check; `I8` + `N10` bound commanded curvature and steering rate	`I13 pure_pursuit` on the corridor centreline; `N7` bounds lateral acceleration, steering rate and jerk
Failure response	Constrain → fault to `N8` → B1, then B3	B3 while the camera still sees the lane, else emergency stop (B4)

Explicit non-goals. B2 is lane *keeping*, not lane *choosing*. The island does not change lane, negotiate a merge, handle an intersection, or select a route. Those need map and planner semantics that are deliberately not on the island (R4, R5). Nor is B2 a *continuation of the mission*: it holds position in the current lane, it does not resume route following, and a returning HPC re-acquires the mission only through the authority hand-back above.

Phase 5 · B2 — Keep the vehicle in its lane after loss of the HPC

When the HPC is unavailable and B1 therefore cannot be used, take authority and hold the vehicle in its current lane — for a bounded 8–15 s window, first down to the ODD's safe speed — while the camera still sees the lane.
The phase-5 drawing, unchanged: what B2 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.


gPTP time source
802.1AS over TSN
HPC — off the page (phase 4)
via H1 si_bridge_hpc · TSN · SI-E2E/A and /B
On-SoC FSI
SPI + SOC_ERROR
(§8.5)
SAFETY ISLAND — standalone ASIL-D ECU · RTOS + nano-ros
objects (opt.)
kinematic state · acceleration
reference, cross-checked against N4
/si/in/odd_status (H4)
/si/in/corridor (H3)
refuge (H8)
trajectory (H5) · operation_mode
trajectory, as the validation reference
/api/operation_mode/state
heartbeat (H7) · capability (H2)
sync · offset
11 signals in (§8.1) · 9 out (§8.2)
ISLAND SENSOR SET — D7
ranked by D8
1 forward camera
(mandatory)
2 forward and
rear-corner radar
3 ultrasonic
4 IMU / GNSS
5 2D lidar
(optional)
own harness,
own power rail
Sensing (island)
N13 si_scan_guard
2D lidar · optional
N6 si_radar_guard
range · range-rate · TTC
N14 si_vision_guard
lane lines · road edge
N15 si_proximity_guard
near-field · rank 3
N17 si_sensor_health
blockage · degradation
N4
si_vehicle_state_estimator
dead reckoning from the
chassis signals, + island
IMU / GNSS where fitted
(+ /vehicle/status/*
from I16)
Localization (island) — all of it
Perception (island)
N16 si_safety_world_model
the fusion point
I19 autonomous_
emergency_braking
I20 collision_detector
N19 si_odd_monitor
visibility · v_{max}
N20 si_object_digest
optional (§3.5), via N1
free space · tracks ·
lane boundaries ·
near-field · validity
degradation cause
veto (opt.)
perceived corridor
Planning (island)
N5 si_corridor_monitor
lateral containment
corridor
free space · nearest range · TTC
N7 si_mrm_planner
bounded Trajectory ≤ 50 pts
refuge · status
N18 si_pull_over_manager
OperateMrm server
N10 si_envelope_monitor
RSS · accel · jerk · curvature
v_{max}
MRM Trajectory ≤ 50 pts
steer hold · rate limit
envelope limits
Control (island) — the Autoware control stack, relocated whole
I9 trajectory_follower_node
I10 mpc_lateral_controller
I11 pid_longitudinal_
controller
I12 trajectory_
follower_base
H5 trajectory
control_cmd
I8 control_validator
I14 shift_decider
I13 pure_pursuit
fallback tier, ASIL-B(D)
nominal control_cmd
fallback control_cmd
gear command
I2 command_mode_decider
command mode
I3 command_mode_switcher
island mode
OperationModeAvailability
I7 mrm_handler
OperateMrm
MRM request
I6 mrm_emergency_
stop_operator
I5 stop_mode_operator
emergency stop
stop · gear P
OperateMrm · MrmBehaviorStatus
every island fault source onto one bus — /diagnostics ERROR from I19 and I20, sensor faults from N17, validator rejects from I8
I1 control_command_gate — ranks the command sources, filters, times out, transitions smoothly. Its built-in stop has no input at all: that is the point of it (§10.4).
source selection
achieved steer · brake · torque
external command (HPC) †
measured motion · standstill (§3.1 B4)
I15
raw_vehicle_cmd_converter
gear · turn · hazard
VehicleInterface (island)
I16 vehicle interface —
pacmod_interface or project equivalent
CAN FD / FlexRay · 100 Hz · SI-E2E/B
Actuator gateway — brake · steer · propulsion · lamps
CAN FD / FlexRay · 100 Hz · SI-E2E/B (§8.3)
System (island)
N2 si_hpc_supervisor
liveness · deadlines
liveness verdict
N3 si_state_machine
single owner of island mode
fault verdict
N8 si_fault_manager
static fault tree · DTC
DTCs
actuator fault
N9 si_actuator_supervisor
N12
si_time_quality_monitor
time base valid / invalid
N11 si_event_recorder
UDS / DTC export
audit (§5.3 step 4)
Diagnostic port
UDS tester (ISO 14229)
HPC log store
/si/out/state · constraints ·
MRM state — 9 signals out (§8.2)
Safety link (island)
N1 si_safety_gateway — verifies Data ID, sequence, CRC, length and age on every HPC message before any island node sees it. A failure is a safety event, never an exception (§8.4)
hardware fault · ≤ 1 ms
request · grant · island status
take-over · release
RCC and local control — §6.5
N23 si_local_control
cable · dead-man · v_{local}
local command (B6) — outranks remote
joystick · dead-man
Joystick — operator in sight
wired pendant on the local port
N21 si_rcc_link
session · latency ·
stream age · E2E
remote command (B5)
commands in · stream, status out
RCC — remote command and control centre
cellular LTE / 5G, the
island's own modem
encoded stream
ego state
N22 si_teleop_streamer
camera · tracks · clearances · ego
forward camera · radar tracks · ultrasonic — read through the same ingest as N14, N6, N15
NEW
MIGRATED
SHARED
B2
Stacks — the colours of ArchitectureMain.pdf
Sensing
Localization
Perception
Planning
Control
System
VehicleInterface
Vehicle
Safety link
Where each node came from
MIGRATED
the same Autoware node, migrated
NEW
new — does not exist in Autoware
Behaviours each node enables (§3.3)
one fixed slot
per behaviour
B2
SHARED
B2
Messages
HPC-sourced, verified by N1
island sensor data
control command into the gate
fault · /diagnostics
actuator feedback
state · mode · decision
optional (§3.5)
In this view — B2, lane-keep hold

What B2 runs on (§3.3) and how it behaves (§3.1)

Detect. N2 HPC supervisor, for entry and for recovery · N14 camera lane lines (required) · N17 camera validity · N6 radar · N16 world model · H3 corridor, buffered through N1.

Decide. N3 state machine — hold budget T_{hold} (8–15 s), ODD class for v_{odd}, hand-back, and the B3/B4 fork · N5 corridor monitor · I8 validator · N10 envelope.

Act. N7 planner → I13 `pure_pursuit` → I1 gate → I15 → I16; I3 switcher for the hand-back. I9–I11 are deselected.

Assure. N9 actuator supervisor; I19 and I20 as the independent last line.

Trigger N2 declares the HPC unavailable and N8 raises `HPC_LIVENESS_LOST`. **Precondition** the camera sees the lane — N14 lane lines, refined by H3 while fresh. **Response** take authority within 70 ms → track the camera's lane centreline → slow to the ODD's safe speed v_{odd} (60 km/h highway, 30 km/h city) → ramp v_{hold} to v_{entry} → hold for at most T_{hold}. **Exit** HPC re-qualified → back to **B1**, the only reversible edge; T_{hold} expired → **B3** if the camera still sees the lane, else **B4**; camera lost, AEB or actuator fault → **B4** at once.

18 migrated + 19 new = **37**, plus the optional N20 — and N21–N23 of §6.5, outside the counts until their detailed design is done. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. † the HPC's external command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table; B5's remote command arrives through N21 instead.

B1 degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

Used by no behaviour — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

full colour — named for B2 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)

half-tone — on B2's path by §3.1, but not named in §3.3

faded — not used by B2. An edge is drawn at the weaker of its two ends. Only B2's own tag is drawn on each box.

Every edge on this page, with the design-document section it is taken from, is catalogued in `../messageflow/`, which draws the same 37 nodes coloured by execution partition instead.

Figure 2. *B2 on the island's node graph: what B2 runs on in full colour, what it only passes through in half-tone, the rest of the island faded*

B2, on the island's node graph. The island steers: I13 with N7's planner, through the gate to I16, on the lane lines N14's camera holds; N6 radar and N17's camera-validity verdict are lit, the optional N13 is not. N1 and N8 are half-tone because the hold's entry and recovery verdicts pass through them; the HPC itself is faded. Full size: `layers/L4SafetyIsland-Layer6-B2-LaneKeepHold.pdf`; all six behaviours on one handout: `layers/L4SafetyIsland-Behaviours.pdf`.

B3 — Pull over if the ODD continues to fail

Trigger	B1 escalation exhausted — the HPC did not comply, or the ODD was not restored within the deadline; **or** B2's hold window `T_hold` expired with the HPC still lost and the camera still holding the lane — the island can see, so it can search; had it lost the camera, B2 would have exited to B4 instead
Precondition	A refuge exists that the island can **verify free** and can still reach at the current speed — from either source in the row below. `N18` reports `pull_over_available = true`, and the candidate refuge is verified free against `N16`'s live free space
Refuge source, in priority order	1. H8's **nominated** refuge — map-validated and law-checked by the HPC, verified free by `N16` before use. 2. `N18`'s own **island-detected** refuge — a lateral clearance found **by the camera**: `N14` tracks the lane lines, finds the outer edge of the running lane and the road edge or kerb beyond it, and measures the drivable space between them; `N6` radar confirms the space is unoccupied and that nothing is closing on it. Admitted only under `legality = EMERGENCY_ONLY`. `N15` ultrasonics (D8 rank 3), where fitted, guard the final metres and let the stop end tight to the kerb rather than within the shoulder; `N13` 2D lidar, where fitted, may confirm the geometry or disqualify the space. Neither is required. 3. Neither — no refuge, and the abort ladder decides what happens next
Authority	**The island steers, brakes and signals.** B2 steers to stay where the vehicle already is; B3 is the only behaviour in which the island initiates lateral *displacement* — deliberately leaving the running lane.
Response	Open the refuge search across **both** sources the moment B3 is entered, and keep it open while decelerating — `N14` holding the lane lines for the in-lane part of the manoeuvre and scanning the road edge for space → **Hazard lamps on the moment the manoeuvre is commanded** → indicator on → bounded lateral displacement to the first refuge that passes verification, across the outer lane line `N14` is tracking, with containment re-armed against the road edge it has observed → deceleration → standstill → secure (gear P, lamps stay on)
Deadlines	The refuge search is bounded by `T_refuge` = **5 s** from B3 entry, and additionally by a distance-travelled-searching cap, for the reason open decision 15 gives about time and distance not being the same constraint at speed (both are open decision 18). Once a refuge is accepted, the §6.4 MRM deadlines apply — begin within **10 s**, complete within **60 s**
Observable output	`MrmState` = `MRM_OPERATING`, behaviour = `PULL_OVER` (see open decision 10); `N18` `MrmBehaviorStatus`
Success criterion	Standstill within the accepted refuge — whichever source offered it —, near field clear, `MrmState` = `MRM_SUCCEEDED`
Abort	Per the ladder in §6.3 — every rung degrades to an **in-lane stop under B2's authority and control law**, then to emergency stop (B4). Two of those rungs are now told apart by *what the search concluded.* A search that **positively determines** that no usable refuge exists — neither source offers one — abandons the pull-over to the **in-lane stop**, which still has a corridor to work with. A search that reaches `T_refuge` **without a determination either way** goes to B4. An inconclusive search is a statement about the island's capability, not about the road, and capability loss goes to the floor (§3.2)
Exit	**`SAFE_STOP`, as B4's is.** B3 is terminal for the drive cycle: one successor, the secured standstill, and no edge back up. The two behaviours differ in *where* they leave the vehicle and in what they needed in order to get there, not in whether the drive continues afterwards (§3.2)
Optional enrichment (§3.5)	The last HPC object digest, to **disqualify** a refuge. It may never qualify one
Minimum sensor rung (§9.4)	**A** — the camera alone — at or below ~30 km/h, the city-road `v_odd`, where a long indicator dwell and a very low lateral rate can stand in for rear-corner radar. **B** above ~30 km/h: forward and rear-corner radar, because the manoeuvre then crosses traffic space a forward camera cannot see

The island may now find a refuge. It still may not choose one. §6.3's split — the HPC nominates, the island verifies and executes — holds wherever the HPC is there to nominate. Under B3 it frequently is not: B3 is reached from B2 precisely when the HPC has been lost, and `H8`'s buffered refuge set ages out like every other buffered input (§8.1). An island with no second source makes the refuge contract a single point of failure for the one manoeuvre it exists to enable.

So the island gets a second source, and it is deliberately weaker than the first:

	`H8` nominated refuge	**`N18` island-detected refuge**
Physically free	verified by `N16`, before use	verified by `N16`, before use
Legal	checked against the map and traffic law by the HPC	**unknown** — the island holds neither
`legality value`	`LEGAL_STOP` or `EMERGENCY_ONLY`, as nominated	`EMERGENCY_ONLY`, always
Admissible	whenever offered and within `validity_distance_m`	only when no nominated refuge is available
Sensors	`N16` free space	`N14` camera — lane lines, road edge, drivable space between them — + `N6` radar occupancy, fused in `N16`; `N15` ultrasonic where fitted, for the final metres and a tight-to-kerb end; `N13` 2D lidar where fitted, only to confirm or disqualify

The asymmetry is the whole argument. An island-detected refuge is a **physically clear lateral space** and nothing more — a verge, a hard shoulder, a wide kerb, and equally a cycle lane, a bus stop or a driveway, which the island cannot tell apart. Admitting one means preferring an unlawful stopping place to a stop in a live running lane. That trade is defensible, but only under the `EMERGENCY_ONLY` semantics the refuge contract already carries, and only in that direction: it is the judgement a human driver makes with a failed engine, and it is why the island-detected refuge sits *second* in the priority order and never displaces a nominated one.

What the island still cannot do is **choose**, in the sense §6.3 rules out. It does not weigh two candidate refuges against each other for legality, does not reason about bridges or bus stops, and does not defer a stop in order to reach a better place further on. It takes the first lateral clearance that passes verification, inside `T_refuge`. **Search, not selection** — the distinction that keeps map and planner semantics off the island (R4, R5).

The camera is what B3 searches with. With the 2D lidar now optional (D8), the island-detected refuge has to be found by the one sensor every island carries. `N14` does three things in B3, and they are the same three things a driver does with their eyes:

1. **Holds the lane lines** — the corridor for the part of the manoeuvre still in the running lane, the same source B2 used, so the hand-over from B2 is seamless.

2. **Finds the pull-over space** — the outer lane line, the road edge or kerb beyond it, and the width and length of drivable surface between them. That is the clearance `N18` tests against open decision 18's geometric criteria.
3. **Bounds the lateral move** — once the vehicle crosses the outer line, the observed road edge is the boundary `N5`'s containment check is re-armed against (§3.2).

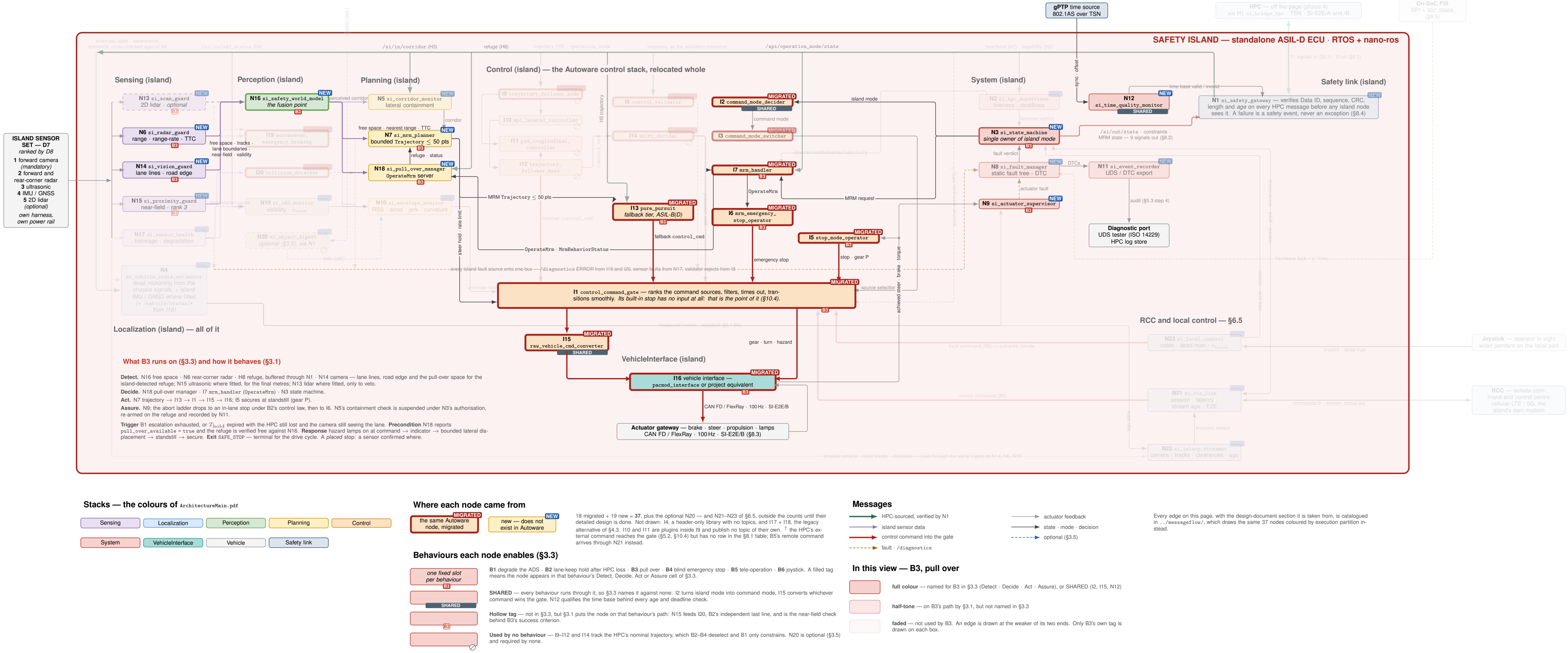
The camera says what is *there*; it is weakest at saying what is *coming*. That is radar's job, and it is why the radar-dependent half of the manoeuvre — the closing-vehicle check on the pull-over side — is the half that sets rung **B** above ~30 km/h. A camera finding that is not confirmed by radar where radar is fitted is not admitted, and a radar or lidar finding can only ever *disqualify* a space the camera found, never qualify one the camera did not.

Both terminal behaviours raise the hazard lamps on entry, not on arrival. B3 raises them when the pull-over is commanded and B4 when the blackout is declared, and in both cases they stay on through the manoeuvre, the standstill and the secure. The lamps exist for the traffic behind, and the traffic behind needs the warning *while* the vehicle is decelerating — and, under B3, moving laterally — not once it has already stopped. B2 already behaves this way, so with this the whole of the island's authority becomes legible from outside within one lamp period of its being taken.

One integration conflict has to be decided rather than discovered. On most production wiring the hazard circuit overrides both direction indicators, so a B3 that raises hazards at entry can no longer signal *which way* it is leaving the running lane — and the indicator is what the §6.3 ladder leans on below ~30 km/h, where a long indicator dwell is what buys back the rear-corner radar. The two candidate resolutions — indicator alone for a fixed dwell with hazards from the start of lateral motion, or hazards throughout and no directional signal at all — are a vehicle-integration question rather than a software one, and the WG should settle which. Nothing else in this design depends on the answer.

Phase 5 · B3 — Pull over if the ODD continues to fail

When degradation does not restore the envelope within a declared deadline, bring the vehicle out of the running lane to a standstill in a refuge — nominated by the HPC, or found by the island's camera.
The phase-5 drawing, unchanged: what B3 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.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own harness, own power rail
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dead reckoning from the chassis signals, + island IMU / GNSS where fitted
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gear · turn · hazard
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commands in · stream, status out
RCC — remote command and control centre
cellular LTE / 5G, the island's own modem
ego state
encoded stream
N22 si_teleop_streamer
camera · tracks · clearances · ego
forward camera · radar tracks · ultrasonic — read through the same ingest as N14, N6, N15

What B3 runs on (§3.3) and how it behaves (§3.1)

Detect. N16 free space · N6 rear-corner radar · H8 refuge, buffered through N1 · N14 camera — lane lines, road edge and the pull-over space for the island-detected refuge; N15 ultrasonic where fitted, for the final metres; N13 lidar where fitted, only to veto.

Decide. N18 pull-over manager · I7 `mrm_handler` (`OperateMrm`) · N3 state machine.

Act. N7 trajectory → I13 → I1 → I15 → I16; I5 secures at standstill (gear P).

Assure. N9; the abort ladder drops to an in-lane stop under B2's control law, then to I6. N5's containment check is suspended under N3's authorisation, re-armed on the refuge and recorded by N11.

Trigger B1 escalation exhausted, or T_{hold} expired with the HPC still lost and the camera still seeing the lane. **Precondition** N18 reports `pull_over_available = true` and the refuge is verified free against N16. **Response** hazard lamps on at command → indicator → bounded lateral displacement → standstill → secure. **Exit** `SAFE_STOP` — terminal for the drive cycle. A *placed* stop: a sensor confirmed where.

Stacks — the colours of `ArchitectureMain.pdf`

Sensing · Localization · Perception · Planning · Control
System · VehicleInterface · Vehicle · Safety link

Where each node came from

MIGRATED — the same Autoware node, migrated
NEW — new — does not exist in Autoware

18 migrated + 19 new = **37**, plus the optional N20 — and N21–N23 of §6.5, outside the counts until their detailed design is done. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. † the HPC's external command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table; B5's remote command arrives through N21 instead.

Behaviours each node enables (§3.3)

one fixed slot per behaviour B3 — **B1** degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED — **SHARED** — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

B3 — **Hollow tag** — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

⊘ — **Used by no behaviour** — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

→	HPC-sourced, verified by N1	→	actuator feedback
→	island sensor data	→	state · mode · decision
→	control command into the gate	- - →	optional (§3.5)
- - →	fault · `/diagnostics`		

In this view — B3, pull over

full colour — named for B3 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)

half-tone — on B3's path by §3.1, but not named in §3.3

faded — not used by B3. An edge is drawn at the weaker of its two ends. Only B3's own tag is drawn on each box.

Every edge on this page, with the design-document section it is taken from, is catalogued in `../messageflow/`, which draws the same 37 nodes coloured by execution partition instead.

Figure 3. *B3 on the island's node graph: what B3 runs on in full colour, what it only passes through in half-tone, the rest of the island faded*

B3, on the island's node graph. N18 selects the refuge from either source — the island-detected one found by N14's camera, which is lit — I7 operates the pull-over slot, N7 plans into it, I13 follows, and I5 secures at standstill; I6 is lit as the abort. Full size: `layers/L4SafetyIsland-Layer6-B3-PullOver.pdf`*; all six behaviours on one handout:* `layers/L4SafetyIsland-Behaviours.pdf`*.*

B4 — Emergency stop when the island is blind

B4 is the **floor of the ladder**, and it exists because B2 and B3 are both conditional behaviours. B2 requires an observable lane; B3 requires a refuge that can be verified free. Both rest on the same sensor — the forward camera, the one D8 makes mandatory and therefore the only one either behaviour is allowed to assume — so both are lost together when it is lost. When the camera has gone — blocked, failed, saturated, or disagreeing beyond what `N16`'s arbitration can resolve — neither behaviour has a defensible precondition left, and the island must not attempt either one on stale evidence. It stops instead, where it is.

"Blind" is defined against the mandatory sensor, not against all of them. An island whose camera is gone but whose radar still reports is blind *to the lane*: it can range a target ahead, but it cannot hold a corridor or verify a refuge, which is everything B2 and B3 need. A sensor the design does not require cannot be what keeps the island out of B4, for the same reason the HPC object digest cannot (§3.5) — but while it lives it still feeds `I19` and may steepen B4's ramp.

Emergency stop is not new machinery. It is already step 3b of the §5.3 graded reaction and the bottom rung of the §6.3 abort ladder. What B4 adds is **declaration**: the rung acquires a trigger, a deadline, a sensor rung and a traceability row like the other three, instead of being reached only as the residue of something else failing. A fallback that is only ever arrived at by accident cannot be argued for in a safety case.

Trigger	`N17` declares **sensing blackout** — the **forward camera** is invalid, blocked or stale, or `N14` has lost the lane lines beyond the bridging budget, so `N16` can publish neither a valid corridor nor valid lateral free space. Radar, ultrasonic, IMU/GNSS or 2D lidar that are still alive do not lift the declaration (D8). Stated in behavioural terms, which is how `N3` evaluates it: **B2 has no corridor *and* B3 has no verifiable refuge.** It is also B2's downward exit whenever `T_hold` expires with the camera lost. Also entered directly on Also entered directly on `N9` (actuators not tracking) or `I19` (AEB fires), per the §6.3 ladder
Precondition	**None, and deliberately none.** A fallback whose entry can be blocked is not a fallback. B4 needs only the proprioceptive channels `N4` already requires — the chassis signals: wheel speed, steering report, ESC yaw rate and acceleration, with the island IMU where it is fitted — and the actuation path to `I16`
Authority	**The island brakes, and declines to steer.** Lateral authority is *surrendered, not exercised*: with no corridor there is nothing to steer against, so the steering command is held at its current angle under `N7`'s rate limit and allowed to relax toward straight. B4 is the only behaviour in which the island deliberately does not steer
Response	**Hazard lamps on the moment the blackout is declared** → `I6` `mrm_emergency_stop_operator` publishes the control command directly → deceleration at the emergency rate `a_emg` → standstill → `I5` secures (gear P). No trajectory is planned, and none could be: planning needs a world model
Deadlines	Authority transfer within the §9.2 fault-reaction budget (≤ 70 ms from the declaring fault). `a_emg` and the steering-relaxation rate are open decision 15

Observable output	`MrmState` = `MRM_OPERATING`, behaviour = `EMERGENCY_STOP` — the one MRM behaviour the existing `MrmState` message can already name without extension (contrast open decision 10). `N3` mode = `EMERGENCY_STOP`; `/si/out/state` carries the blackout fault code and which sources were lost
Success criterion	Standstill, secured, hazard lamps on. **Nothing is claimed about *where*.** B4 stops the vehicle in the running lane, and that is the honest statement of what a blind island can promise
Failure	`N9` reports the actuators still not tracking → the gate's built-in stop (§5.3 step 3c) and the FSI hardware fault line (§8.5), both of which sit outside the ROS-node path by construction
Exit	**`SAFE_STOP`, and nothing else.** B4 has exactly one successor — the secured standstill — and no edge back up the ladder. It stays terminal for the drive cycle even if sensing returns mid-stop: a source that has just been declared invalid has not earned authority back inside a stop that is already under way
Optional enrichment (§3.5)	The last HPC object digest, to shape the ramp only. It may not delay entry, offer an exit, or count as a valid source — `N20` never contributes to the blackout determination. **A surviving radar is treated the same way:** `N6` may keep `I19` armed and steepen the ramp toward the upper bound of `a_emg` for a target in path, and may do nothing else
Minimum sensor rung (§9.4)	**0 — none.** B4 is the behaviour that must survive the loss of the entire island sensor set, the camera included, on the chassis signals alone. It is what makes rung A a *design* choice rather than a *safety* floor

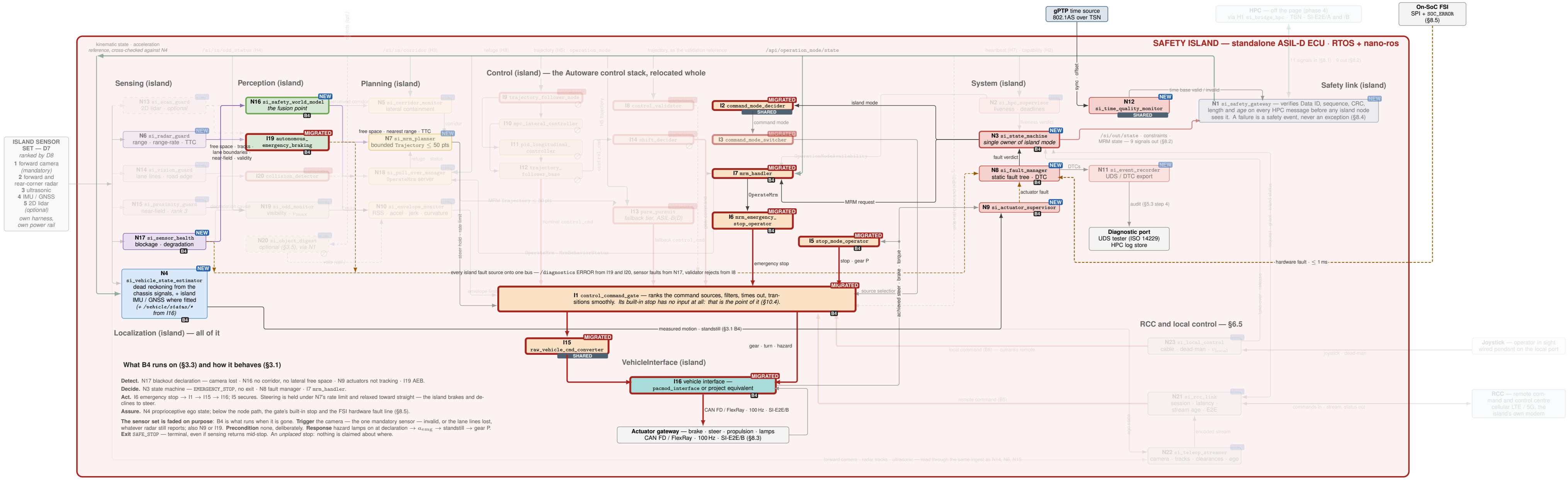
Why stopping in the lane is accepted. Stopping in a running lane is dangerous, and no design should reach for it casually. The argument for B4 is comparative rather than absolute: the only other things a blind island can do are continue on stale evidence, or attempt a lateral manoeuvre it cannot verify. Both are worse, and both fail *unboundedly* — the error grows with every metre travelled. A stop is bounded: the vehicle's own kinetic energy is the whole of it, and the manoeuvre is legible to everything around the vehicle through brake lights and hazard lamps. The residual risk is real, and it belongs in the Part-4 HARA as a **named accepted risk**, not as a gap.

B4 is not "AEB fired". `I19` autonomous emergency braking is a *reflex* against a specific detected obstacle, and it presupposes a working forward sensor. B4 is a *state*: the island no longer knows what is ahead of it at all. The two share the actuator path (`I6`) and nothing else. An AEB event escalates *into* B4 — the §6.3 ladder already says so — but the reverse is impossible, because a blind island cannot detect the obstacle that would trigger AEB.

B4 is not the in-lane stop either. The in-lane stop that B3 degrades to when its search finds no refuge is *obstacle-aware*: it decelerates along the corridor the camera still sees, shortening the ramp against `N16`'s free-space distance. It is a reduced behaviour, not a blind one. B4 is what remains when the camera no longer sees a corridor to stop along, and `N8` distinguishes the two by fault code so the event recorder and the HARA can tell them apart.

Phase 5 · B4 — Emergency stop when the island is blind

When the island has lost the sensing that B2 and B3 both depend on — the camera — decelerate to a standstill where the vehicle already is — no lane keeping, no refuge, no lateral manoeuvre.
The phase-5 drawing, unchanged: what B4 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.


gPTP time source
802.1AS over TSN
HPC — off the page (phase 4)
via H1 si_bridge_hpc · TSN · SI-E2E/A and /B
On-SoC FSI
SPI + SOC_ERROR
(§8.5)
SAFETY ISLAND — standalone ASIL-D ECU · RTOS + nano-ros
kinematic state · acceleration
reference, cross-checked against N4
/si/in/odd_status (H4)
/si/in/corridor (H3)
refuge (H8)
trajectory (H5) · operation_mode
trajectory, as the validation reference
/api/operation_mode/state
heartbeat (H7) · capability (H2)
11 signals in (§8.1) · 9 out (§8.2)
objects (opt.)
ISLAND SENSOR SET — D7
ranked by D8
1 forward camera
(mandatory)
2 forward and rear-corner radar
3 ultrasonic
4 IMU / GNSS
5 2D lidar
(optional)
own harness, own power rail
Sensing (island)
N13 si_scan_guard
2D lidar · optional
N6 si_radar_guard
range · range-rate · TTC
N14 si_vision_guard
lane lines · road edge
N15 si_proximity_guard
near-field · rank 3
N17 si_sensor_health
blockage · degradation
NEW
B4
Perception (island)
N16 si_safety_world_model
the fusion point
I19 autonomous_emergency_braking
MIGRATED
free space · tracks · lane boundaries · near-field · validity
I20 collision_detector
degradation cause
N19 si_odd_monitor
visibility · v_{max}
N20 si_object_digest
optional (§3.5), via N1
veto (opt.)
Planning (island)
perceived corridor
N5 si_corridor_monitor
lateral containment
corridor
free space · nearest range · TTC
N7 si_mrm_planner
bounded Trajectory ≤ 50 pts
refuge · status
N18 si_pull_over_manager
OperateMrm server
N10 si_envelope_monitor
RSS · accel · jerk · curvature
MRM Trajectory ≤ 50 pts
steer hold · rate limit
envelope limit
Control (island) — the Autoware control stack, relocated whole
I9 trajectory_follower_node
I10 mpc_lateral_controller
I11 pid_longitudinal_controller
I12 trajectory_follower_base
H5 trajectory
control_cmd
nominal control_cmd
I8 control_validator
I14 shift_decider
I13 pure_pursuit
fallback tier, ASIL-B(D)
fallback control_cmd
gear command
OperateMrm · MrmBehaviorStatus
I2 command_mode_decider
SHARED
command mode
I3 command_mode_switcher
OperationModeAvailability
I7 mrm_handler
OperateMrm
MRM request
I6 mrm_emergency_stop_operator
emergency stop
I5 stop_mode_operator
stop · gear P
island mode
every island fault source onto one bus — /diagnostics ERROR from I19 and I20, sensor faults from N17, validator rejects from I8
I1 control_command_gate — ranks the command sources, filters, times out, transitions smoothly. Its built-in stop has no input at all: that is the point of it (§10.4).
source selector
achieved steer · brake · torque
external command (HPC) †
measured motion · standstill (§3.1 B4)
N4
si_vehicle_state_estimator
dead reckoning from the chassis signals, + island IMU / GNSS where fitted
(+ /vehicle/status/* from I16)
Localization (island) — all of it
I15
raw_vehicle_cmd_converter
gear · turn · hazard
VehicleInterface (island)
I16 vehicle interface — pacmod_interface or project equivalent
CAN FD / FlexRay · 100 Hz · SI-E2E/B
Actuator gateway — brake · steer · propulsion · lamps
CAN FD / FlexRay · 100 Hz · SI-E2E/B (§8.3)
System (island)
N2 si_hpc_supervisor
liveness · deadlines
liveness verdict
sync · offset
N12
si_time_quality_monitor
time base valid / invalid
N3 si_state_machine
single owner of island mode
fault verdict
N8 si_fault_manager
static fault tree · DTC
actuator fault
N9 si_actuator_supervisor
DTCs
N11 si_event_recorder
UDS / DTC export
audit (§5.3 step 4)
Diagnostic port
UDS tester (ISO 14229)
HPC log store
Safety link (island)
N1 si_safety_gateway — verifies Data ID, sequence, CRC, length and age on every HPC message before any island node sees it. A failure is a safety event, never an exception (§8.4)
/si/out/state · constraints · MRM state — 9 signals out (§8.2)
request · grant · island status
take-over · release
hardware fault · ≤ 1 ms
RCC and local control — §6.5
N23 si_local_control
cable · dead-man · v_{local}
local command (B6) — outranks remote
remote command (B5)
N21 si_rcc_link
session · latency · stream age · E2E
encoded stream
ego state
N22 si_teleop_streamer
camera · tracks · clearances · ego
forward camera · radar tracks · ultrasonic — read through the same ingest as N14, N6, N15
Joystick — operator in sight
wired pendant on the local port
joystick · dead-man
RCC — remote command and control centre
cellular LTE / 5G, the island's own modem
commands in · stream, status out

What B4 runs on (§3.3) and how it behaves (§3.1)

Detect. N17 blackout declaration — camera lost · N16 no corridor, no lateral free space · N9 actuators not tracking · I19 AEB.

Decide. N3 state machine — `EMERGENCY_STOP`, no exit · N8 fault manager · I7 `mrm_handler`.

Act. I6 emergency stop → I1 → I15 → I16; I5 secures. Steering is held under N7's rate limit and relaxed toward straight — the island brakes and declines to steer.

Assure. N4 proprioceptive ego state; below the node path, the gate's built-in stop and the FSI hardware fault line (§8.5).

The sensor set is faded on purpose: B4 is what runs when it is gone. **Trigger** the camera — the one mandatory sensor — invalid, or the lane lines lost, whatever radar still reports; also N9 or I19. **Precondition** none, deliberately. **Response** hazard lamps on at declaration → a_{emg} → standstill → gear P. **Exit** `SAFE_STOP` — terminal, even if sensing returns mid-stop. An *unplaced* stop: nothing is claimed about where.

Stacks — the colours of `ArchitectureMain.pdf`

Sensing · Localization · Perception · Planning · Control

System · VehicleInterface · Vehicle · Safety link

Where each node came from

MIGRATED — the same Autoware node, migrated

NEW — new — does not exist in Autoware

18 migrated + 19 new = **37**, plus the optional N20 — and N21–N23 of §6.5, outside the counts until their detailed design is done. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. † the HPC's external command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table; B5's remote command arrives through N21 instead.

Behaviours each node enables (§3.3)

one fixed slot per behaviour B4	**B1** degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.
SHARED	**SHARED** — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.
▭	**Hollow tag** — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.
⊘	**Used by no behaviour** — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

⟶	HPC-sourced, verified by N1
⟶	island sensor data
⟶	control command into the gate
⇢	fault · `/diagnostics`
⟶	actuator feedback
⟶	state · mode · decision
⇢	optional (§3.5)

Every edge on this page, with the design-document section it is taken from, is catalogued in `../messageflow/`, which draws the same 37 nodes coloured by execution partition instead.

In this view — B4, blind emergency stop

🟥	**full colour** — named for B4 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)
▨	**half-tone** — on B4's path by §3.1, but not named in §3.3
▭	**faded** — not used by B4. An edge is drawn at the weaker of its two ends. Only B4's own tag is drawn on each box.

Figure 4. *B4 on the island's node graph: what B4 runs on in full colour, what it only passes through in half-tone, the rest of the island faded*

B4, on the island's node graph. Nothing exteroceptive is lit: N17 declares the blackout, I6 brakes, I5 secures, and N4's dead reckoning — fed by the chassis signals, and the island IMU where fitted — is what N9 confirms the stop against. The sensor set, N6 and N1 are half-tone: the AEB trigger — now radar's, since the 2D lidar is optional — and the blackout declaration come through them, and `/si/out/state` still leaves through N1. Full size: `layers/L4SafetyIsland-Layer6-B4-EmergencyStop.pdf`; all six behaviours on one handout: `layers/L4SafetyIsland-Behaviours.pdf`.

B5 — Tele-operation from the remote command centre

Detailed design in §6.6: entered from standstill, or taken over from a moving vehicle in B1–B3 when the ADS requests it. Its communication — the module, the session and its security, the video transport — is deferred to the next design (§8.6). Open decision 19 holds what remains.

B5 hands the driving task to a person at the **remote command and control centre (RCC)**. It is how a vehicle that has ended its drive in B3 or B4 — or that the ADS can no longer continue — is moved on without sending someone to it. The remote driver sees the road through **the island's own sensors**, streamed over **the island's own communication module**, so B5 survives the loss of the HPC: nothing on its path runs there. That is the reason D9 puts the modem on the island and not on the HPC.

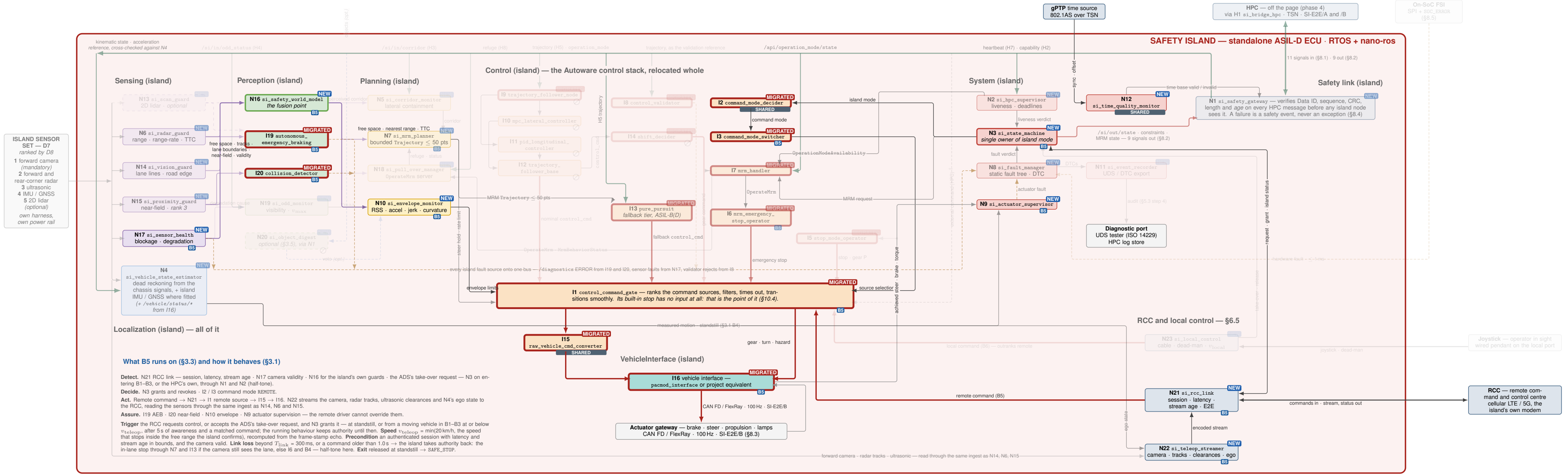
What the remote driver is shown follows D8's ranking, for the same reason the behaviours do: the camera stream is the one B5 may require, and radar tracks and ultrasonic clearances enrich it where fitted. A remote driver watching a stream the island cannot itself validate would be driving blind with extra steps, so B5 is admissible only while `N17` holds the camera valid.

Trigger	The RCC requests control and `N3` grants it. Two origins: `SAFE_STOP` — recovery after B3 or B4 — and a **moving vehicle in B1–B3**, after the ADS has raised a remote take-over request because the HPC behaves abnormally and the RCC has accepted it. B5 is never entered from `ADS_ACTIVE` or from B4. The handshake and the take-over conditions are in §6.6
Precondition	An authenticated RCC session on `N21`, with measured round-trip latency and stream age inside their bounds · the camera valid (`N17`) · no B6 active · **from motion, also**: speed at or below `v_teleop`, 5 s of situational awareness, a matched command, and no lateral manoeuvre in progress
Authority	The remote driver commands; the island executes and bounds. Commands arrive on `N21` and enter `I1` as the **remote** command source, validated by `I8` and `N10` like any other (§10.4)
Streamed to the RCC	`N22`: forward camera, compressed to a bounded bit rate — required · radar tracks — rung B up · ultrasonic near-field clearance — rung C up · `N4` ego state · the island's state and fault summary
Speed	`v_teleop`, bounded by the measured end-to-end latency — camera to RCC screen, and command back to the actuator — and by the range the stream shows: the vehicle must be able to stop inside what the remote driver could see, *after* the latency has elapsed: `v_teleop` = min(20 km/h, `v_vis`), recomputed at 10 Hz from the measured command age (§6.6). The values are for ratification in open decision 19(b)
Guards that stay armed	`I19` AEB · `I20` near-field, rung C up · `N10` envelope · `N9` actuator supervision. The remote driver cannot override them
Link loss	No valid command or heartbeat for `T_link` = 300 ms, or a command more than 1.0 s old by the frame-stamp echo → the island takes authority back and stops: the **in-lane stop** under B2's control law if the camera still sees the lane, **B4** if it does not — the same capability-first test as every other downward edge. Never "continue on the last command"
Exit	The RCC releases control at standstill → `SAFE_STOP`. From there the RCC may also re-engage the ADS, once the HPC is re-qualified and the ODD is satisfiable (§6.6, open decision 19(c))
Observable output	`OperationModeState` = `REMOTE` — a mode Autoware already defines

Minimum sensor rung (§9.4)	**A** — the camera, which is what the remote driver watches. Radar and ultrasonic are streamed where fitted
Still open	Link technology and redundancy; who owns the keys and certificates; the take-over numbers — `T_sa` and the matching tolerances; stage 2 — a path-based remote-assistance mode; the RCC's human-factors requirements (open decision 19)

Phase 5 · B5 — Tele-operation from the remote command centre

A remote driver at the RCC drives the vehicle over the island's own communication module, on the camera, radar and ultrasonic the island streams — while the island keeps its guards armed and stops the vehicle itself if the link fails.
The phase-5 drawing, unchanged: what B5 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded. *Detailed design: §6.6 — from standstill, or taken over from a moving vehicle in B1–B3 on the ADS's request.*


gPTP time source
802.1AS over TSN
HPC — off the page (phase 4)
via H1 si_bridge_hpc · TSN · SI-E2E/A and /B
On-SoC FSI
SPI + SOC_ERROR
(§8.5)
SAFETY ISLAND — standalone ASIL-D ECU · RTOS + nano-ros
objects (opt.)
kinematic state · acceleration
reference, cross-checked against N4
/si/in/odd_status (H4)
/si/in/corridor (H3)
refuge (H8)
trajectory (H5) · operation_mode
trajectory, as the validation reference
/api/operation_mode/state
heartbeat (H7) · capability (H2)
sync · offset
11 signals in (§8.1) · 9 out (§8.2)
ISLAND SENSOR SET — D7
ranked by D8
1 forward camera
(mandatory)
2 forward and rear-corner radar
3 ultrasonic
4 IMU / GNSS
5 2D lidar
(optional)
own harness, own power rail
Sensing (island)
N13 si_scan_guard
2D lidar · optional
NEW
N6 si_radar_guard
range · range-rate · TTC
NEW
N14 si_vision_guard
lane lines · road edge
NEW
N15 si_proximity_guard
near-field · rank 3
NEW
N17 si_sensor_health
blockage · degradation
NEW
B5
N4
si_vehicle_state_estimator
dead reckoning from the chassis signals, + island IMU / GNSS where fitted
(+ /vehicle/status/* from I16)
NEW
Localization (island) — all of it
Perception (island)
N16 si_safety_world_model
the fusion point
NEW
B5
I19 autonomous_emergency_braking
MIGRATED
B5
free space · tracks · lane boundaries · near-field · validity
I20 collision_detector
MIGRATED
B5
degradation cause
N19 si_odd_monitor
visibility · v_{max}
N20 si_object_digest
optional (§3.5), via N1
veto (opt.)
Planning (island)
perceived corridor
N5 si_corridor_monitor
lateral containment
corridor
free space · nearest range · TTC
N7 si_mrm_planner
bounded Trajectory ≤ 50 pts
NEW
B5
refuge · status
N18 si_pull_over_manager
OperateMrm server
v_{max}
N10 si_envelope_monitor
RSS · accel · jerk · curvature
NEW
B5
steer hold · rate limit
envelope limits
Control (island) — the Autoware control stack, relocated whole
I9 trajectory_follower_node
I10 mpc_lateral_controller
I11 pid_longitudinal_controller
I12 trajectory_follower_base
H5 trajectory
I8 control_validator
I14 shift_decider
control_cmd
MRM Trajectory ≤ 50 pts
I13 pure_pursuit
fallback tier, ASIL-B(D)
MIGRATED
nominal control_cmd
fallback control_cmd
gear command
OperateMrm · MrmBehaviorStatus
I2 command_mode_decider
MIGRATED
SHARED
command mode
I3 command_mode_switcher
MIGRATED
B5
island mode
OperationModeAvailability
I7 mrm_handler
MIGRATED
OperateMrm
MRM request
I6 mrm_emergency_stop_operator
MIGRATED
B5
I5 stop_mode_operator
stop · gear P
emergency stop
every island fault source onto one bus — /diagnostics ERROR from I19 and I20, sensor faults from N17, validator rejects from I8
I1 control_command_gate — ranks the command sources, filters, times out, transitions smoothly. Its built-in stop has no input at all: that is the point of it (§10.4).
MIGRATED
B5
source selection
achieved steer · brake · torque
external command (HPC)
measured motion · standstill (§3.1 B4)
I15
raw_vehicle_cmd_converter
MIGRATED
SHARED
gear · turn · hazard
VehicleInterface (island)
I16 vehicle interface — pacmod_interface or project equivalent
MIGRATED
B5
CAN FD / FlexRay · 100 Hz · SI-E2E/B
Actuator gateway — brake · steer · propulsion · lamps
CAN FD / FlexRay · 100 Hz · SI-E2E/B (§8.3)
remote command (B5)
local command (B6) — outranks remote
System (island)
N2 si_hpc_supervisor
liveness · deadlines
NEW
liveness verdict
N3 si_state_machine
single owner of island mode
NEW
B5
fault verdict
N8 si_fault_manager
static fault tree · DTC
NEW
DTCs
N11 si_event_recorder
UDS / DTC export
actuator fault
N9 si_actuator_supervisor
NEW
B5
audit (§5.3 step 4)
Diagnostic port
UDS tester (ISO 14229)
HPC log store
hardware fault · ≤ 1 ms
N12
si_time_quality_monitor
NEW
SHARED
time base valid / invalid
Safety link (island)
N1 si_safety_gateway — verifies Data ID, sequence, CRC, length and age on every HPC message before any island node sees it. A failure is a safety event, never an exception (§8.4)
NEW
/si/out/state · constraints · MRM state — 9 signals out (§8.2)
request · grant · island status
take-over · release
RCC and local control — §6.5
N23 si_local_control
cable · dead-man · v_{local}
joystick · dead-man
Joystick — operator in sight
wired pendant on the local port
N21 si_rcc_link
session · latency · stream age · E2E
NEW
B5
commands in · stream, status out
RCC — remote command and control centre
cellular LTE / 5G, the island's own modem
encoded stream
ego state
N22 si_teleop_streamer
camera · tracks · clearances · ego
NEW
B5
forward camera · radar tracks · ultrasonic — read through the same ingest as N14, N6, N15

What B5 runs on (§3.3) and how it behaves (§3.1)

Detect. N21 RCC link — session, latency, stream age · N17 camera validity · N16 for the island's own guards · the ADS's take-over request — N3 on entering B1–B3, or the HPC's own, through N1 and N2 (half-tone).

Decide. N3 grants and revokes · I2 / I3 command mode `REMOTE`.

Act. Remote command → N21 → I1 remote source → I15 → I16. N22 streams the camera, radar tracks, ultrasonic clearances and N4's ego state to the RCC, reading the sensors through the same ingest as N14, N6 and N15.

Assure. I19 AEB · I20 near-field · N10 envelope · N9 actuator supervision — the remote driver cannot override them.

Trigger the RCC requests control, or accepts the ADS's take-over request, and N3 grants it — at standstill, or from a moving vehicle in B1–B3 at or below v_{teleop}, after 5 s of awareness and a matched command; the running behaviour keeps authority until then. **Speed** v_{teleop} = min(20 km/h, the speed that stops inside the free range the island confirms), recomputed from the frame-stamp echo. **Precondition** an authenticated session with latency and stream age in bounds, and the camera valid. **Link loss** beyond T_{link} = 300 ms, or a command older than 1.0 s → the island takes authority back: the in-lane stop through N7 and I13 if the camera still sees the lane, else I6 and B4 — half-tone here. **Exit** released at standstill → `SAFE_STOP`.

Stacks — the colours of `ArchitectureMain.pdf`

Sensing · Localization · Perception · Planning · Control
System · VehicleInterface · Vehicle · Safety link

Where each node came from

MIGRATED — the same Autoware node, migrated

NEW — new — does not exist in Autoware

18 migrated + 19 new = **37**, plus the optional N20 — and N21–N23 of §6.5, outside the counts until their detailed design is done. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. † the HPC's external command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table; B5's remote command arrives through N21 instead.

Behaviours each node enables (§3.3)

one fixed slot per behaviour B5

B1 degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED

SHARED — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

⊘ **Used by no behaviour** — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

- HPC-sourced, verified by N1
- island sensor data
- control command into the gate
- fault · `/diagnostics`
- actuator feedback
- state · mode · decision
- optional (§3.5)

Every edge on this page, with the design-document section it is taken from, is catalogued in `../messageflow/`, which draws the same 37 nodes coloured by execution partition instead.

In this view — B5, tele-operation from the remote command centre

full colour — named for B5 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)

half-tone — on B5's path by §3.1, but not named in §3.3

faded — not used by B5. An edge is drawn at the weaker of its two ends. Only B5's own tag is drawn on each box.

Figure 5. *B5 on the island's node graph: what B5 runs on in full colour, what it only passes through in half-tone, the rest of the island faded*

B5, on the island's node graph. The RCC, outside the boundary, talks to N21 over the island's own modem; the remote command enters the gate as its remote source, and N3 grants and revokes it. N22 packs the stream from the island sensors (read through the same ingest as N14, N6 and N15) and N4's ego state. I19, I20, N10 and N9 stay lit as the guards the remote driver cannot override; N7, I13 and I6 are half-tone, the link-loss stop. Full size: *`layers/L4SafetyIsland-Layer6-B5-TeleOperation.pdf`.*

B6 — Manual joystick control without streamed sensing

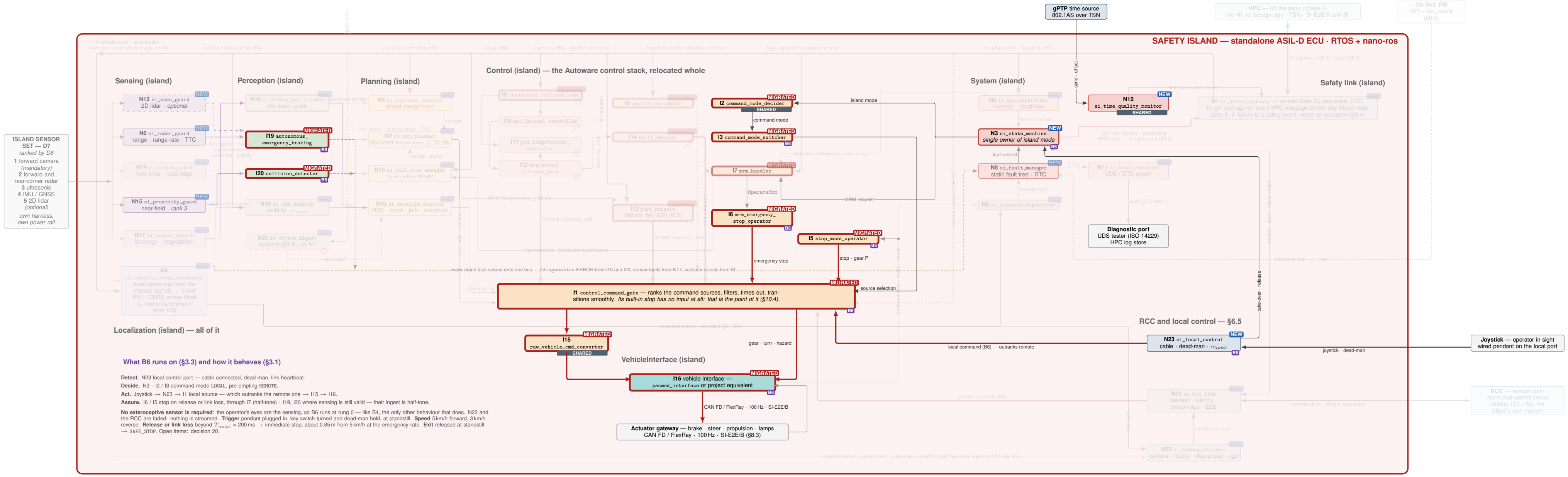
Detailed design in §6.6: stage 1, entered from standstill. Open decision 20 holds what remains.

B6 is the manual fallback for when there is no stream to watch. An operator **within sight of the vehicle** drives it with a joystick connected to the island's **local control port** — **by cable** in this design; a short-range radio link is deferred to the next design. The operator's own eyes are the sensing, so B6 needs no exteroceptive sensor and no RCC link: with B4, it is one of only two behaviours that run at rung 0. It is how a vehicle is moved off a carriageway, onto a recovery truck or into a depot when automation and tele-operation are both unavailable.

Trigger	The local operator takes control on `N23` — pendant plugged in, key switch turned, dead-man held. Proposed origins: `SAFE_STOP`, and B5 — **local presence pre-empts the remote driver**. A moving vehicle is never taken over: a moving B5 is stopped first (§6.6, open decision 20(a))
Precondition	A pendant connected to `N23` with its link heartbeat inside `T_local` · the vehicle at standstill on entry
Authority	The operator commands through `I1`'s **local** command source, which outranks the remote one. `I8` and `N10` still validate
Sensing	None required and none streamed. Island sensors that are still valid keep guarding — `I19`, and `I20` on `N15`'s near-field clearance
Speed	Creep only: `v_local`, proposed at walking pace (~5 km/h), so that the vehicle stays inside the operator's sight and the cable's reach: 5 km/h forward, 3 km/h in reverse, 0.5 m/s^2 (§6.6, open decision 20(b))
Release or link loss	Dead-man released, the joystick's stop pressed, or the local link silent beyond `T_local` = 200 ms → immediate stop at the emergency rate on B4's actuator path (`I6`, then `I5`). From 5 km/h the stop is about 0.85 m, all latencies included
Exit	The operator releases control at standstill → `SAFE_STOP`
Observable output	`OperationModeState` = `LOCAL` — a mode Autoware already defines
Minimum sensor rung (§9.4)	**0** — no island sensor is required
Still open	Where the wired port sits on the vehicle, and the cable's length; the grade of the enabling switch and the emergency stop; the hardwired emergency-stop line (open decision 20). A short-range radio link is deferred to the next design

Phase 5 · B6 — Manual joystick control without streamed sensing

An operator in sight of the vehicle drives it with a joystick on the island's local control port — by cable; a radio link is deferred to the next design — with no sensor stream; the island holds the speed to a creep and stops on release or on loss of the link. The phase-5 drawing, unchanged: what B6 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded. *Stage-1 detailed design: §6.6, entered from standstill.*


gPTP time source
802.1AS over TSN
HPC — off the page (phase 4)
via H1 si_bridge_hpc · TSN · SI-E2E/A and /B
On-SoC FSI
SPI + SOC_ERROR
(§8.5)
SAFETY ISLAND — standalone ASIL-D ECU · RTOS + nano-ros
ISLAND SENSOR SET — D7
ranked by D8
1 forward camera
(mandatory)
2 forward and rear-corner radar
3 ultrasonic
4 IMU / GNSS
5 2D lidar
(optional)
own harness, own power rail
kinematic state · acceleration
reference, cross-checked against N4
/si/in/odd_status (H4)
/si/in/corridor (H3)
refuge (H8)
trajectory (H5) · operation_mode
trajectory, as the validation reference
/api/operation_mode/state
heartbeat (H7) · capability (H2)
11 signals in (§8.1) · 9 out (§8.2)
objects (opt.)
Sensing (island)
N13 si_scan_guard
2D lidar · optional
NEW
N6 si_radar_guard
range · range-rate · TTC
N14 si_vision_guard
lane lines · road edge
N15 si_proximity_guard
near-field · rank 3
N17 si_sensor_health
blockage · degradation
N4
si_vehicle_state_estimator
dead reckoning from the chassis signals, + island IMU / GNSS where fitted
(+ /vehicle/status/* from I16)
Localization (island) — all of it
Perception (island)
N16 si_safety_world_model
the fusion point
MIGRATED
I19 autonomous_emergency_braking
B6
I20 collision_detector
free space · tracks · lane boundaries · near-field · validity
degradation cause
N19 si_odd_monitor
visibility · v_{max}
N20 si_object_digest
optional (§3.5), via N1
veto (opt.)
perceived corridor
Planning (island)
N5 si_corridor_monitor
lateral containment
corridor
free space · nearest range · TTC
N7 si_mrm_planner
bounded Trajectory ≤ 50 pts
refuge · status
N18 si_pull_over_manager
OperateMrm server
v_{max}
N10 si_envelope_monitor
RSS · accel · jerk · curvature
steer hold · rate limit
envelope limit
MRM Trajectory ≤ 50 pts
Control (island) — the Autoware control stack, relocated whole
I9 trajectory_follower_node
I10 mpc_lateral_controller
I11 pid_longitudinal_controller
I12 trajectory_follower_base
H5 trajectory
control_cmd
I8 control_validator
I14 shift_decider
I13 pure_pursuit
fallback tier, ASIL-B(D)
nominal control_cmd
fallback control_cmd
gear command
OperateMrm · MrmBehaviorStatus
I2 command_mode_decider
SHARED
command mode
I3 command_mode_switcher
OperationModeAvailability
I7 mrm_handler
OperateMrm
MRM request
I6 mrm_emergency_stop_operator
I5 stop_mode_operator
emergency stop
stop · gear P
every island fault source onto one bus — /diagnostics ERROR from I19 and I20, sensor faults from N17, validator rejects from I8
I1 control_command_gate — ranks the command sources, filters, times out, transitions smoothly. Its built-in stop has no input at all: that is the point of it (§10.4).
source selection
measured motion · standstill (§3.1 B4)
I15
raw_vehicle_cmd_converter
gear · turn · hazard
VehicleInterface (island)
I16 vehicle interface —
pacmod_interface or project equivalent
CAN FD / FlexRay · 100 Hz · SI-E2E/B
Actuator gateway — brake · steer · propulsion · lamps
CAN FD / FlexRay · 100 Hz · SI-E2E/B (§8.3)
achieved steer · brake · torque
external command (HPC)
System (island)
island mode
N2 si_hpc_supervisor
liveness · deadlines
liveness verdict
N3 si_state_machine
single owner of island mode
fault verdict
N8 si_fault_manager
static fault tree · DTC
DTCs
actuator fault
N9 si_actuator_supervisor
N11 si_event_recorder
UDS / DTC export
audit (§5.3 step 4)
Diagnostic port
UDS tester (ISO 14229)
HPC log store
sync · offset
N12
si_time_quality_monitor
time base valid / invalid
Safety link (island)
N1 si_safety_gateway — verifies Data ID, sequence, CRC, length and age on every HPC message before any island node sees it. A failure is a safety event, never an exception (§8.4)
/si/out/state · constraints · MRM state — 9 signals out (§8.2)
hardware fault · ≤ 1 ms
request · grant · island status
take-over · release
RCC and local control — §6.5
N23 si_local_control
cable · dead-man · v_{local}
local command (B6) — outranks remote
joystick · dead-man
Joystick — operator in sight
wired pendant on the local port
remote command (B5)
N21 si_rcc_link
session · latency · stream age · E2E
commands in · stream, status out
RCC — remote command and control centre
cellular LTE / 5G, the island's own modem
ego state
encoded stream
N22 si_teleop_streamer
camera · tracks · clearances · ego
forward camera · radar tracks · ultrasonic — read through the same ingest as N14, N6, N15

What B6 runs on (§3.3) and how it behaves (§3.1)

Detect. N23 local control port — cable connected, dead-man, link heartbeat.

Decide. N3 · I2 / I3 command mode `LOCAL`, pre-empting `REMOTE`.

Act. Joystick → N23 → I1 local source — which outranks the remote one → I15 → I16.

Assure. I6 / I5 stop on release or link loss, through I7 (half-tone) · I19, I20 where sensing is still valid — their ingest is half-tone.

No exteroceptive sensor is required: the operator's eyes are the sensing, so B6 runs at rung 0 — like B4, the only other behaviour that does. N22 and the RCC are faded: nothing is streamed. **Trigger** pendant plugged in, key switch turned and dead-man held, at standstill. **Speed** 5 km/h forward, 3 km/h reverse. **Release or link loss** beyond T_{local} = 200 ms → immediate stop, about 0.85 m from 5 km/h at the emergency rate. **Exit** released at standstill → `SAFE_STOP`. Open items: decision 20.

Stacks — the colours of `ArchitectureMain.pdf`

Sensing · Localization · Perception · Planning · Control
System · VehicleInterface · Vehicle · Safety link

Where each node came from

MIGRATED	the same Autoware node, migrated
NEW	new — does not exist in Autoware

18 migrated + 19 new = **37**, plus the optional N20 — and N21–N23 of §6.5, outside the counts until their detailed design is done. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. † the HPC's external command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table; B5's remote command arrives through N21 instead.

Behaviours each node enables (§3.3)

one fixed slot per behaviour B6	**B1** degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.
SHARED	**SHARED** — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.
	Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.
⊘	**Used by no behaviour** — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

→	HPC-sourced, verified by N1	→	actuator feedback
→	island sensor data	→	state · mode · decision
→	control command into the gate	- - →	optional (§3.5)
- - →	fault · `/diagnostics`		

In this view — B6, manual joystick control without streamed sensing

🟥	**full colour** — named for B6 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)
▨	**half-tone** — on B6's path by §3.1, but not named in §3.3
☐	**faded** — not used by B6. An edge is drawn at the weaker of its two ends. Only B6's own tag is drawn on each box.

Every edge on this page, with the design-document section it is taken from, is catalogued in `../messageflow/`, which draws the same 37 nodes coloured by execution partition instead.

Figure 6. *B6 on the island's node graph: what B6 runs on in full colour, what it only passes through in half-tone, the rest of the island faded*

B6, on the island's node graph. The joystick, outside the boundary, reaches N23 on the local control port; the local command enters the gate as its local source, which outranks the remote one, and N3 holds the mode. Nothing is streamed, so N22, N21 and the RCC are faded. I6 and I5 stop the vehicle on release or link loss; I19 and I20 stay armed where sensing is still valid. Full size: *`layers/L4SafetyIsland-Layer6-B6-Joystick.pdf`.*

3.2 Precedence: four automated episodes, two edges back up, two human-driven exits

The four automated behaviours, B1–B4, are episodes on a ladder, and what distinguishes them is **who holds authority** and **whether the situation is still recoverable**.

- **B1 is the first response** to a degrading ODD, and it is cooperative: the HPC stays in control and the island only constrains it. B1 therefore *requires a working HPC*. Its object is an **operating domain**, not a speed: the island declares the current domain unsatisfiable and the HPC moves to one it can meet. B1 is also the only behaviour with an **upward** exit — the drive continues, in a narrower domain — and the only one whose downward exit can land on any of the other three.
- **B2 is what replaces B1 when that requirement fails.** The intent is the same — buy time in a safe state rather than commit to a terminal manoeuvre — but the authority is inverted: the island now steers. B2 is admissible only while the lane remains observable by the camera. It first brings the vehicle down to the safe speed of the current ODD, and it is bounded: after 8–15 s without the HPC it hands over to B3 if the camera still sees the lane and to B4 if it does not — never to anything else.
- **B3 is the terminal response to a worsening *situation***, taken when the envelope cannot be restored: either B1 exhausted its deadlines with the HPC alive but non-compliant, or B2's hold window expired with the HPC still gone. It brings the vehicle to a standstill and does not give it back.
- **B4 is the floor**, taken when the island can no longer *see*. B2 needs a lane and B3 needs a verifiable refuge; both are served by the same mandatory sensor, the camera, so both are lost together when it fails. B4 is what is left, and it too brings the vehicle to a standstill and does not give it back; it claims nothing about where the vehicle comes to rest.

Two behaviours end the drive, not one. B3 and B4 are *both* terminal. Neither has an edge back up, and in the state diagram below each has exactly one successor, `SAFE_STOP`. What separates them is not *whether* the vehicle stops but **where it stops, and on what evidence it chose the place**:

- B3 is a **placed** stop. The island can still see, so it nominates a refuge, verifies it free against `N16`'s live free space, leaves the running lane, and stops there. `MRM_SUCCEEDED` under B3 therefore asserts a *location*, and the claim is defensible because a sensor confirmed it.

- B4 is an **unplaced** stop. The island cannot see, so the only thing it is entitled to promise is a standstill — in whichever lane the vehicle already occupies. §3.1's success criterion says exactly that, and says nothing more.

Reading B3 as "the" terminal behaviour, and B4 as an abort path that merely fell out of it, is the misreading this subsection exists to prevent. The island has **two** ways to end a drive, and they differ in precondition (a verified refuge against none), in minimum sensor rung (§9.4: **B** against **0**), and in residual risk — a stop in a refuge and a stop in a running lane are not the same event, and the Part-4 HARA must carry them as two.

What takes over when B1 runs out of time. `T_odd` expiring is the most common way the island leaves B1 downward, and the successor is **not** fixed. Which behaviour takes over is decided by what the island still *has* — never by what the HPC did wrong, because the island cannot always tell a non-compliant HPC from a dead one, and does not need to:

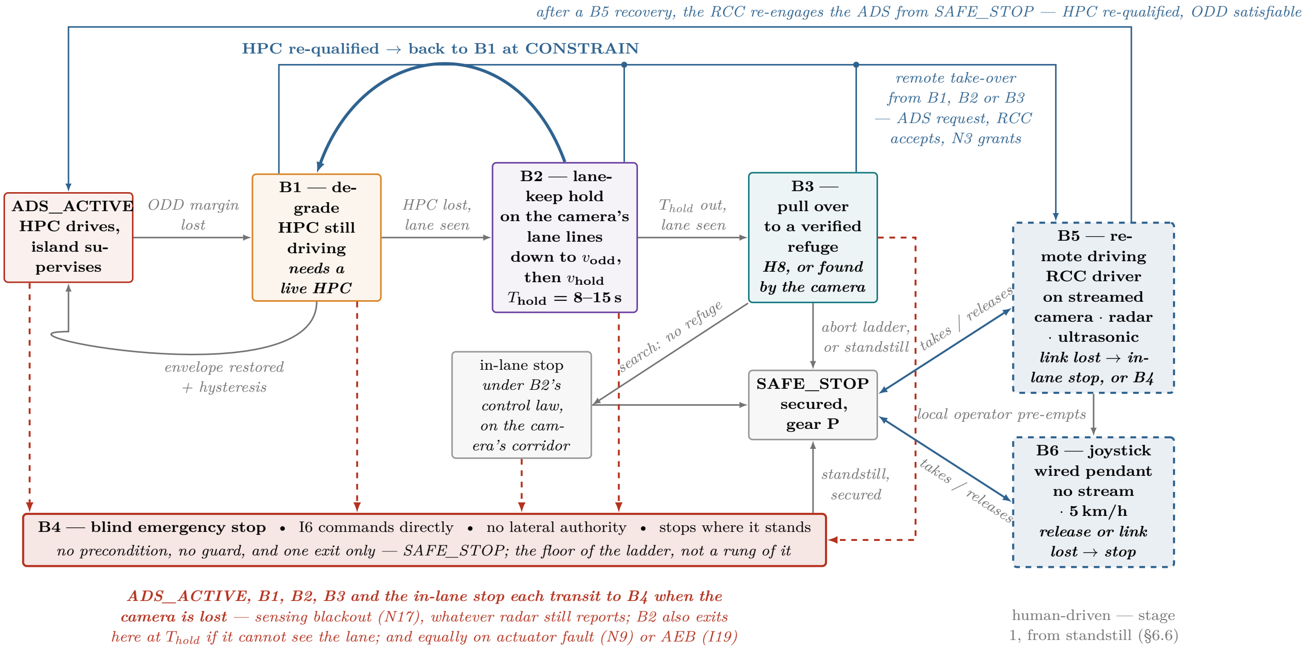
Condition at `T_odd` expiry	**Takes over**	**Why**
Sensing blackout — `N17` reports the camera invalid, or the lane lines lost	**B4**	Capability, not situation. Neither B2 nor B3 has a precondition left
HPC unavailable — `N2` reports liveness lost	**B2**	The executor of every B1 rung is gone, and the corridor is still observable
HPC alive, and either non-compliant or out of satisfiable domains	**B3**	The situation has not been recovered and will not be. The drive ends in a refuge

The tests are applied **in that order**, and the order is not arbitrary: it runs from capability loss to executor loss to situation loss, which is the order in which each removes the preconditions of the options below it. Applying them in any other order can select a behaviour whose precondition has already failed — choosing B3 on a blackout, for instance, commits the island to a refuge it cannot verify.

The ladder is one-way, with two exceptions. B1 → B2 → B3 are entered on worsening evidence and are not undone by wishful thinking. B2's downward exit is a fork, not a step: at `T_hold` it goes to B3 or to B4, applying the same capability-first test as the `T_odd` table above, and it has no third downward edge — the in-lane stop is reached only through B3's search, never straight from B2. Two edges climb back up, and they are different in kind. The reversible *authority* edge is **B2** → **B1**: if the HPC comes back and proves itself over a confirmation window, the island hands authority back and returns to supervising rather than driving. The other is **B1** → `ADS_ACTIVE`: the domain the HPC declared through `H4` has passed both acceptance tests and held for the §6.4 recovery window, so the HPC is driving normally again in the narrower domain and the island releases its cap. No authority moves on that edge — the HPC was driving throughout. Nothing else climbs back up — in particular, B3 never returns to B2 or B1, because a pull-over commits the vehicle to leaving the running lane.

B4 is not a fourth rung of that ladder; it is underneath all of them. B1–B3 escalate on a worsening *situation* — the envelope is not held, the HPC is gone, the deadline has run out. B4 is entered on a loss of *capability*, and capability can be lost at any rung. The two axes are independent, and the design is easier to argue once they are drawn that way: every state in the island has an edge into B4, including the in-lane stop, and B4's **only** edge out is to `SAFE_STOP` — the secured standstill, which is a terminal *state*, not another behaviour. Nothing climbs back up out of it. That is why it carries no precondition. A floor with an entry condition is a trapdoor.

B5 and B6 sit outside the ladder: they are how a person takes the vehicle back. Every automated behaviour ends at `SAFE_STOP`, and none of them leaves it. B5 and B6 are the only edges out of `SAFE_STOP` short of the end of the drive cycle, and each needs a person to take it — the remote driver over the RCC link, or the operator at the vehicle. Both return to `SAFE_STOP` when released, and both fall back onto the automated floor when their link fails: B5 to the in-lane stop or to B4, by the same capability-first test; B6 straight to a stop. B6 pre-empts B5. B6 starts only from standstill. **B5 has three more entries, from B1, B2 and B3**: when the HPC behaves abnormally the ADS asks for a remote take-over, and once the RCC accepts and `N3` grants, the remote driver takes the moving vehicle over from whichever of the three is running — at or below `v_teleop`, after 5 s of situational awareness, with a matched command (§6.6). Until then the current behaviour keeps authority and its timers run. B5 also carries the one edge from `SAFE_STOP` back to automated driving: once the HPC is re-qualified and the ODD is satisfiable, the RCC may re-engage the ADS.


after a B5 recovery, the RCC re-engages the ADS from SAFE_STOP — HPC re-qualified, ODD satisfiable
HPC re-qualified → back to B1 at CONSTRAIN
remote take-over from B1, B2 or B3 — ADS request, RCC accepts, N3 grants
ADS_ACTIVE HPC drives, island supervises
ODD margin lost
B1 — degrade HPC still driving needs a live HPC
HPC lost, lane seen
B2 — lane-keep hold on the camera's lane lines down to v_{odd}, then v_{hold} $T_{\text{hold}} = 8\text{–}15\,\text{s}$
T_{hold} out, lane seen
B3 — pull over to a verified refuge H8, or found by the camera
B5 — remote driving RCC driver on streamed camera · radar · ultrasonic link lost → in-lane stop, or B4
envelope restored + hysteresis
in-lane stop under B2's control law, on the camera's corridor
search: no refuge
abort ladder, or standstill
takes / releases
SAFE_STOP secured, gear P
local operator pre-empts
takes / releases
standstill, secured
B6 — joystick wired pendant no stream · 5 km/h release or link lost → stop
B4 — blind emergency stop • I6 commands directly • no lateral authority • stops where it stands
no precondition, no guard, and one exit only — SAFE_STOP; the floor of the ladder, not a rung of it
human-driven — stage 1, from standstill (§6.6)

ADS_ACTIVE, B1, B2, B3 and the in-lane stop each transit to B4 when the camera is lost *— sensing blackout (N17), whatever radar still reports; B2 also exits here at T_{hold} if it cannot see the lane; and equally on actuator fault (N9) or AEB (I19)*

Figure 7. *The behaviour ladder. One-way except the B2 → B1 hand-back; B2 exits to B3 or B4 only — and every state transits to B4. B5 and B6, human-driven, are the only exits from SAFE_STOP; B5 may also take over a moving vehicle from B1, B2 or B3 — and after a B5 recovery the RCC may re-engage the ADS.*

Two consequences must be written down rather than discovered during integration.

> **B2 must be able to give authority back.** A behaviour that can only be entered is not a hold, it is a one-way MRM with extra steps. The hand-back path is `I3`'s smooth-transition mechanism, it re-enters B1 at `CONSTRAIN` rather than `ALLOW`, and it is gated on evidence of the HPC's health, not merely on its heartbeat returning. The confirmation window and the acceptance criteria are open decision 14.

> **B3 deliberately and temporarily violates lateral containment.** A pull-over leaves the running lane on purpose. The containment check (`N5`) must therefore be *suspended under authorisation* for the duration of the manoeuvre, and re-armed against the refuge boundary instead. An unsuppressed containment check would abort every pull-over it was asked to protect.

Authorisation for that suspension is granted by `N3` and by nothing else, is bounded in time and in lateral extent by `N7`, and is recorded by `N11`.

3.3 Behaviour-to-node traceability

Every island node exists to serve at least one declared behaviour. Nodes that serve none do not belong on the island.

Behaviour	**Detect**	**Decide**	**Act**	**Assure**
B1	N19 ODD monitor · N17 sensor health · N6/N14 ingest (N13 where fitted) · N16 world model · H4 declared active domain	N10 envelope · N3 state machine (demand, `T_ack`/`T_odd`, takeover selection) · N8 fault manager	N1/H1 constraint egress → HPC switches domain and re-plans	Compliance supervision in N2 and the two-part acceptance test in N19; N11 records the SOTIF event
B2	N2 HPC supervisor (entry and recovery) · N14 camera lane lines (required) · N17 camera validity · N6 radar · N16 world model · H3 corridor · N19 for the `v_max` term of `v_hold` · H4's last `active_odd` for the `v_odd` term	N3 state machine (hold budget, hand-back) · N5 corridor monitor · I8 validator · N10 envelope · N7 owns the `v_odd` descent and the `v_hold` ramp · N3 forks the exit to B3 or B4	I13 pure_pursuit + N7 planner → I1 gate → I16; I3 switcher for hand-back	N9 actuator supervisor; I19/I20 as independent last line
B3	N16 free space · N6 rear-corner radar · H8 refuge · N14 camera — lane lines, road edge and pull-over space for the island-detected refuge · N15 near-field, where fitted · N13 only where fitted	N18 pull-over manager (both refuge sources, `T_refuge`) · I7 mrm_handler · N3	N7 trajectory → I13 → I1 → I16; I5 secures at standstill	N9; abort ladder to the in-lane stop under B2's control law, then I6
B4	N17 blackout declaration — camera invalid · N16 no corridor, no lateral free space · N9 actuator supervisor · I19 AEB	N3 state machine (`EMERGENCY_STOP`, no exit) · N8 fault manager · I7 mrm_handler	I6 emergency stop → I1 gate → I16; I5 secures	N4 proprioceptive ego state — the only assurance left; below the node path, the gate's built-in stop and the §8.5 FSI fault line

Behaviour	**Detect**	**Decide**	**Act**	**Assure**
B5	N21 RCC link — session, latency, stream age · N17 camera validity · N16 for the island's own guards · the ADS's take-over request — N3 on entering B1–B3, or the HPC through `HpcCapability` (N1, N2)	N3 grants and revokes · I2/I3 mode `REMOTE`	remote command → N21 → I1 remote source → I16; N22 streams camera, radar and ultrasonic to the RCC	I19 · I20 · N10 · N9; link-loss stop through N7 or I6 — stage 1, §6.6
B6	N23 local control port — cable connected, dead-man, link heartbeat	N3 · I2/I3 mode `LOCAL`, pre-empting `REMOTE`	joystick → N23 → I1 local source → I16	I6/I5 stop on release or link loss · I19, I20 where sensing is valid — stage 1, §6.6

Every **Detect** cell above may additionally carry the optional §3.5 object digest from N20. It is left out of the table deliberately: a table that traces behaviours to the nodes they *require* must not list a node that no behaviour requires.

3.4 The four internal functions

B1–B4 are what the island does as seen from outside — and B5, B6 what it lets a person do through it. Internally, deck p.19's features expand into four functional blocks:

1. **Supervise** — is the HPC alive, on time, in-ODD, and internally healthy?
2. **Validate** — is the command it just sent physically plausible and inside the safety envelope?
3. **Control** — turn the accepted trajectory into actuator commands, at cadence, without the HPC in the loop for a single control period.
4. **Fall back** — when 1 or 2 fail, take authority and execute a minimal risk manoeuvre.

Everything else — how the trajectory was produced, what the world looks like, where the vehicle is on the map — belongs to the HPC.

3.5 Optional enrichment: the HPC object list

Everything above is written so that the island delivers all six behaviours without any HPC-produced perception. That is the floor, and it does not move. Above the floor there is one HPC-produced input worth taking, because it is the only one that tells the island what the HPC *believes about the world* rather than what it *intends to do*: `/perception/object_recognition/objects`, the output of the HPC's object-recognition and prediction stack.

While the HPC is functioning the island may subscribe to it, and when the message is available all four automated behaviours may use it. It is an option, never a dependency.

The rule that makes it admissible

> **Monotone conservatism.** An optional input may only *narrow* what the island permits. It may shorten a deadline, lower a speed ceiling, tighten an envelope, disqualify a refuge or steepen a deceleration ramp. It may never widen a bound, extend a hold, qualify a manoeuvre, hand back authority, or serve as the precondition of any behaviour. **Remove the input entirely and every behaviour must remain reachable and correct — only less well informed.**

Three consequences follow at once, and they are what keep the rest of this document true:

1. **No behaviour acquires a precondition.** B4 in particular keeps its defining property — no precondition, no guard, no exit. A fresh object list may not delay it, and no object list is ever consulted to decide whether the island is blind (§6.2).
2. **No integrity is inherited.** The list comes from a QM-to-ASIL-B pipeline on the far side of the link. It cannot raise the integrity of an island decision; it can only add a conservative term to a decision the island was already entitled to make on its own evidence.
3. **Its failure mode is nuisance, not collision.** Because the input can only make the island *more* cautious, a corrupted, phantom or persistent object list produces unnecessary braking — an availability problem, to be argued as such in the Part-4 HARA — rather than a missed hazard. That is the whole reason a QM-sourced input is admissible at all.

Why this input, when R5 excludes the others

R5 keeps HD maps, point clouds and images off the link, and the reasoning applies to a *raw* object list as well: `PredictedObjects` is variable-length, and a variable-length message on the safety link offends R4 as surely as it offends R5. So the island does not consume it raw. `N20 si_object_digest` truncates it at the boundary into a **bounded digest** — a fixed maximum object count ordered by threat, a fixed prediction horizon, a fixed field set — and everything downstream sees only the digest. What crosses the link is an Autoware topic; what enters the island is a bounded one. R5's consequence line is amended accordingly in §2.1.

The reason to take it at all is that it is **diverse in the useful direction**. R7 prefers a different channel to a copy of the HPC's, and this is not a copy: it is the HPC's own conclusion, which lets the island check the HPC *against itself*. The island does not have to out-perceive the HPC in order to catch it planning through an object it has just published.

What each behaviour may do with it

Behaviour	What the digest may do	What it may never do
B1	**The self-consistency veto** — `N10` and `I8` may raise `CONSTRAIN`, or escalate to `MRM`, when the HPC's trajectory is inconsistent with the HPC's *own* object list: the planned path passes through a published object, or the commanded speed leaves no safe following distance to one. Separately, `N10`'s RSS longitudinal envelope takes the *smaller* of the island-perceived and digest-derived safe speeds	Relax any envelope, raise `v_max`, or satisfy a compliance deadline. **A quiet object list is never evidence that the road is clear**
B2	Within the carry-over budget below, lower `v_hold` or shorten `N7`'s deceleration ramp when a digest object was predicted to enter the corridor	Extend `T_hold`, substitute for a lost corridor, or keep the hold alive when `N5` has no corridor source. B2's corridor precondition is unchanged
B3	**Disqualify** a refuge — an object predicted to occupy it, or a vehicle predicted to close on the pull-over side — and abort to the in-lane stop	Qualify a refuge. `pull_over_available` remains a function of `N16` free space, rear-corner radar and `N18` alone
B4	Shape the ramp only: permit deceleration up to the upper bound of `a_emg` when a digest object was last seen in path	Delay entry, offer an exit, or count as a valid source. B4 keeps no precondition

Freshness, and what survives the HPC

Two budgets, not one. **While the HPC is alive** the digest is age-checked like every other §8.1 row (300 ms). **After the HPC goes silent** the last valid digest is carried forward under a **carry-over budget bounded in both time and distance**; past either bound `N20` marks it invalid and it leaves the fusion entirely. Both numbers are open decision 16.

The carry-over is deliberately short and deliberately double-bounded, for the same reason decision 15 bounds blindness by distance as well as by time: at 60 km/h a "small" 500 ms is 8 m of road, and a predicted path that was right when it was published is not right after the vehicle has travelled past the geometry it was predicted against.

Where the digest matters least is exactly where one might hope for most. B2, B3 and B4 are all entered *after* the HPC has failed, so in the ordinary case the digest is already stale or gone by the time they execute. Its real value is in B1, where the HPC is alive by definition and the list is fresh. The B2–B4 uses are worth specifying anyway, because the boundary case — the HPC has just gone silent and the last digest is 200 ms old — is exactly the moment the island is deciding how hard to brake, and a 200 ms-old prediction is genuinely informative about the next two seconds. What it is not, ever, is a reason to wait.

---

4 Autoware nodes allocated to the Safety Island

The allocation, in five phases. The tables in this section answer *which* nodes move and *why*. The five figures below answer *what the graph looks like while it happens*: the same picture, redrawn five times, each step one further than the last. They are worth reading before §4.1, because a migration list is far easier to check against a picture than against a table alone.

Phase	What it shows	What to take from it
1	Autoware today, every node	The starting point. Nothing on it is changed by this design
2	The same graph, one box per stack	The picture the island modifies, with the detail folded away
3	The same boxes, after the island	**The change itself**: Control and the Vehicle Interface migrate bodily; the island grows reduced versions of the other stacks; one narrow link joins the two
4	Only what changed, at node level	The delta, opened up: removed, re-sourced and new on the HPC; migrated on the island
5	The Safety Island alone	All 37 island nodes — the 18 of §4.1 and the 19 of §6 — each tagged **? B1–B6** with the behaviours it enables, per the §3.3 traceability table — **? SHARED** where every behaviour runs through it, **?** where none uses it. `layers/` also redraws it once per behaviour (B1–B6) — with N21–N23 of §6.5 drawn for B5 and B6

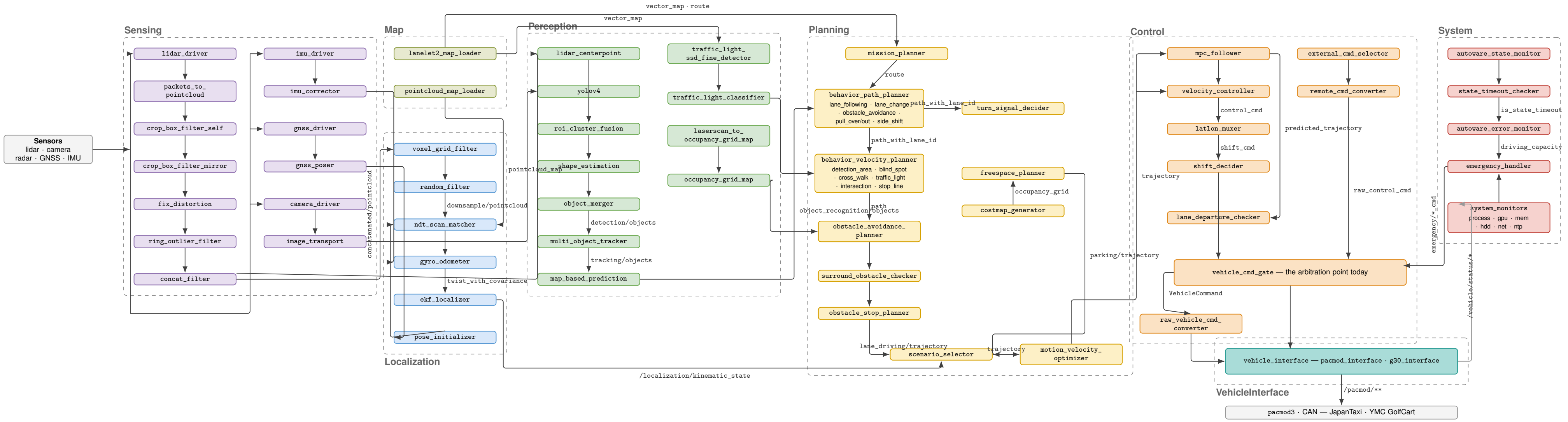
The progression is the argument of this section in visual form. Phase 3 is where the allocation is decided; phase 4 is where it is justified node by node; phase 5 is what the island then contains.

Each phase is reproduced **whole, sideways, on its own page**, so that nothing is cropped and each step of the argument is a single picture rather than a pair to be assembled by eye. Every phase is also available at full size in `layers/L4SafetyIsland-Layers.pdf`.

Colour is the **Autoware stack** a node belongs to, taken from the legend of `ArchitectureMain.pdf`, and it means the same thing on all five. What changes between phases is only how far the picture is zoomed and — from phase 3 onward — which ECU each box runs on. The five diagrams follow in the order of the table above.

Phase 1 — Autoware today, every node

The published Autoware node graph (`ArchitectureMain.pdf`), redrawn here in the same visual language as phases 2–5 so that the five read as one progression.
Nothing on this page is changed by the Safety Island — this is the picture the rest of the series modifies. Colour is the Autoware stack, from the original's own legend.


vector_map · route
vector_map
Sensing
Map
Perception
Planning
Control
System
Sensors
lidar · camera
radar · GNSS · IMU
lidar_driver
packets_to_pointcloud
crop_box_filter_self
crop_box_filter_mirror
fix_distortion
ring_outlier_filter
concat_filter
imu_driver
imu_corrector
gnss_driver
gnss_poser
camera_driver
image_transport
concatenated/pointcloud
lanelet2_map_loader
pointcloud_map_loader
voxel_grid_filter
random_filter
downsample/pointcloud
ndt_scan_matcher
gyro_odometer
twist_with_covariance
ekf_localizer
pose_initializer
Localization
pointcloud_map
lidar_centerpoint
yolov4
roi_cluster_fusion
shape_estimation
object_merger
detection/objects
multi_object_tracker
tracking/objects
map_based_prediction
traffic_light_ssd_fine_detector
traffic_light_classifier
laserscan_to_occupancy_grid_map
occupancy_grid_map
mission_planner
route
behavior_path_planner
lane_following · lane_change · obstacle_avoidance · pull_over/out · side_shift
path_with_lane_id
turn_signal_decider
path_with_lane_id
behavior_velocity_planner
detection_area · blind_spot · cross_walk · traffic_light · intersection · stop_line
freespace_planner
occupancy_grid
costmap_generator
object_recognition/objects
path
obstacle_avoidance_planner
surround_obstacle_checker
obstacle_stop_planner
lane_driving/trajectory
scenario_selector
trajectory
motion_velocity_optimizer
parking/trajectory
/localization/kinematic_state
mpc_follower
velocity_controller
control_cmd
latlon_muxer
shift_cmd
shift_decider
trajectory
lane_departure_checker
predicted_trajectory
external_cmd_selector
remote_cmd_converter
raw_control_cmd
vehicle_cmd_gate — the arbitration point today
VehicleCommand
raw_vehicle_cmd_converter
autoware_state_monitor
state_timeout_checker
is_state_timeout
autoware_error_monitor
driving_capacity
emergency_handler
system_monitors
process · gpu · mem · hdd · net · ntp
emergency/*_cmd
/vehicle/status/*
vehicle_interface — pacmod_interface · g30_interface
VehicleInterface
/pacmod/**
pacmod3 · CAN — JapanTaxi · YMC GolfCart

Stacks — the colours of `ArchitectureMain.pdf`

Sensing
Localization
Perception
Planning
Control
System
VehicleInterface
Vehicle
Map

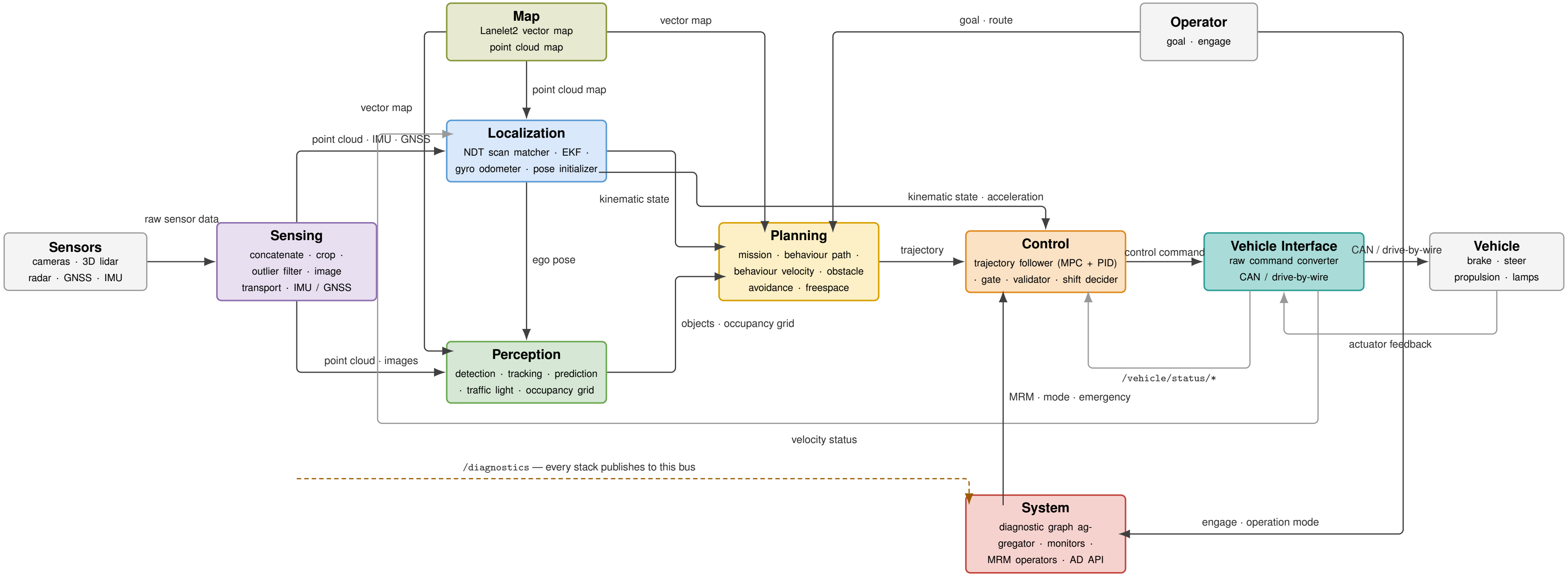
This is the unmodified picture. Every node here stays exactly where it is until phase 3; phase 2 is this same graph with each stack merged into one box. Two condensations, for the page: the per-sensor lidar pre-processing chain is drawn once rather than once per sensor, and the two behaviour planners' module lists are given inside their box rather than as separate boxes.

Names are those of `ArchitectureMain.pdf`, which predates the `autoware_universe` renaming — `mpc_follower` is today's `mpc_lateral_controller`, `vehicle_cmd_gate` today's `control_command_gate`, and so on. The mapping is in `../nodediagram/L4SafetyIsland-NodeDiagram.md` §2.

Figure 8. *Phase 1 of 5 — Autoware today, every node. The published node graph, redrawn in the visual language the other four phases use, so the series reads as one progression. Nothing on this page is changed by the Safety Island: this is the picture the remaining phases modify.*

Phase 2 — Autoware today, one box per stack

Every node of phase 1 merged into the stack it belongs to, and every link merged with it.
This is the picture the Safety Island changes; phase 3 shows the same boxes after the change.


Map
Lanelet2 vector map
point cloud map
vector map
goal · route
Operator
goal · engage
point cloud map
vector map
point cloud · IMU · GNSS
Localization
NDT scan matcher · EKF ·
gyro odometer · pose initializer
kinematic state · acceleration
kinematic state
raw sensor data
Sensors
cameras · 3D lidar
radar · GNSS · IMU
Sensing
concatenate · crop ·
outlier filter · image
transport · IMU / GNSS
ego pose
Planning
mission · behaviour path ·
behaviour velocity · obstacle
avoidance · freespace
trajectory
Control
trajectory follower (MPC + PID)
· gate · validator · shift decider
control command
Vehicle Interface
raw command converter
CAN / drive-by-wire
CAN / drive-by-wire
Vehicle
brake · steer
propulsion · lamps
objects · occupancy grid
actuator feedback
point cloud · images
Perception
detection · tracking · prediction
· traffic light · occupancy grid
/vehicle/status/*
MRM · mode · emergency
velocity status
/diagnostics — every stack publishes to this bus
System
diagnostic graph aggregator · monitors ·
MRM operators · AD API
engage · operation mode

Stacks — the colours of `ArchitectureMain.pdf`

Sensing	Localization	Perception	Map	Planning
Control	System	Vehicle Interface	Vehicle / actor	

Merged links

→ data, command or decision

→ feedback from the vehicle

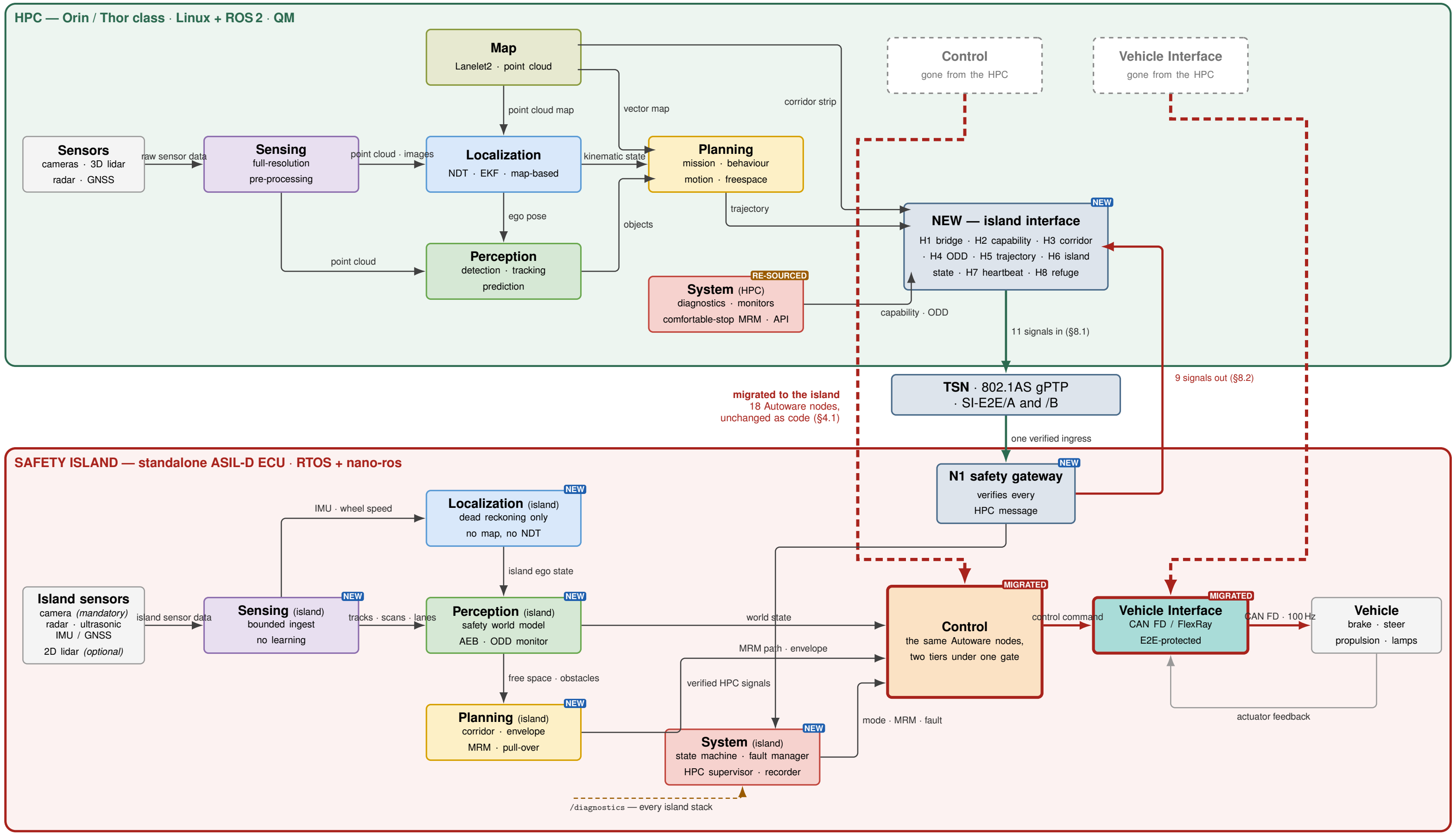
⇢ diagnostics

One box per stack, one arrow per merged link. The diagnostics bus is drawn once rather than as one stub per stack; every stack publishes to it. Phase 1 is the same graph with all of its nodes shown.

Figure 9. *Phase 2 of 5 — the same graph with one box per stack. Every node of phase 1 is merged into the stack it belongs to, and every link merged with it: sensors through perception, planning, control, the vehicle interface and System.*

Phase 3 — the same picture, after the Safety Island

Same boxes and same colours as phase 2. **The Control stack and the Vehicle Interface move bodily to the island** — dashed red, from the ghost they leave behind. The island grows reduced versions of the other stacks (blue tag), the HPC gains an interface block, and one narrow link joins the two. Node-level detail is phase 4.


HPC — Orin / Thor class · Linux + ROS 2 · QM
Map
Lanelet2 · point cloud
point cloud map
vector map
corridor strip
Sensors
cameras · 3D lidar
radar · GNSS
raw sensor data
Sensing
full-resolution
pre-processing
point cloud · images
Localization
NDT · EKF · map-based
kinematic state
Planning
mission · behaviour
motion · freespace
trajectory
ego pose
objects
point cloud
Perception
detection · tracking
prediction
RE-SOURCED
System (HPC)
diagnostics · monitors
comfortable-stop MRM · API
capability · ODD
Control
gone from the HPC
Vehicle Interface
gone from the HPC
NEW
NEW — island interface
H1 bridge · H2 capability · H3 corridor · H4 ODD · H5 trajectory · H6 island state · H7 heartbeat · H8 refuge
11 signals in (§8.1)
9 signals out (§8.2)
TSN · 802.1AS gPTP · SI-E2E/A and /B
migrated to the island
18 Autoware nodes,
unchanged as code (§4.1)
one verified ingress
SAFETY ISLAND — standalone ASIL-D ECU · RTOS + nano-ros
NEW
N1 safety gateway
verifies every HPC message
IMU · wheel speed
NEW
Localization (island)
dead reckoning only
no map, no NDT
island ego state
Island sensors
camera (mandatory)
radar · ultrasonic
IMU / GNSS
2D lidar (optional)
island sensor data
NEW
Sensing (island)
bounded ingest
no learning
tracks · scans · lanes
NEW
Perception (island)
safety world model
AEB · ODD monitor
world state
MRM path · envelope
verified HPC signals
free space · obstacles
NEW
Planning (island)
corridor · envelope
MRM · pull-over
NEW
System (island)
state machine · fault manager
HPC supervisor · recorder
mode · MRM · fault
/diagnostics — every island stack
MIGRATED
Control
the same Autoware nodes,
two tiers under one gate
control command
MIGRATED
Vehicle Interface
CAN FD / FlexRay
E2E-protected
CAN FD · 100 Hz
Vehicle
brake · steer
propulsion · lamps
actuator feedback

What changed

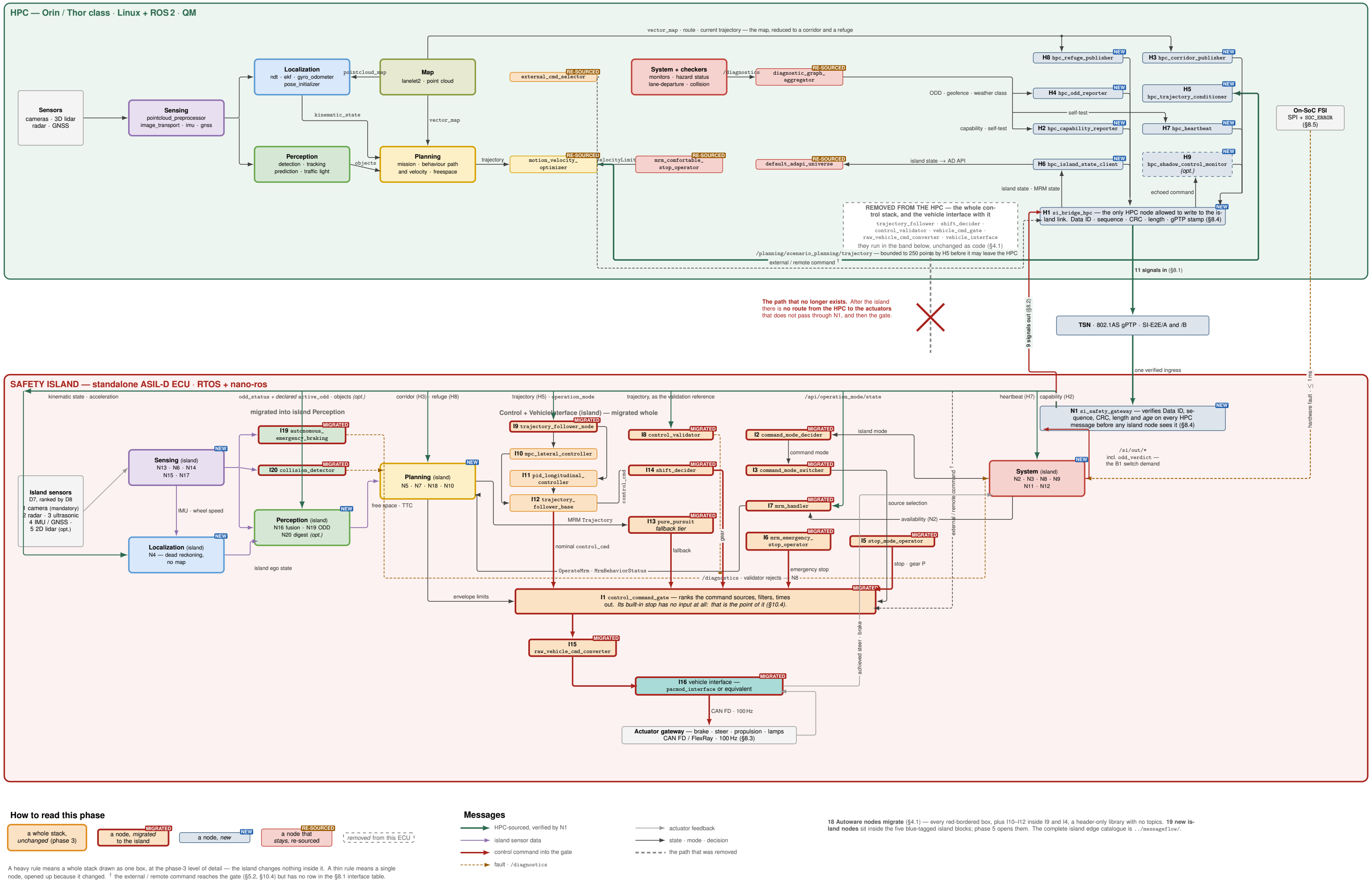
- MIGRATED: same Autoware stack, now on the island
- NEW: new — does not exist in Autoware today
- RE-SOURCED: stays, but its role or sources change
- no longer on this ECU
- - - -▶ the migration

The island is not a monitor bolted on the side: it runs a *reduced Autoware*. Its localization is one node, its planning is four bounded map-free ones — and its control stack is not a reimplementation at all, it is Autoware's, relocated. Phase 4 opens what changed to node level.

Figure 10. *Phase 3 of 5 — the same picture after the Safety Island, at the same level of abstraction. The HPC band above keeps its stacks; the island band below grows reduced versions of them. The Control stack and the Vehicle Interface migrate to the island (dashed red, from the ghost they leave behind), the HPC gains the interface block, and one narrow link joins the two ECUs.*

Phase 4 — what the Safety Island changes, at node level

A **selective** zoom, between phase 3 (one box per stack) and phase 5 (the island, node by node).
On the HPC only the nodes that change are opened up — what was **removed**, what is **re-sourced**, and the **new** interface block. Every stack that the island leaves alone stays as one box, exactly as in phase 3.
On the island only the nodes migrated from the HPC are opened up. The island's own new stacks stay as one box each; phase 5 opens them.


HPC — Orin / Thor class · Linux + ROS 2 · QM
Sensors
cameras · 3D lidar
radar · GNSS
Sensing
pointcloud_preprocessor
image_transport · imu · gnss
Localization
ndt · ekf · gyro_odometer
pose_initializer
pointcloud_map
Map
lanelet2 · point cloud
kinematic_state
vector_map
Perception
detection · tracking
prediction · traffic light
objects
Planning
mission · behaviour path
and velocity · freespace
trajectory
RE-SOURCED
external_cmd_selector
RE-SOURCED
motion_velocity_optimizer
VelocityLimit
System + checkers
monitors · hazard status
lane-departure · collision
/diagnostics
RE-SOURCED
diagnostic_graph_aggregator
RE-SOURCED
mrm_comfortable_stop_operator
RE-SOURCED
default_adapi_universe
vector_map · route · current trajectory — the map, reduced to a corridor and a refuge
NEW
H8 hpc_refuge_publisher
NEW
H3 hpc_corridor_publisher
ODD · geofence · weather class
NEW
H4 hpc_odd_reporter
NEW
H5
hpc_trajectory_conditioner
self-test
capability · self-test
NEW
H2 hpc_capability_reporter
NEW
H7 hpc_heartbeat
island state → AD API
NEW
H6 hpc_island_state_client
NEW
H9
hpc_shadow_control_monitor
(opt.)
island state · MRM state
echoed command
NEW
H1 si_bridge_hpc — the only HPC node allowed to write to the island link. Data ID · sequence · CRC · length · gPTP stamp (§8.4)
On-SoC FSI
SPI + SOC_ERROR
(§8.5)
REMOVED FROM THE HPC — the whole control stack, and the vehicle interface with it
trajectory_follower · shift_decider ·
control_validator · vehicle_cmd_gate ·
raw_vehicle_cmd_converter · vehicle_interface
they run in the band below, unchanged as code (§4.1)
/planning/scenario_planning/trajectory — bounded to 250 points by H5 before it may leave the HPC
external / remote command †
11 signals in (§8.1)
The path that no longer exists. After the island there is no route from the HPC to the actuators that does not pass through N1, and then the gate.
9 signals out (§8.2)
TSN · 802.1AS gPTP · SI-E2E/A and /B
one verified ingress
hardware fault · ≤ 1 ms
SAFETY ISLAND — standalone ASIL-D ECU · RTOS + nano-ros
kinematic state · acceleration
odd_status + declared active_odd · objects (opt.)
corridor (H3) · refuge (H8)
trajectory (H5) · operation_mode
trajectory, as the validation reference
/api/operation_mode/state
heartbeat (H7)
capability (H2)
NEW
N1 si_safety_gateway — verifies Data ID, sequence, CRC, length and age on every HPC message before any island node sees it (§8.4)
Island sensors
D7, ranked by D8
1 camera (mandatory)
2 radar · 3 ultrasonic
4 IMU / GNSS ·
5 2D lidar (opt.)
NEW
Sensing (island)
N13 · N6 · N14
N15 · N17
IMU · wheel speed
NEW
Localization (island)
N4 — dead reckoning,
no map
island ego state
migrated into island Perception
MIGRATED
I19 autonomous_emergency_braking
MIGRATED
I20 collision_detector
NEW
Perception (island)
N16 fusion · N19 ODD
N20 digest (opt.)
NEW
Planning (island)
N5 · N7 · N18 · N10
free space · TTC
envelope limits
Control + VehicleInterface (island) — migrated whole
MIGRATED
I9 trajectory_follower_node
I10 mpc_lateral_controller
I11 pid_longitudinal_controller
I12 trajectory_follower_base
control_cmd
MIGRATED
I8 control_validator
MIGRATED
I14 shift_decider
gear
MRM Trajectory
MIGRATED
I13 pure_pursuit
fallback tier
nominal control_cmd
fallback
OperateMrm · MrmBehaviorStatus
/diagnostics · validator rejects — N8
MIGRATED
I2 command_mode_decider
island mode
command mode
MIGRATED
I3 command_mode_switcher
MIGRATED
I7 mrm_handler
source selection
availability (N2)
MIGRATED
I6 mrm_emergency_stop_operator
emergency stop
MIGRATED
I5 stop_mode_operator
stop · gear P
external / remote command †
NEW
System (island)
N2 · N3 · N8 · N9
N11 · N12
/si/out/*
incl. odd_verdict —
the B1 switch demand
MIGRATED
I1 control_command_gate — ranks the command sources, filters, times out. Its built-in stop has no input at all: that is the point of it (§10.4).
MIGRATED
I15
raw_vehicle_cmd_converter
MIGRATED
I16 vehicle interface —
pacmod_interface or equivalent
achieved steer · brake
CAN FD · 100 Hz
Actuator gateway — brake · steer · propulsion · lamps
CAN FD / FlexRay · 100 Hz (§8.3)
How to read this phase
a whole stack, unchanged (phase 3)
MIGRATED
a node, migrated to the island
NEW
a node, new
RE-SOURCED
a node that stays, re-sourced
removed from this ECU
Messages
HPC-sourced, verified by N1
island sensor data
control command into the gate
fault · /diagnostics
actuator feedback
state · mode · decision
the path that was removed

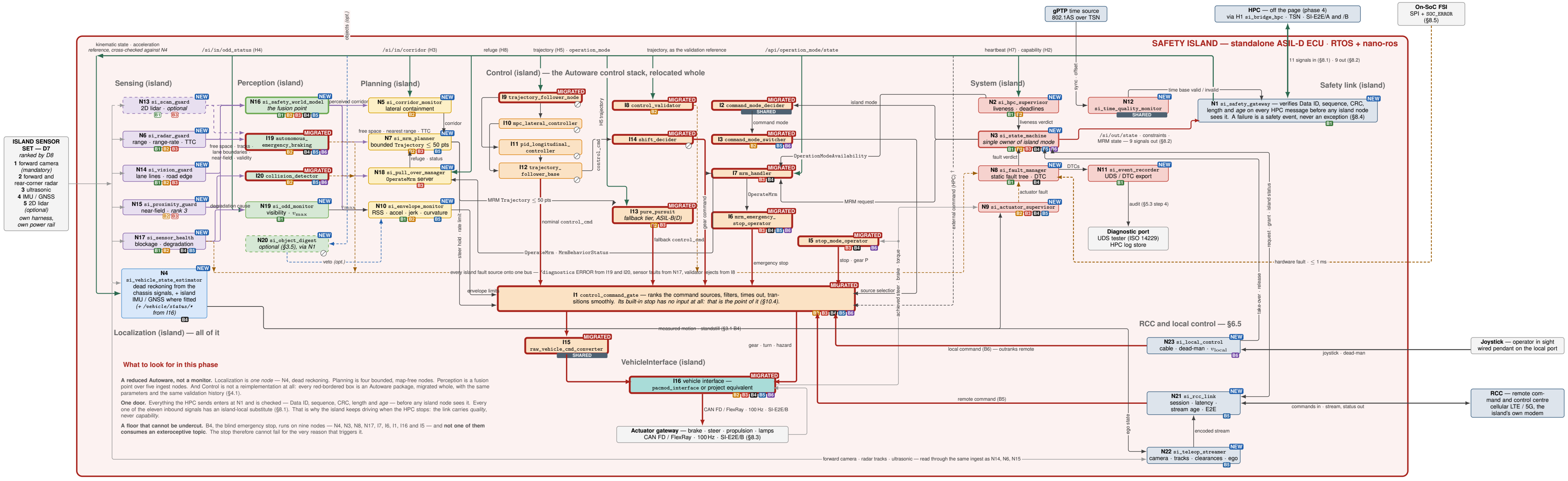
A heavy rule means a whole stack drawn as one box, at the phase-3 level of detail — the island changes nothing inside it. A thin rule means a single node, opened up because it changed. † the external / remote command reaches the gate (§5.2, §10.4) but has no row in the §8.1 interface table.

18 Autoware nodes migrate (§4.1) — every red-bordered box, plus I10–I12 inside I9 and I4, a header-only library with no topics. **19 new island nodes** sit inside the five blue-tagged island blocks; phase 5 opens them. The complete island edge catalogue is `../messageflow/`.

Figure 11. *Phase 4 of 5 — what changed, at node level. On the HPC, only the nodes removed, re-sourced or newly designed are opened up; on the island, only the nodes migrated from it. A heavy box rule means a whole stack, unchanged at phase-3 level; a thin rule means a single node, opened up because it changed.*

Phase 5 — the Safety Island alone, every node

The last zoom. All 37 island nodes — 18 migrated Autoware **I**-nodes and 19 new **N**-nodes — plus the optional N20, and N21–N23 of §6.5 for B5 and B6 (§6.6).
Colour is still the Autoware stack, as in `ArchitectureMain.pdf`; the island runs a reduced version of nearly every stack. The HPC is off the page: it reaches this diagram only through N1.


objects (opt.)
SAFETY ISLAND — standalone ASIL-D ECU · RTOS + nano-ros
kinematic state · acceleration
reference, cross-checked against N4
/si/in/odd_status (H4)
/si/in/corridor (H3)
refuge (H8)
trajectory (H5) · operation_mode
trajectory, as the validation reference
/api/operation_mode/state
heartbeat (H7) · capability (H2)
gPTP time source 802.1AS over TSN
HPC — off the page (phase 4) via H1 si_bridge_hpc · TSN · SI-E2E/A and /B
On-SoC FSI SPI + SOC_ERROR (§8.5)
11 signals in (§8.1) · 9 out (§8.2)
ISLAND SENSOR SET — D7
ranked by D8
1 forward camera (mandatory)
2 forward and rear-corner radar
3 ultrasonic
4 IMU / GNSS
5 2D lidar (optional)
own harness, own power rail
Sensing (island)
N13 si_scan_guard 2D lidar · optional
N6 si_radar_guard range · range-rate · TTC
N14 si_vision_guard lane lines · road edge
N15 si_proximity_guard near-field · rank 3
N17 si_sensor_health blockage · degradation
N4 si_vehicle_state_estimator dead reckoning from the chassis signals, + island IMU / GNSS where fitted (+ /vehicle/status/* from I16)
Localization (island) — all of it
Perception (island)
N16 si_safety_world_model the fusion point
I19 autonomous_emergency_braking
I20 collision_detector
N19 si_odd_monitor visibility · $v_{\max}$
N20 si_object_digest optional (§3.5), via N1
free space · tracks · lane boundaries · near-field · validity
degradation cause
perceived corridor
veto (opt.)
Planning (island)
N5 si_corridor_monitor lateral containment
corridor
free space · nearest range · TTC
N7 si_mrm_planner bounded Trajectory ≤ 50 pts
refuge · status
N18 si_pull_over_manager OperateMrm server
N10 si_envelope_monitor RSS · accel · jerk · curvature
$v_{\max}$
MRM Trajectory ≤ 50 pts
steer hold · rate limit
envelope limits
Control (island) — the Autoware control stack, relocated whole
I9 trajectory_follower_node
I10 mpc_lateral_controller
I11 pid_longitudinal_controller
I12 trajectory_follower_base
H5 trajectory
control_cmd
I8 control_validator
I14 shift_decider
I13 pure_pursuit fallback tier, ASIL-B(D)
I2 command_mode_decider
command mode
I3 command_mode_switcher
OperationModeAvailability
I7 mrm_handler
OperateMrm
MRM request
I6 mrm_emergency_stop_operator
I5 stop_mode_operator
nominal control_cmd
fallback control_cmd
gear command
emergency stop
stop · gear P
OperateMrm · MrmBehaviorStatus
every island fault source onto one bus — /diagnostics ERROR from I19 and I20, sensor faults from N17, validator rejects from I8
I1 control_command_gate — ranks the command sources, filters, times out, transitions smoothly. Its built-in stop has no input at all: that is the point of it (§10.4).
source selection
achieved steer · brake · torque
external command (HPC) †
measured motion · standstill (§3.1 B4)
I15 raw_vehicle_cmd_converter
gear · turn · hazard
VehicleInterface (island)
I16 vehicle interface — pacmod_interface or project equivalent
CAN FD / FlexRay · 100 Hz · SI-E2E/B
Actuator gateway — brake · steer · propulsion · lamps CAN FD / FlexRay · 100 Hz · SI-E2E/B (§8.3)
forward camera · radar tracks · ultrasonic — read through the same ingest as N14, N6, N15
System (island)
N2 si_hpc_supervisor liveness · deadlines
liveness verdict
island mode
N3 si_state_machine single owner of island mode
/si/out/state · constraints · MRM state — 9 signals out (§8.2)
fault verdict
N8 si_fault_manager static fault tree · DTC
DTCs
actuator fault
N9 si_actuator_supervisor
sync · offset
N12 si_time_quality_monitor
time base valid / invalid
Safety link (island)
N1 si_safety_gateway — verifies Data ID, sequence, CRC, length and age on every HPC message before any island node sees it. A failure is a safety event, never an exception (§8.4)
N11 si_event_recorder UDS / DTC export
audit (§5.3 step 4)
Diagnostic port UDS tester (ISO 14229) HPC log store
hardware fault · ≤ 1 ms
request · grant · island status
take-over · release
RCC and local control — §6.5
N23 si_local_control cable · dead-man · v_{local}
local command (B6) — outranks remote
joystick · dead-man
Joystick — operator in sight wired pendant on the local port
N21 si_rcc_link session · latency · stream age · E2E
remote command (B5)
commands in · stream, status out
RCC — remote command and control centre cellular LTE / 5G, the island's own modem
ego state
encoded stream
N22 si_teleop_streamer camera · tracks · clearances · ego
NEW
MIGRATED
SHARED
B1
B2
B3
B4
B5
B6

What to look for in this phase

A reduced Autoware, not a monitor. Localization is *one node* — N4, dead reckoning. Planning is four bounded, map-free nodes. Perception is a fusion point over five ingest nodes. And Control is not a reimplementation at all: every red-bordered box is an Autoware package, migrated whole, with the same parameters and the same validation history (§4.1).

One door. Everything the HPC sends enters at N1 and is checked — Data ID, sequence, CRC, length and *age* — before any island node sees it. Every one of the eleven inbound signals has an island-local substitute (§8.1). That is why the island keeps driving when the HPC stops: the link carries *quality*, never *capability*.

A floor that cannot be undercut. B4, the blind emergency stop, runs on nine nodes — N4, N3, N8, N17, I7, I6, I1, I16 and I5 — and **not one of them consumes an exteroceptive topic**. The stop therefore cannot fail for the very reason that triggers it.

Stacks — the colours of `ArchitectureMain.pdf`

Sensing · Localization · Perception · Planning · Control
System · VehicleInterface · Vehicle · Safety link

Where each node came from

MIGRATED — the same Autoware node, migrated
NEW — new — does not exist in Autoware

18 migrated + 19 new = **37**, plus the optional N20 — and N21–N23 of §6.5, outside the counts until their detailed design is done. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. † the HPC's external command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table; B5's remote command arrives through N21 instead.

Behaviours each node enables (§3.3)

one fixed slot per behaviour B1 B2 B3 B4 B5 B6 — **B1** degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED — **SHARED** — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

B2 B3 — **Hollow tag** — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

⊘ — **Used by no behaviour** — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

→ HPC-sourced, verified by N1	→ actuator feedback
→ island sensor data	→ state · mode · decision
→ control command into the gate	⇢ optional (§3.5)
⇢ fault · `/diagnostics`	

Every edge on this page, with the design-document section it is taken from, is catalogued in `../messageflow/`, which draws the same 37 nodes coloured by execution partition instead.

Figure 12. *Phase 5 of 5 — the Safety Island alone, every node: all 37 island nodes, the 18 migrated Autoware I-nodes of S4.1 and the 19 new N-nodes of S6, plus the optional N20. The HPC is off the page; it reaches this diagram only through N1.*

All five are built from `layers/` (`make` there, or `latexmk -pdf` per file); `layers/L4SafetyIsland-Layers.pdf` collects them onto uniform landscape pages for presentation, and `layers/L4SafetyIsland-Layers.md` records the drawing conventions.

4.1 Migration list

Package names are as of `autoware_universe` `main`, August 2026.

#	Package	Function on the island	Tier	Notes / migration effort
I1	`autoware_control_command_gate`	**The arbiter.** Selects among command sources, applies nominal filtering, generates the built-in stop when the selected source times out	T0	The single most important migration. Its built-in stop is the last-resort MRM.
I2	`autoware_command_mode_decider` (+ `_plugins`)	Decides the command mode: which source has authority (autonomous / MRM / stop)	T0	Plugin logic must be replaced with a static, table-driven decider for certifiability.
I3	`autoware_command_mode_switcher` (+ `_plugins`)	Applies the decided mode; drives transitions	T0	Same static-configuration requirement.
I4	`autoware_command_mode_types`	Shared type/constant library	—	Header-only; shared with HPC.
I5	`autoware_stop_mode_operator`	Publishes stop commands; optional auto-park at standstill	T0	Explicitly **does not depend on localization** — reads vehicle status telemetry only. Ideal island resident.
I6	`autoware_mrm_emergency_stop_operator`	Emergency-stop MRM: publishes control commands directly	T0	Independent of planning; survives HPC loss.
I7	`autoware_mrm_handler`	Selects the MRM behaviour from the availability state	T0	Must be island-resident (R1/R3): its *reason to act* is HPC loss. Availability input is re-sourced — see §6, N2.
I8	`autoware_control_validator`	Checks the control command against the trajectory and vehicle limits	T1	Small, deterministic, no map dependency.
I9	`autoware_trajectory_follower_node`	Host node for the two controller plugins	T2	**Already ported and closed-loop-validated on nano-ros.**
I10	`autoware_mpc_lateral_controller`	Nominal lateral control	T2	Already ported. OSQP iteration cap + time budget required (§4.2).
I11	`autoware_pid_longitudinal_controller`	Nominal longitudinal control	T2	Already ported.
I12	`autoware_trajectory_follower_base`	Controller plugin base / utilities	T2	Already ported.
I13	`autoware_pure_pursuit`	**Fallback lateral control** — deterministic, closed-form, bounded time	T1	*New to the island.* This is the T1 controller that replaces MPC when the MPC misses its budget.
I14	`autoware_shift_decider`	Derives the gear command	T2	Small; keeps gear in the island's command set.
I15	`autoware_raw_vehicle_cmd_converter`	Converts generic accel/steer to vehicle-specific pedal/torque	T2	Must be **downstream of the gate**, therefore island-side.

#	Package	Function on the island	Tier	Notes / migration effort
I16	*vehicle interface* (`pacmod_interface` or project equivalent)	CAN FD / FlexRay TX/RX to the actuator gateway	T0	Vehicle-specific; not in `autoware_universe`. E2E-protected.
I19	`autoware_autonomous_emergency_braking`	**Island AEB.** Collision check against the ego predicted path; raises an ERROR diagnostic that escalates to emergency stop	T1s	*Moved from the HPC by D7.* Runs on the island's own point cloud — radar tracks rendered as `PointCloud2` by `N6`, merged with the 2D-lidar scan where one is fitted — with `use_pointcloud_data=true`, `use_predicted_object_data=false`, `use_imu_path=true`. May additionally consume `use_predicted_trajectory` — the MPC predicted path is on the island (I10). See §4.4.
I20	`autoware_collision_detector`	**Near-field guard.** Geometric collision check against the current ego footprint within ~5 m	T1s	*Moved from the HPC by D7.* Runs point-cloud-only with `use_dynamic_object=false`. Fed by ultrasonic + 2D lidar — ultrasonic ranked third under D8, the lidar optional, so I20 is deployed from rung C up (§9.4) and on any island that carries a lidar. Governs standstill, creep and low-speed MRM. See §4.4.
I17	`autoware_operation_mode_transition_manager`	Engage/disengage transition management	T0	Required only on the **legacy** path (§4.3). Superseded by I2/I3.
I18	`autoware_vehicle_cmd_gate`	Legacy gate	T0	**Alternative to I1** on the legacy path only. Do not deploy both.

Count: 18 nodes on the target path (I1–I16, I19–I20), or 17 on the legacy path (I17+I18 replacing I1–I3). This is a realistic figure for a Cortex-R52-class MCU when grouped as in §9.

4.2 Why a two-tier controller

D2 puts "the control module" on the island. Taken literally, that places the full MPC — with an iterative QP solver — inside the safety-critical ECU, and the certification argument then has to cover OSQP's worst-case execution time. That argument is expensive and fragile.

The recommendation is therefore to migrate the control module as **two tiers under one gate**:

	Nominal tier (T2)	Fallback tier (T1)
Lateral	`autoware_mpc_lateral_controller`	`autoware_pure_pursuit`
Longitudinal	`autoware_pid_longitudinal_controller`	Fixed deceleration profile (`si_mrm_profile`, §6 N7)
Integrity	QM, monitored	ASIL-B(D)
Time budget	Soft — may be dropped	Hard — always meets deadline
Trigger to switch	MPC budget overrun, validator reject, solver non-convergence, tracking error above bound	—

Switching is a T0 decision made by the gate (I1) via the decider (I2), inside one control period. The behaviour is already demonstrated in the existing port: the MPC commands an emergency stop on excessive tracking error, so the escalation path is live today — this proposal formalises it and adds a graceful intermediate step instead of going straight to a stop.

This satisfies R8 and directly answers open question 3 in `SaftyIsland_ReferenceDesign.md` ("real-time ROS 2 nodes on Cortex-R52 for MRM trajectory selection"): the MRM tier is closed-form and bounded; only the comfort tier is iterative.

4.3 Target architecture vs. legacy architecture

Autoware is mid-transition. Two consistent sets exist; the island must pick one.

	Target (recommended)	**Legacy**
Gate	`autoware_control_command_gate`	`autoware_vehicle_cmd_gate`
Mode decision	`autoware_command_mode_decider`	`autoware_operation_mode_transition_manager`
Mode application	`autoware_command_mode_switcher`	(in gate)
Stop source	`autoware_stop_mode_operator`	(in gate)

The target set is recommended because it separates *decide* / *switch* / *gate* into three nodes, and that separation is exactly the decomposition a safety argument needs — the decider can be a static table, the switcher a state machine, the gate a filter, each independently reviewable. The legacy `vehicle_cmd_gate` fuses all three and is correspondingly harder to certify.

4.4 Why AEB and the collision detector can now migrate

In the previous revision both stayed on the HPC because they were assumed to need perception object lists. Checking their actual interfaces shows otherwise, and D7 removes the remaining obstacle.

	`autoware_autonomous_emergency_braking`	**`autoware_collision_detector`**
Perception objects	**Optional** — `use_predicted_object_data=false`	**Optional** — `use_dynamic_object=false`
Point cloud	`use_pointcloud_data=true`	`use_pointcloud=true`
Ego path source	IMU-based kinematic prediction (velocity + yaw rate), *or* the MPC predicted trajectory	none — current footprint only
Map dependency	none	none
Output	ERROR on `/diagnostics`	ERROR on `/diagnostics`
Island sensor that feeds it	radar tracks → `PointCloud2` (`N6`); 2D lidar scan where fitted	2D lidar + ultrasonic → `PointCloud2` — ultrasonic ranked third, the lidar optional, so a node deployed from rung C up

Both are therefore **island-viable without any HPC input at all**: AEB's IMU path comes from `N4`, its point cloud from `N6`'s radar tracks (and `N13` where a 2D lidar is fitted), and — usefully — its optional MPC path from `I10`, which is already on the island. Neither node needs the planner, the map, or the perception stack.

This is the concrete payoff of D7. Migrating them means the island's emergency braking is not a bespoke reimplementation but **the same Autoware node, with the same tuning parameters and the same validation history**, running on a diverse sensor channel. `autoware_spheric_collision_detector` stays on the HPC — it is object-list based.

Their `/diagnostics` ERROR is consumed by `N8` (fault manager) and escalated through `N3` to the gate, exactly like any other island fault source. Neither node commands the actuators directly.

5 Autoware nodes that remain on the HPC

5.1 Whole stacks — HPC only

`sensing`, `perception`, `localization`, `map`, `planning` execute entirely on the HPC. Every node in `AutowareSW.md` §1–§5 stays: `ndt_scan_matcher`, `ekf_localizer`, `bevfusion`, `multi_object_tracker`, `lanelet2_map_loader`, `behavior_path_planner`, `behavior_velocity_planner`, `mission_planner`, `freespace_planner`, and the rest. They fail R4 (unbounded memory, learning-based, map-sized state) and R5 (their inputs and outputs would flood the link).

Consequence that must be handled, not assumed away: the island's controllers currently subscribe to `/localization/kinematic_state` and `/localization/acceleration`, both HPC-produced. Under R2 the island cannot depend on them after HPC loss. This is resolved by N4 in §6.

5.2 Control and system packages that stay

Package	Decision	Reason
`autoware_spheric_collision_detector`	**HPC**	Object-list based. Its point-cloud siblings moved to the island — see §4.4.
`autoware_obstacle_collision_checker`, `autoware_predicted_path_checker`	**HPC**	Depend on predicted paths and map.
`autoware_lane_departure_checker`	**HPC**	Requires the Lanelet2 vector map (R4). Island equivalent is the corridor monitor N5 — now dual-sourced from H3's map corridor *and* the island's own camera lane detection (N14).
`autoware_external_cmd_selector`, `autoware_joy_controller`, `autoware_external_cmd_converter`	**HPC**	Remote/manual sources. They reach the island as one E2E-protected command source into the gate, not as island nodes. Under D9 the remote and local paths of B5 and B6 terminate on the island instead, so `autoware_external_cmd_selector` — whose *local* versus *remote* choice is exactly B6 versus B5 — and `autoware_external_cmd_converter` become candidates to migrate. The allocation is settled with the detailed design (open decisions 19, 20); until then they stay here.

Package	**Decision**	**Reason**
`autoware_control_performance_analysis`, `autoware_smart_mpc_trajectory_follower`	**HPC**	Development and analysis only; not in the operational path.
`autoware_mrm_comfortable_stop_operator`	**HPC**	It publishes an `autoware_internal_planning_msgs/VelocityLimit` to the **planner** — it cannot function without planning. It is therefore an *HPC-dependent MRM*, valid only while the HPC is degraded-but-alive. See §5.3.
`autoware_diagnostic_graph_aggregator`, `autoware_diagnostic_graph_utils`, `autoware_hazard_status_converter`	**HPC**	Large, config-driven fault trees over HPC-internal diagnostics. Island runs its own static fault manager (N8).
`autoware_system_monitor`, `autoware_component_monitor`, `autoware_component_state_monitor`, `autoware_topic_state_monitor`, `autoware_duplicated_node_checker`, `autoware_processing_time_checker`, `autoware_pipeline_latency_monitor`, `autoware_bluetooth_monitor`, `autoware_velodyne_monitor`	**HPC**	They monitor HPC resources. Under R3 they are *evidence sources*, not the safety monitor — their verdict is condensed by H2 and shipped to the island, and the island never relies on them being alive.
`autoware_default_adapi_universe`, `autoware_default_adapi_helpers`, `autoware_topic_relay_controller`	**HPC**	API and orchestration surface.
`autoware_accel_brake_map_calibrator`, `autoware_steer_offset_estimator`	**HPC**	Offline/estimation tooling; produce parameters consumed by I15.
`autoware_dummy_diag_publisher`, `autoware_dummy_infrastructure`	**HPC**	Test only.

5.3 The two-class MRM split

This is a design consequence worth stating explicitly, because it changes the graded-reaction ladder in `SaftyIsland_ReferenceDesign.md`:

Graded reaction	**Implemented by**	**Requires HPC alive?**
1. Notify & request self-recovery	Island → HPC constraint/veto message; HPC re-plans	**Yes**
1b. **Demand an ODD switch**	Island declares the current domain unsatisfiable; the HPC selects one it can meet and declares it through `H4`; the island verifies the declaration and then the behaviour (§6.4)	**Yes**
2. Restrict speed / enlarge margins	`autoware_mrm_comfortable_stop_operator` (HPC) + island envelope limits — held underneath step 1b for its duration	**Yes**
3a$_0$. Take authority — **lane-keep hold** (B2)	Island: fallback tier + `I13 pure_pursuit` on the camera's lane lines, decelerating first to the ODD's safe speed `v_odd` and then along the `v_hold` ramp to `v_entry` for at most `T_hold` (8–15 s) while awaiting HPC recovery ; exits to 3a via B3, or to 3b (B4) if the camera is lost	No
3a. Take authority — **pull-over** (B3), degrading to the **obstacle-aware** in-lane stop	Island: `N18` on the camera's road-edge and pull-over-space evidence, or fallback tier + `si_mrm_profile`, decelerating along the camera-perceived corridor to a free-space target	No
3b′. Emergency braking on imminent collision	Island: `autoware_autonomous_emergency_braking` (I19) on island sensors	No
3b. Take authority — **emergency stop** (B4)	Island: `autoware_mrm_emergency_stop_operator` (I6), steering surrendered, no world model required	No
3c. Last resort	Island: gate built-in stop (`autoware_control_command_gate`)	No
4. Audit	Island `si_event_recorder` + HPC log store	No

Steps 1 to 2 are cooperative degradation; $3a_0$–3c are island-autonomous. **The island must never depend on step 2 succeeding before it can reach step 3.** Step $3a_0$ is the only rung that can be *left upward*: if the HPC recovers and is confirmed healthy during the hold, authority returns to it and the ladder resets to step 1. Every rung below $3a_0$ is terminal for the drive cycle.

Step 3b is B4, and it is not only reached by falling through 3a. Every rung from 1 to 3a has a direct edge into it, because the condition that declares it — the island can no longer see — is independent of which rung the vehicle happens to be on (§3.2). Since D8, *can no longer see* means *has lost the camera.* Reading the table as a strict descent is the one misreading that matters: it would let an implementation guard 3b behind 3a, and a guarded floor is not a floor.

That ladder is for *HPC failure*. §6.4 defines a second, slower ladder for *ODD exit*, where the HPC is healthy and the correct first move is to constrain it rather than replace it.

6 New nodes to add on the Safety Island

These do not exist in Autoware today. They are what turns a ported trajectory follower into a safety island.

6.1 Supervision, state and fallback

#	Node	Purpose	Tier	Rate
N1	`si_safety_gateway`	In-house E2E ingress/egress (§8.4). Verifies Data ID, sequence counter, CRC, length and age on every message from the HPC before it reaches any island node; publishes island state outward. Rejects become **safety events**, not transport errors.	T0	per-message
N2	`si_hpc_supervisor`	HPC liveness and deadline supervision: heartbeat timeout, per-topic minimum rate, end-to-end latency against the declared budget. **Produces the island-local `OperationModeAvailability` that feeds I7** — this is what makes `mrm_handler` work when the HPC is silent. Also owns the **recovery verdict**: during a B2 hold it requalifies a returning HPC over the declared confirmation window — heartbeat restored *and* trajectories valid, fresh and in-envelope — and only then permits hand-back.	T0	100 Hz

#	Node	Purpose	Tier	Rate
N3	`si_state_machine`	The island safety state machine: `BOOT` → `SELF_TEST` → `STANDBY` → `ADS_ACTIVE` → `{DEGRADED, LANE_KEEP_HOLD, MRM_PREPARE, MRM_EXECUTE, MRM_COMPLETE, EMERGENCY_STOP}` → `SAFE_STOP`. Single owner of island mode. `LANE_KEEP_HOLD` is B2 (§3.1): N3 owns its hold budget `T_hold`, is the only node that may authorise the hand-back edge `LANE_KEEP_HOLD` → `ADS_ACTIVE` on N2's recovery verdict, and latches the ODD class for `v_odd` at entry. When the budget expires it forks: to `MRM_PREPARE` (B3) if `N17` holds the camera valid and the lane lines are seen, to `EMERGENCY_STOP` (B4) if not. `EMERGENCY_STOP` is B4: it is entered from **every** other state including `LANE_KEEP_HOLD` and `MRM_EXECUTE`, it is the one transition with no guard and no precondition, and its only successor is `SAFE_STOP`. Every other transition remains one-way.	T0	100 Hz
N4	`si_vehicle_state_estimator`	**Island-local odometry.** Dead-reckons pose, twist and acceleration from wheel speed, steering angle and the chassis ESC yaw rate and acceleration; the island-attached IMU and GNSS (D8 rank 4), where fitted, tighten the estimate and give it a channel independent of the chassis bus. Publishes island-domain `Odometry` and `AccelWithCovarianceStamped`. Resolves the R2 orphan-input problem for I9–I13.	T0	100 Hz
N5	`si_corridor_monitor`	Lateral containment. **Dual-sourced:** while the HPC drives, prefers the compact map-derived corridor from H3 while it is fresh and cross-checks it against the camera's lane boundaries from N14; once the island drives, the camera is the required source and H3 only refines it. There is no heading-tube fallback: with no camera corridor, N5 declares the corridor lost, which is a B4 trigger. Under B3 it is re-armed against the road edge N14 observes. Island-side equivalent of `lane_departure_checker`.	T1	50 Hz
N6	`si_radar_guard`	Radar track ingest: range, range-rate, closing speed, conservative stopping-distance check. Weather-robust longitudinal channel; see §6.2.	T1s	20–50 Hz
N7	`si_mrm_profile`	Generates the MRM motion profile when planning is unavailable: deceleration to the ODD's safe speed `v_odd` and then to the B2 hold speed `v_hold` or to standstill, target standstill point, hazard-lamp and gear sequencing — the lamps are raised when the manoeuvre is commanded, not when it completes (§3.1). **Obstacle-aware** — shortens the ramp against N16's free-space distance and nearest-obstacle range, and reverts to a fixed conservative ramp when N16 is stale (§2.2 staleness rule). Feeds the fallback tier.	T1	50 Hz
N8	`si_fault_manager`	Static fault tree + DTC store + fault reaction table. Aggregates N2, N5, N17, I8, the I19/I20 `/diagnostics` ERRORs, actuator faults, and the FSI hardware fault line into a single fault verdict for N3. Holds the distinct fault codes for **sensing blackout** (B4) — the camera lost — and for a search that found no refuge (in-lane stop under B2's control law, from B3), so the two are separable in the DTC store and in the HARA. A blackout record also names which lower-ranked sensors were still alive.	T0	100 Hz
N9	`si_actuator_supervisor`	Actuator-feedback plausibility: commanded vs. achieved steering/brake/torque, response-time check, brake-pressure and EPS availability. Confirms MRM is actually being executed; non-response is a direct B4 trigger, and non-response *during* B4 escalates below the node path to the gate's built-in stop and the §8.5 fault line.	T1	100 Hz
N10	`si_envelope_monitor`	Kinematic/RSS safety envelope: bounds on acceleration, jerk, curvature, curvature rate, yaw rate, and **speed against N16's stopping-distance result** — the island can now enforce a true RSS longitudinal envelope rather than a kinematic-only one. Emits the constraint/veto vector.	T1	50 Hz

#	Node	Purpose	Tier	Rate
N11	`si_event_recorder`	Protected ring buffer of the pre/post-event window; UDS/DTC export (ISO 14229). Non-real-time, physically separated from the control path.	T3	1–10 Hz
N12	`si_time_quality_monitor`	gPTP sync state and offset; declares message-freshness checking valid or invalid. A lost time base invalidates every age check in N1.	T1	1 Hz

Optional / phase 2: `si_odd_monitor` (island-side geofence and ODD validity from a dedicated GNSS receiver). Merged into N10 in phase 1, because the HPC's ODD verdict (H4) is sufficient while the HPC is alive, and after HPC loss the island's response is B2 and then a bounded MRM regardless of ODD — the island's own ODD sets `v_hold` and the admissible hold duration, not whether the response happens. The road class that sets B2's `v_odd` is therefore taken from H4's last declaration, not measured; GNSS, which D8 ranks fourth and does not require, may only corroborate it downward.

6.2 Island safety sensing

D7 attaches a low-resolution, high-robustness sensor set directly to the island. The selection principle is **robustness and boundedness over resolution**: every sensor must degrade gracefully, emit a small fixed-size payload, and be processable without a neural network.

Not every island carries all of it. D8 ranks the set, and the ranking is what the behaviours are written against: a behaviour may *require* only the camera, and every sensor below it buys speed or ODD width, never the behaviour itself.

Rank	Sensor	Physical link to island	Data class	Rate	What it gives the MRM
1 — mandatory	**Forward camera** (low resolution, mono)	GMSL or 100BASE-T1, direct	Lane boundaries + road edge / kerb + drivable lateral space	10–20 Hz	Lane geometry for lateral containment under B2; lane lines and the pull-over space for B3; the contrast and lane-continuity evidence of §6.4. **The sensor whose loss is B4. See open decision 8** — classical CV vs. a certified smart camera emitting lane/object lists.
2	**Radar** — forward, and rear-corner on the pull-over side	CAN FD (native track output)	Object tracks — range, range-rate, azimuth	20 Hz	Longitudinal obstacle range and closing speed — AEB's point cloud when no lidar is fitted. Works in rain, fog, darkness, low sun. Rear-corner: closing traffic on the pull-over side. Lifts B2 to the highway `v_odd` and B3 above ~30 km/h.

Rank	Sensor	Physical link to island	Data class	Rate	What it gives the MRM
3	**Ultrasonic array**	LIN / DSI3 via a sensor interface, or CAN FD	Echo distance per sector, near field, up to about 5 m	10–20 Hz	Near-field clearance for the final metres of B3 and a tight-to-kerb end; creep and standstill release; `I20`'s near-field footprint through `N15`. Blind beyond a few metres: it buys a terminal condition, not speed.
4	**Island IMU / GNSS**	direct / CAN FD	Vehicle dynamics ; position and road-class corroboration	100 Hz / 10 Hz	A dynamics channel independent of the chassis bus for N4; the friction estimate of §6.4; GNSS corroborates the ODD class downward.
5 — optional	**2D lidar** (single-plane scanner)	100BASE-T1 or RS-422, direct	Planar scan, ~1080 points	10–40 Hz	Free-space distance ahead, in-path occupancy, lateral clearance — as a *confirming* channel: it may disqualify a space or add a hazard, never qualify a refuge or supply a corridor. Deterministic, classical processing, no learning.
— *(vehicle)*	**Chassis signals** — wheel speed, steering report, ESC yaw rate and acceleration	vehicle bus, via `I16`	Vehicle dynamics	100 Hz	Not an island sensor and not ranked: present on every vehicle. The floor N4 — and therefore B4 — stands on.

Two wiring rules follow from R7 and from RFC design principle 1:

1. **Independent harness and power.** Island sensors must not share a power rail, a harness bundle, or a switch with the HPC's sensor set. A common-cause failure that blinds the HPC must not blind the island. This is a hardware requirement that belongs in the Part-2 device design, but it is decided here.
2. **One-way data flow.** Sensors feed the island. The island *may* forward its safety world state to the HPC for cross-validation (§8.2), but no island sensing node may ever *depend* on an HPC-produced topic. §3.5's object digest is admissible under exactly that wording — it is *consumed*, never depended on — and `N20` below is the only node in this section with an HPC-side input.

The processing chain adds five nodes — two of which, `N15` and `N13`, exist only on an island that carries their sensor, the ranked-third ultrasonic or the optional lidar — and one optional sixth:

#	Node	Purpose	Tier	Rate
N13	`si_scan_guard` *(optional — only where a 2D lidar is fitted)*	2D lidar ingest. Fixed-size planar occupancy over the ego corridor; extracts free-space distance ahead and nearest in-path return. Publishes a bounded `PointCloud2` for I19/I20. Classical, no learning, static memory. Contributes to N16 as a confirming channel only.	T1s	10–40 Hz

#	Node	Purpose	Tier	Rate
N14	`si_vision_guard`	Forward-camera ingest — **the one mandatory sensing node**. Extracts left/right lane boundary polylines and a free-space edge. Also extracts the **road edge or kerb** beyond the outer lane line and the **drivable lateral space** between them — the evidence N18 searches for an island-detected refuge under B3. Feeds N5 as the required corridor source once the island drives. Either classical IPM + line fitting, or pass-through of a certified smart camera's lane/object output.	T1s	10–20 Hz
N15	`si_proximity_guard` *(rank 3 — where the ultrasonic array is fitted)*	Ultrasonic ingest. Near-field clearance per sector; gates creep, standstill release and the final metres of a pull-over. Feeds I20. Like every source below the camera, it may narrow N16's free space, never create it.	T1s	10–20 Hz
N16	`si_safety_world_model`	**The fusion point.** Combines N6, N13, N14, N15 into the single bounded safety world state defined in §2.2 — nearest in-path obstacle, range, range-rate, TTC, free-space distance, perceived corridor, validity per source. Conservative fusion: a hazard reported by *any* source is taken as present; a corridor is trusted only when sources agree — and only when the camera is one of them: corridor and lateral free space are anchored on N14, and the lower-ranked sources may narrow them but never create them. May additionally take `N20`'s optional digest as a **conservative-only** term — it can add a hazard or shorten a distance, never remove one and never make a corridor valid. Publishes the **no-valid-source** condition explicitly rather than an empty world state, so that a blackout is distinguishable from an empty road — the distinction B4 turns on.	T1s	20–50 Hz
N17	`si_sensor_health`	Per-sensor availability, frame-rate, blockage and degradation detection — dirt, spray, fog, low sun, radar interference. Sets each source's validity flag for N16 and drives the island's own ODD verdict. This is deck p.19's "monitoring sensor status, frame rates, weather conditions", now for island-attached sensors. **Declares sensing blackout** — the camera invalid, whatever the lower-ranked sources report — which is the trigger for B4; the declaration is debounced, and the debounce is open decision 15. **§3.5's object digest is deliberately not an input to this determination:** an island that cannot see is blind whether or not the HPC is still publishing objects, and letting a live HPC mask a blackout would put a guard under B4.	T1s	5–10 Hz
N20	`si_object_digest` *(optional — §3.5)*	Bounded ingest of the HPC's `/perception/object_recognition/objects`, admitted through `N1` like every other cross-domain message. Truncates to a fixed maximum object count ordered by threat, a fixed prediction horizon and a fixed field set; stamps age and distance travelled since receipt using `N4`; plausibility-checks against `N16` where the fields of view overlap; marks the digest invalid once either carry-over bound is exceeded. Publishes `SafetyObjectDigest` on `/si/obj/digest`. **Contributes a conservative term only, and never a validity flag.** Bounds and budgets are open decision 16.	T1s	10 Hz

What this buys, stated precisely. Without D7 the island could execute a *blind* MRM: decelerate on a fixed ramp along a remembered trajectory and hope the road ahead is clear. With D7 it can execute an **obstacle-aware** MRM — brake harder for a detected obstacle, stop short of it, hold at standstill only when the near field is clear, and keep lane using perceived boundaries after the map corridor has expired. The graded reaction in §5.3 becomes genuinely autonomous rather than open-loop.

The island's own ODD. The sensor set is deliberately weaker than the HPC's, so the island's fallback capability has its own, narrower ODD — forward-looking only, bounded speed, no complex merging or intersection negotiation. It also depends on *which* of the ranked sensors are fitted: a camera-only island (rung A, §9.4) claims a city-road fallback at ~30 km/h, and it is radar that extends the claim to the highway class. `N17` and `N10` enforce it. Part 4 must state that ODD explicitly: it is the domain over which the MRM is claimed to work, and it is not the vehicle's ODD.

6.3 Pull-over MRM after HPC loss

A pull-over is a materially harder manoeuvre than an in-lane stop, and the difference is not the control law — it is *knowing where to go*. Three capabilities are new: a lateral target that is legal and physically free, a bounded lateral displacement path, and awareness of the traffic space the vehicle is about to cross.

The island cannot decide *where* to pull over. Choosing a legal stopping place means knowing about intersections, crosswalks, bus stops, bridges, tunnels and local traffic law — that is map and planner work, and it stays on the HPC. So the split is:

> **The HPC continuously nominates refuges. The island verifies and executes one.**

That split holds wherever the HPC is there to nominate, and under B3 it often is not: B3 is reached from B2 exactly when the HPC has been lost, and the buffered refuge set ages out like every other buffered input (§8.1). The island therefore carries a **second, weaker refuge source of its own**, used only when the first offers nothing — the full argument is in §3.1. The split above is unchanged in the part that matters: **the island may now *find* a refuge, but it still may not *choose* one.** It takes the first lateral clearance that passes verification, and it compares no candidates for legality, because it has no map and no traffic law to compare them with.

The refuge contract

New HPC node `H8` (§7) streams a rolling **pull-over refuge** that it has already validated against the map and traffic law. While one is on offer, the island only accepts, shifts within, or rejects it — it does not invent an alternative. It falls back on its own detection only when the contract below cannot be honoured at all, and what it then produces is an `EMERGENCY_ONLY` clearance, never a `LEGAL_STOP`.

Field	**Meaning**
`side`	LEFT / RIGHT — which side the refuge is on
`entry_s`, `exit_s`	Longitudinal window along the buffered trajectory, as arc length
`lateral_offset_m`	Target lateral displacement from lane centre
`refuge_width_m`	Usable width, for clearance checking
`surface_class`	PAVED / GRAVEL / UNKNOWN
`legality`	LEGAL_STOP / EMERGENCY_ONLY / PROHIBITED

Field	**Meaning**
`validity_distance_m`	How far ahead this refuge remains valid
`fallback_behavior`	What to do if the refuge proves unusable — normally IN_LANE_STOP

The contract's one hard guarantee: at any moment, the buffered refuge set must cover at least the vehicle's current worst-case stopping distance plus the island's fault-reaction budget. When the HPC cannot honour that — a tunnel, a bridge, a barriered section, roadworks — it must say so explicitly rather than fall silent.

That guarantee has a consequence worth stating plainly, because it links this feature to the next one:

> **The island always knows, *before* any failure, whether a pull-over is currently available.** `N18` maintains a continuous `pull_over_available` flag. Fallback capability is therefore an observable, and §6.4 treats it as an ODD dimension.

Two nodes carry it

#	**Node**	**Purpose**	**Tier**	**Rate**
N18	`si_pull_over_manager`	Implements Autoware's **existing `pull_over_manager`** MRM slot — `OperateMrm` service server, `MrmBehaviorStatus` publisher — so `I7 mrm_handler` selects it natively with no new interface. Owns refuge go/no-go against `N16`'s live free space, longitudinal shifting within the window, the **`pull_over_available`** flag, and the abort ladder below. Also owns the **island-detected refuge**: when `H8` offers none, it searches `N16`'s free space for a lateral clearance wide enough and long enough to hold the vehicle, found by the camera — `N14`'s lane lines, road edge and the drivable space between them — and confirmed unoccupied by `N6` radar; `N15`, where fitted, guards the final metres and permits a tight-to-kerb end, and `N13`, where fitted, may only disqualify it, and emits it as a refuge with `legality = EMERGENCY_ONLY` and `surface_class = UNKNOWN`. The search is bounded by `T_refuge`, and its three outcomes are distinct: **found**, **positively none**, and **inconclusive** — only the last goes to B4 (§3.1). `pull_over_available` is true if *either* source currently offers a refuge, and carries which one.	T1	20 Hz
N7	`si_mrm_planner` *(extended)*	Previously emitted a longitudinal ramp only. Now emits a bounded `autoware_planning_msgs/Trajectory` ($\leq$ 50 points) carrying **both** the lateral displacement path and its velocity profile, for in-lane stop or pull-over alike. Hard bounds on lateral acceleration, steering rate and jerk are applied here.	T1	50 Hz

Extending `N7` to emit a standard `Trajectory` rather than adding a separate path planner is the cheaper move: `I13 pure_pursuit` then consumes the fallback path through exactly the same interface it would use for a nominal one, unmodified, and the fallback tier looks identical to the nominal tier from the gate's point of view.

`mrm_handler` already defines `/system/mrm/pull_over_manager/{status,operate}`. **Nothing new is needed for MRM selection.** What *is* missing upstream is the reporting enum: `autoware_adapi_v1_msgs/MrmState` carries only `NONE`, `EMERGENCY_STOP` and `COMFORTABLE_STOP`, with no pull-over value. See open decision 10.

The abort ladder

Pull-over must be abandonable at every point into a strictly simpler manoeuvre. This is the property that makes it safe to attempt at all.

Trigger	Response
Refuge occupied or narrower than needed	Shift longitudinally within `entry_s`…`exit_s`
No usable window remains	Abandon pull-over → **in-lane stop**
Rear-corner radar detects a closing vehicle on the pull-over side	Hold lateral position, continue decelerating → **in-lane stop**
Lateral path bounds violated, or steering not tracking	Freeze lateral command → **in-lane stop**
`I19` AEB fires	**Emergency stop** (B4), abandon the manoeuvre
Actuator not responding (`N9`)	**Emergency stop** (B4)
Sensing blackout (`N17`) — camera lost, so the refuge can no longer be verified free and the lane lines are gone	**Emergency stop** (B4); the in-lane stop is *not* available either, because it too needs the camera's corridor
Refuge search concludes: neither source offers a usable refuge	Abandon pull-over → **in-lane stop** — the corridor is still there to use
`T_refuge` elapsed with the search still inconclusive	**Emergency stop** (B4). An inconclusive search is a capability statement, not a road statement
Standstill reached, near field not clear	Hold, do not release; hazard lamps stay on — they were raised at entry

Every rung degrades toward something the island can definitely do. The manoeuvre never escalates in complexity under fault.

Drawn as a ladder, the property the table asserts becomes visible: **each rung down removes a requirement, and never adds one.**

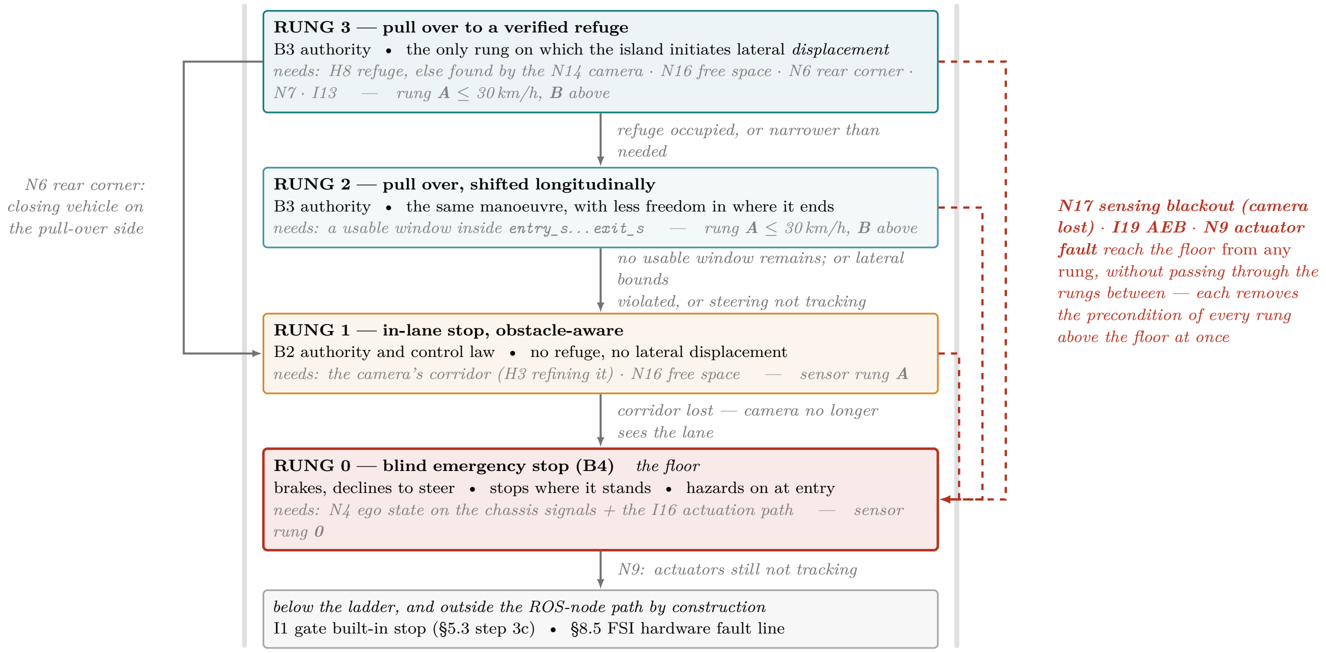

RUNG 3 — pull over to a verified refuge
B3 authority • the only rung on which the island initiates lateral displacement
needs: H8 refuge, else found by the N14 camera · N16 free space · N6 rear corner · N7 · I13 — rung A ≤ 30 km/h, B above
refuge occupied, or narrower than needed
RUNG 2 — pull over, shifted longitudinally
B3 authority • the same manoeuvre, with less freedom in where it ends
needs: a usable window inside entry_s…exit_s — rung A ≤ 30 km/h, B above
no usable window remains; or lateral bounds violated, or steering not tracking
N6 rear corner: closing vehicle on the pull-over side
RUNG 1 — in-lane stop, obstacle-aware
B2 authority and control law • no refuge, no lateral displacement
needs: the camera's corridor (H3 refining it) · N16 free space — sensor rung A
corridor lost — camera no longer sees the lane
RUNG 0 — blind emergency stop (B4) the floor
brakes, declines to steer • stops where it stands • hazards on at entry
needs: N4 ego state on the chassis signals + the I16 actuation path — sensor rung 0
N9: actuators still not tracking
below the ladder, and outside the ROS-node path by construction
I1 gate built-in stop (§5.3 step 3c) • §8.5 FSI hardware fault line
N17 sensing blackout (camera lost) · I19 AEB · N9 actuator fault reach the floor from any rung, without passing through the rungs between — each removes the precondition of every rung above the floor at once

Figure 13. *The pull-over abort ladder. Each rung down removes a requirement and never adds one — and the dashed edges reach the floor from any rung, without walking down.*

Three things are worth reading off the drawing rather than the table.

The solid edges only ever go down, and each one is a *loss*. Rung 3 → 2 gives up the chosen stopping place but keeps the manoeuvre; 2 → 1 gives up leaving the running lane; 1 → 0 gives up knowing what is ahead — which, since D8, is losing the camera. Nothing in the ladder moves upward, and no rung requires anything the rung above it did not already have. That is what makes attempting rung 3 acceptable: the cost of failing at it is bounded by what is underneath, and what is underneath is always strictly simpler.

The dashed edges are the ones that do not walk down the ladder. A sensing blackout, an `I19` AEB event or an `N9` actuator fault reaches rung 0 *from wherever the vehicle is*, without passing through the rungs in between — because each of them removes the precondition of every rung above the floor at once. An implementation that walks the ladder one rung at a time under those conditions is wrong, and §5.3 makes the same point about step 3b from the other direction.

The floor is shared with a different ladder. Rung 0 is B4, and it is also rung 0 of the *sensor* ladder in §9.4 — the configuration with no exteroceptive sensor at all — the configuration every island is effectively in once its camera is lost, whatever else it still carries. The two ladders measure different things, the manoeuvre and the sensor set, and they meet at exactly one place. That coincidence is not accidental: the floor of the manoeuvre ladder is defined as the manoeuvre that survives the floor of the sensor ladder.

6.4 ODD degradation on weather and visibility

This is a **different trigger class** from everything above, and the difference drives the design.

	HPC failure (§6.3)	**ODD exit — weather / visibility**
Standard	ISO 26262 — something malfunctioned	**ISO 21448 / SOTIF** — nothing malfunctioned
HPC state	Dead, hung, or lying	**Healthy and correct by its own lights**
Time constant	Milliseconds	Seconds to minutes
Correct first response	Seize authority	**Constrain, and let the HPC act**
Who should pull over	The island, because the HPC cannot	**The HPC, because it has the map**

The last row is the important inversion. When visibility collapses, the *best* outcome is that the HPC executes a planned, comfortable, legal pull-over while it still can. The island's job is to **demand that, then verify compliance within a deadline**, and to fall back to its own cruder manoeuvre only if the HPC does not act. Seizing authority first would replace a good pull-over with a worse one.

Why the island must detect this independently

An island that only reads the HPC's self-reported perception confidence cannot detect the case that matters: **the HPC is confidently wrong.** That is the central SOTIF failure mode, and no amount of health monitoring reaches it. The only thing that does is an independent sensing channel that degrades *differently*. This is the strongest argument for the D7 sensor set — stronger than the fallback argument.

Detection: concrete signals, no learning required

Signal	Sensor	D8 rank	Indicates
Contrast / MTF collapse; sky-to-scene luminance ratio; flare area; drop and blur detection	camera	1	Fog, glare, low sun, dirty or wet lens
Lane-detection continuity and confidence	camera	1	Markings obscured by snow, water film, wear
Radar-vs-camera detection divergence — radar reports targets the camera cannot confirm at the same range	radar + camera	2	**Visibility collapse, measured by physical diversity rather than self-report**
Longitudinal slip ratio vs. commanded acceleration	wheel speed + IMU (chassis ESC below rank 4)	4	**Low friction — ice, standing water**
Yaw-rate vs. steering-angle model residual	IMU + steering	4	Reduced lateral grip
Max return range collapse; intensity drop; dense near-range returns	2D lidar	5 — optional	Fog, heavy rain, spray, snow
Multi-echo / dual-return ratio rise	2D lidar	5 — optional	Airborne particulates
Radar-vs-lidar detection divergence	radar + lidar	5 — optional	Visibility collapse — a second, independent instance of the test above, where a lidar is fitted
Wiper state, exterior temperature, rain sensor	vehicle CAN	—	Cheap corroboration

The radar-versus-camera divergence test is the one to build first, because it uses the two highest-ranked sensors and is therefore available on every island from rung B up. It is a *physical* diversity argument: radar penetrates what optics do not, so a persistent disagreement between them at the same range **is** reduced visibility. Nothing about it depends on a model of the weather. Where a 2D lidar is fitted, radar-versus-lidar is the same argument a second time; it may tighten the verdict, and its absence never loosens it.

Friction matters as much as visibility and is usually forgotten. It bounds achievable deceleration, which sets stopping distance, which sets both the speed limit below and the refuge window in §6.3.

The node

#	Node	Purpose	Tier	Rate
N19	`si_odd_monitor`	Environmental ODD verdict: visibility range estimate, precipitation class, surface-friction estimate, ambient-light class, and the resulting speed ceiling and ODD margin. Applies asymmetric hysteresis. Promoted from the phase-2 option noted in §6.1. Also raises the **ODD switch demand** — `odd_satisfiable = false` with the failed dimension and the `T_odd` budget — and runs the two-part acceptance test on the answer: the domain `H4` declares must lie inside the measured envelope, *and* the vehicle's measured behaviour must converge on that domain's bounds within `T_odd`. It declares `T_odd` per situation rather than as a constant, from the rate at which the margin is falling (B1). It also supplies the `v_max` term of B2's `v_hold` ramp.	T1s	2–5 Hz

`N17` and `N19` answer different questions and must not be merged:

- **`N17 si_sensor_health` — "is my sensor working?"** A device question. Blockage, dirt, frame rate, internal fault. Its answer is a per-source validity flag, and its escalation path is the **fault manager**: a blinded sensor is a *fault*.
- **`N19 si_odd_monitor` — "is the world outside my operating envelope?"** An environment question. Its answer is an ODD verdict with a margin, and its escalation path is the **graded degradation ladder** below: reduced visibility is not a fault.

The speed law

The island's speed ceiling is closed-form, auditable, and derived from two measured quantities rather than a lookup table:

$$v_{\text{max}} \;=\; \sqrt{2\,a_{\text{brake}}\,(d_{\text{visible}} - d_{\text{margin}})}$$

d_{visible}	measured sensing range, minimum over the active modalities	(`N19`)
a_{brake}	achievable deceleration, bounded by the friction estimate	(`N19`)
d_{margin}	reaction and actuation margin at the current speed	

You may drive only as fast as you can stop within what you can see. Worked example: `d_visible` 30 m in fog, `a_brake` 3 m/s^2 on a wet surface, `d_margin` 5 m → `v_max` = 12.2 m/s ≈ 44 km/h. Both inputs are measured on the island, so the limit is defensible without trusting the HPC.

The graded degradation ladder

ODD margin	Island action	Mechanism	Deadline for HPC compliance
Nominal	none	—	—
Marginal	Notify. Publish the reduced verdict; the HPC re-plans conservatively	`OddVerdict` advisory + `SafetyEnvelopeCommand` mode `ALLOW`	—

ODD margin	Island action	Mechanism	Deadline for HPC compliance
Unsatisfiable	**Demand an ODD switch.** Declare the current domain unsatisfiable; require the HPC to move to one it can meet. The island names no target	`OddVerdict` with `odd_satisfiable = false`, the failed dimension, and the `T_odd` budget	`T_ack` = 2 s to declare a new domain or declare exhaustion; `T_odd` = 10 s to satisfy it
Degraded	Restrict. Hard speed cap from the law above; enlarge margins — held underneath the switch, not instead of it	`SafetyEnvelopeCommand` mode `CONSTRAIN`, `speed_limit = v_max`	3 s to reach compliance
Beyond ODD, HPC compliant	Request MRM. Ask the HPC to pull over *while it still can*	mode `MRM` + refuge request	10 s to begin, 60 s to complete
Beyond ODD, HPC non-compliant	Take authority — island pull-over per §6.3	`I7` → `N18`	—

Compliance supervision is the load-bearing mechanism. The island issues a constraint, then watches whether actual speed and behaviour converge on it within the declared deadline. *Non-compliance is itself a fault*, raised to `N8` — and it is a fault the island can detect without any HPC cooperation, using only vehicle status it already receives.

The ODD switch demand, and what the island verifies

The **Unsatisfiable** rung is the one that changes the character of the ladder. Every other rung constrains a *quantity*; this one demands a change of *domain*. The vehicle is being driven under a set of assumptions — sensing range, friction, lighting, road class — which the island now measures to be false, and the remedy is not to drive the same way more slowly but to drive under a different set of assumptions that the measurements do support.

The island does not name that set. It holds no catalogue of the HPC's operating domains, no model of which L4 behaviours each of them admits, and no means of choosing between them — all three are planning, which R4 and R5 keep off the island. What it holds is the *measurement*, and a measurement is enough to reject an answer without being able to produce one:

Step	Who	What crosses the link
1. Declare unsatisfiable	island, `N19`	`OddVerdict.odd_satisfiable = false` + the measured envelope + the failed dimension + the `T_odd` budget
2. Select a domain	HPC	nothing — this is planning, and it happens entirely on the HPC
3. Declare the new active domain	HPC, `H4`	`OddStatus.active_odd` + its bounds + `odd_switch_state`, within `T_ack`
4. Verify the declaration	island, `N19`	nothing — the bounds are compared against the island's own envelope
5. Verify the behaviour	island, `N2`/`N19`	nothing — measured speed, acceleration and curvature are compared against the declared bounds, within `T_odd`
6. Accept, and release the cap	island, `N3`	`SafetyEnvelopeCommand` mode returns to `ALLOW`; `OddVerdict` carries the accepted domain

Steps 4 and 5 are separate on purpose. A declaration the island can check against its own measurements catches an HPC that reports a domain it *could not* be in. Checking the vehicle's behaviour afterwards catches an HPC that reports a domain it *is not* in — a different failure, and the more dangerous one, because the declaration is plausible. A domain that passes step 4 and fails step 5 is recorded by `N11` as a SOTIF event under the same rule as verdict disagreement below: it is evidence about the HPC that no other mechanism produces.

The cap does not come off for the declaration. It comes off for the behaviour. Between steps 1 and 6 the vehicle is held at `v_max` by the `CONSTRAIN` rung regardless of what the HPC has declared, so a false declaration buys the HPC no speed — only time, and only `T_odd` of it.

Exhaustion is a valid answer. If the HPC declares at step 3 that none of its domains is satisfiable, the ladder moves straight to **Request MRM** — an HPC-executed pull-over, with the map and the traffic law available to it, which is strictly better than the island-executed one under B3. `T_odd` is not run out on an HPC that answered honestly.

Hysteresis, and why it is asymmetric

Weather flickers; a vehicle that oscillates between speed limits is both uncomfortable and unsafe. `N19` degrades on ~2 s of consistent evidence and recovers only after ~30 s of clear evidence, with a margin band between the thresholds. **Degrade fast, recover slowly.** Recovery must additionally require that the HPC's own ODD verdict agrees.

Verdict arbitration

The HPC also publishes an ODD verdict (`H4`). Two rules:

1. **The more restrictive verdict binds.** Either domain may restrict; neither may relax the other.
2. **Disagreement is itself a signal.** HPC says in-ODD, island says out-of-ODD → that is precisely the confidently-wrong case. It binds to the island's verdict, and it is recorded by `N11` as a SOTIF event for field monitoring, because it is evidence about the HPC's perception that no other mechanism produces.

Where the two features meet

Fallback availability and environmental ODD are not independent. If `N18` reports `pull_over_available = false` — a tunnel, a barriered section, no shoulder — **and** `N19` reports falling visibility, then the vehicle is entering a stretch where the degraded fallback is *in-lane stop on a road it cannot see.* Neither signal alone justifies much; together they justify a substantially tighter speed ceiling, or declining to enter the section at all.

`N19` therefore takes `pull_over_available` as an input and applies a reduction factor to `v_max` when it is false. This is the concrete payoff of making fallback capability observable in §6.3.

6.5 Remote command and control link and local control port

New by D9. Their detailed design is proposed in §6.6; these three nodes stay outside the counts of §4 and §9.4 until the WG accepts it.

The island carries its own path to the remote command and control centre (RCC), and its own port for a local joystick, so that a person can drive the vehicle when the HPC cannot — B5 from the RCC on streamed island sensing, B6 from beside the vehicle on none.

#	Node	Purpose	Tier	Rate
N21	`si_rcc_link`	**The communication module's island end.** Owns the cellular modem and the RCC session: mutual authentication, heartbeat, measured round-trip latency and stream age, and the E2E check of every remote command before it reaches `I1`'s remote source. Declares the link lost on `T_link`, which is B5's exit to the in-lane stop or B4. Also carries the RCC's status uplink — island state, fault summary, the ADS's **remote take-over request** in B1–B3, and an assistance request after B3 or B4. *How it does so — modem, session, security — is deferred to the next design (§8.6)*	T1 for the command and liveness path · T3 for the status uplink	50 Hz commands · 10 Hz heartbeat
N22	`si_teleop_streamer`	B5's **sensor stream**. Compresses the forward camera to a bounded bit rate and packs radar tracks, ultrasonic clearances where fitted and `N4`'s ego state alongside, with the capture timestamps the RCC needs to show their age. Reads the sensors through the same ingest as `N14`, `N6` and `N15`; never writes to them	T3, in its own partition, below P2	10–30 Hz video · 20 Hz tracks
N23	`si_local_control`	**The local control port** for B6: the wired joystick input. Connection detection, dead-man, link heartbeat, and the creep limit `v_local`; publishes the **local** command source into `I1`, which outranks the remote one, and requests the stop on release or on `T_local`	T1	50 Hz

The streamer is the heaviest load on the island and the least critical, which is the combination the partitioning exists to handle. Video compression must not take time from P0–P2, so `N22` runs in a partition of its own below safety sensing; when it overruns, the stream degrades and B5's latency monitor sees it — the island does not. Where the island SoC cannot carry the encoder at all, the open alternative is a hardware encoder in the modem module, fed the camera directly (open decision 19(d)).

`N21` terminates the RCC link on the island, not on the HPC. A teleoperation path that ran through the HPC would fail with it, which is exactly when B5 is wanted. The HPC may still use its own link for fleet services; what crosses `N21` is only what B5 and the RCC's view of the island need.

6.6 Detailed design of B5 and B6

This section turns the two declarations of D9 into a design the WG can review. It covers how each behaviour starts, what the person may command, how fast the vehicle may go, how the island notices a failing link or an absent operator, and how it stops. B6 starts only from a standstill. B5 starts from a standstill, or takes a moving vehicle over from B1, B2 or B3 when the ADS requests it. A path-based remote-assistance mode is left for stage 2 (open decision 19(e)). **Two things are deferred to the next design**: the RCC communication that B5 runs over — module, session, security, video transport — and a radio link for B6, which in this design is **wired only**. Their requirements are stated here; their realisation is not. Everything here reuses the island's existing command path: the gate `I1`, validation by `I8` and `N10`, and the stop operators `I6` and `I5`. The only additions are the three nodes of §6.5 and the interface in §8.6.

Principles both behaviours follow

1. **The person decides, the island bounds.** Every remote or local command passes through `I8` and `N10` and the gate's rate limits, like any other command source. AEB `I19`, the near-field guard `I20` and actuator supervision `N9` stay armed, and no command can override them.
2. **A silent link means stop, never "keep going".** Each command is valid for a short, fixed time. When commands stop arriving, the island first brings the speed setpoint to zero, then takes authority back and stops.
3. **Stopping uses the paths that already exist.** B5 falls back to the in-lane stop if the camera still sees the lane, and to B4 if it does not. That is the same test every other downward edge in §3.2 applies. B6 stops at once through `I6` and `I5`.
4. **Local presence beats remote presence.** An operator standing at the vehicle outranks a driver at the RCC. The gate's source ranking already encodes this (§10.4).

B5 — remote driving from the RCC

What this design fixes, and what it defers. It fixes B5's behaviour: how it starts, what the driver commands, how fast the vehicle may go, and how the island stops it when the link fails. It does not design the link. The cellular module and its redundancy, the RCC session and its keys, the protection profile and the video transport are deferred to the next design. Until then the link figures below — 50 Hz commands, `T_link` = 300 ms, the command-age bounds, the frame-stamp echo — are the requirements that design must meet.

What the remote driver sends. A steering-angle setpoint and a signed speed setpoint at 50 Hz, plus a gear request, the indicators, the hazard lamps and the horn. Every command carries a dead-man flag and a **frame stamp**: the island's capture time of the newest video frame the driver had on screen when the command was issued. `N21` checks each command (see below), turns it into an Autoware `Control` message and publishes it as the gate's **remote** source. The command is a *setpoint*, not a pedal position: `I1` applies its own rate limits, and `I8` and `N10` validate it, exactly as they do for the HPC's commands.

The frame-stamp echo. The island and the RCC do not share a clock. They do not need to: the frame stamp is a time the island itself wrote, so `N21` can compute

$$\textit{command age} = \text{island time now} - \text{frame stamp echoed in the command}$$

on its own clock alone. That one number covers the whole loop the safety argument depends on: capture, encoding, the uplink, display, the driver's reaction, and the downlink. `N21` checks it on every command, so the island knows how old the picture was that each command was decided on.

Session state. A **B5** session has six states. `N3` owns the transitions; `N21` supplies the evidence for each of them.

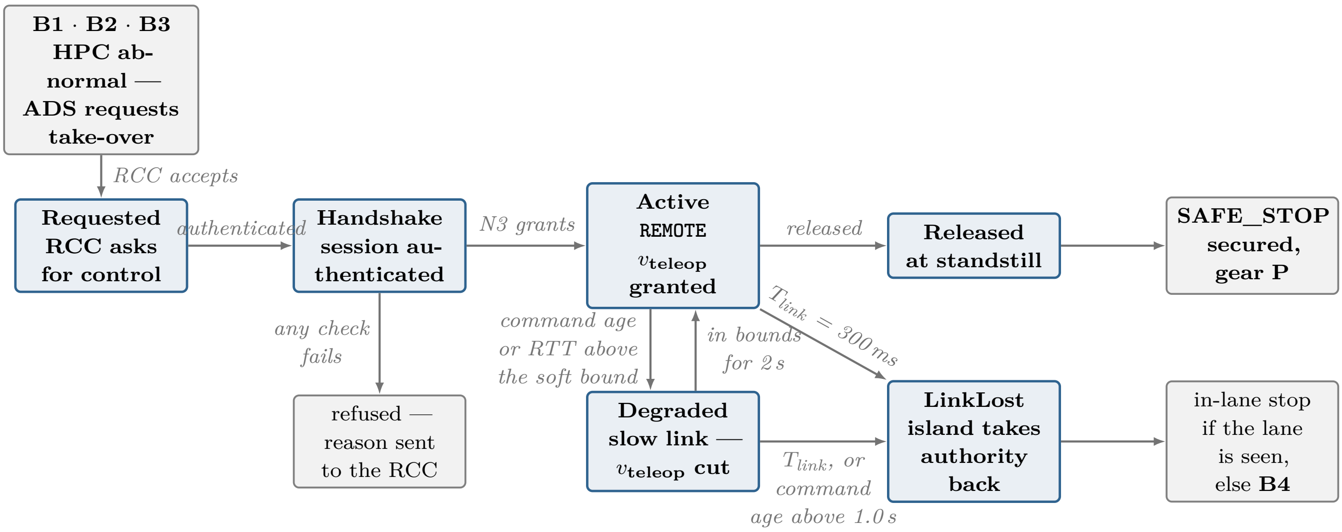

B1 · B2 · B3 HPC abnormal — ADS requests take-over
RCC accepts
Requested RCC asks for control
authenticated
Handshake session authenticated
N3 grants
Active REMOTE v_{teleop} granted
released
Released at standstill
SAFE_STOP secured, gear P
any check fails
refused — reason sent to the RCC
command age or RTT above the soft bound
in bounds for 2 s
$T_{\text{link}} = 300\,ms$
Degraded slow link — v_{teleop} cut
T_{link}, or command age above 1.0 s
LinkLost island takes authority back
in-lane stop if the lane is seen, else B4

Figure 14. *The B5 session, owned by N3 with N21's evidence: granted at standstill, or from a moving vehicle in B1–B3 on the ADS's request once speed, awareness and the command match; always with the camera valid and the link in bounds; a slow link derates, a lost link stops.*

- **Requested → Handshake.** The RCC opens an authenticated session on `N21`, on its own initiative or in answer to the ADS's take-over request. The request names the vehicle and the operator.
- **Handshake → Active.** `N3` grants control only when all the grant conditions below hold. On the grant, `I2` and `I3` switch the command mode to `REMOTE`. The brake stays applied until the first valid command arrives with the dead-man held.
- **Degraded.** The link still works but is slow. The island cuts the speed limit `v_teleop` (below) and tells the RCC why. It does not stop: a slow link is a reason to drive slower, not a reason to strand the vehicle.
- **LinkLost.** No valid command or heartbeat for `T_link`, or a command whose age exceeds the hard bound. The island takes authority back and stops: the in-lane stop through `N7` and `I13` if the camera still sees the lane, B4 if it does

not.

- **Released.** The RCC releases control at standstill. `I5` secures the vehicle, gear P, and the island is in `SAFE_STOP`.

The grant conditions. `N3` grants **B5** only when all of these hold:

- the vehicle is at standstill — or, for a take-over from a moving vehicle, the four further conditions given below hold;
- `N17` holds the camera valid;
- the round-trip time over the last 2 s is within the bound, and the stream is up and fresh;
- no **B6** session is active;
- the remote driver has confirmed the vehicle's identity, shown on the stream as an overlay that the island generates.

Checks on every remote command. `N21` drops any command that fails one of the following, and counts the drops. A run of drops longer than `T_link` is a link loss.

Check	**Bound (proposal)**	**On failure**
Session ID, sequence number, authentication and CRC — the RCC profile, deferred to the next design (§8.6)	exact; replay window 64	drop, and raise a security event to `N8`
Dead-man held	every command	speed setpoint to zero; brake held
Command validity	100 ms per command	setpoint decays to zero speed; then `T_link`
Command age (frame-stamp echo)	soft 600 ms · hard 1.0 s	soft: Degraded, `v_teleop` cut · hard: LinkLost
Heartbeat	10 Hz; `T_link` = 300 ms	LinkLost
Steering and speed range	within `v_teleop` and the vehicle's limits	clamp, and report it to the RCC

How fast — `v_teleop`. The vehicle must be able to stop inside the free road ahead, counting everything that happens before the remote driver's braking reaches the wheels:

> `v_teleop` = min(`v_cap`, `v_vis`), where `v_vis` solves $\texttt{v} \cdot \texttt{t\_loop} + \texttt{v}^2 / (2 \cdot \texttt{a\_tele}) = \texttt{d\_free} - 2\ \texttt{m}$

- `t_loop` is the measured command age plus the island and actuator time. As a design value it is 1.8 s: 0.23 s from capture to display, 1.2 s driver reaction, 0.12 s downlink, 0.04 s through the island (§9.2) and 0.2 s actuator response.
- `a_tele` = 3 m/s^2 is a firm stop, not an emergency one.
- `d_free` is the free distance ahead that the island itself has verified: `N16`'s free-space range, capped at the range at which the camera stream can still

show an obstacle. The driver is never allowed to drive faster than the island can confirm is clear.

- `v_cap` is **20 km/h**.

`d_free`	**10 m**	**15 m**	**20 m**	**30 m**	**40 m**
`v_vis` at `t_loop` = 1.8 s	12 km/h	18 km/h	23 km/h	31 km/h	38 km/h

`N21` recomputes `v_teleop` at 10 Hz from the *measured* command age, so a slow link lowers the speed automatically. The limit falls at no more than 2 m/s^2 and rises only after 2 s of stable link. The two branches of §9.5 differ here only through `d_free`. On Least-Cost, the camera and the 2D lidar confirm about 15–20 m, which gives roughly 18–23 km/h, and the vehicle drives **forward only**. On Maximum-Safety, `v_cap` binds, and the rear radars allow reversing at creep speed.

What the remote driver sees. `N22` streams the forward camera, compressed to a bounded bit rate: about 2 Mbit/s on Least-Cost, where the camera encodes it, and up to 6 Mbit/s on Maximum-Safety. Each frame carries its capture stamp. `N22` also sends radar tracks and ultrasonic clearances where fitted, `N4`'s ego state, the granted `v_teleop`, and which island guards are armed. The vehicle-identity overlay and the `v_teleop` readout are generated by the island, not by the RCC, so the driver sees what the island will enforce.

Reaction budget. Link loss is detected after `T_link` = 300 ms. The decision and the stop command take 40 ms more (§9.2). From 20 km/h, stopping at `a_tele` then takes about 7 m in total, inside the free range `v_teleop` was computed for. An obstacle the island sees itself is handled faster still, by `I19`, whatever the link is doing.

==**Leaving B5.** The normal exit is release at standstill into `SAFE_STOP`. **B5** may also hand the drive back to the ADS, but only from `SAFE_STOP` and only with all of these:

- the RCC requests it;
- `N2` has re-qualified the HPC over the same confirmation window as **B2**'s hand-back (open decision 14);
- `N19` reports the ODD satisfiable, and `H4` has declared an `active_odd`.==

The island then engages `AUTONOMOUS` through the normal Autoware path. Without this edge, every remote recovery would end the drive cycle, including one that followed a transient fault (open decision 19(c)).

B5 taken over from a moving vehicle — B1, B2 or B3

B5 can also start while the vehicle is still moving under **B1**, **B2** or **B3**. This is the path that lets a person at the RCC step in *before* the vehicle has to stop.

Who asks. The **ADS** raises a **remote take-over request** when the HPC behaves abnormally. It comes from one of two places:

- `N3`, automatically, as the island enters **B1**, **B2** or **B3**. Each of those is the island's own finding that the HPC has stopped meeting its obligations.
- The HPC itself, when its own diagnostics find it abnormal, through a request flag in `HpcCapability` (§8.1), admitted by `N1`. The flag travels in the capability vector, so a request is by construction a **capability drop**: `N2` treats it as a **B1** trigger. An HPC that asks to be replaced has, on its own report, lost the capability it was driving on.

==`N21` sends the request to the RCC with:

- the reason;
- the current behaviour;
- the speed;
- the budget that behaviour has left: `T_odd`, `T_hold` or `T_refuge`.==

A request never stops the clock. Until the grant, the current behaviour keeps authority and every one of its timers runs. If `T_odd`, `T_hold` or `T_refuge` expires first, the ladder moves on exactly as §3.2 says, and the request follows the vehicle down: the RCC may still take over from the next behaviour. If the camera is lost, the island goes to **B4**, and the request is withdrawn. **B5** cannot be granted without a camera the island can validate.

Who grants. The RCC accepts the request and asks for control. `N3` grants it when the standstill conditions above hold, except standstill itself, and **all** of these hold as well:

1. **Speed at or below `v_teleop`.** The island brings the vehicle there.

- In **B2** and **B3** the island is driving, so it lowers its own speed target to `v_teleop`.
- In **B1** the HPC is still driving, so `N3` puts `v_teleop` into the speed limit of `SafetyEnvelopeCommand`. That is **B1**'s own constraint channel (§8.2), and the HPC slows the vehicle itself.

1. **Situational awareness.** The remote driver has watched the live stream for at least `T_sa` = **5 s**, with every command inside its age bound, before the grant.
2. **A matched command.** For 0.5 s, the driver's setpoints are within ?1.5° of the current steering angle and ?2 km/h of the current speed. The island shows both as a target on the stream. The hand-over is then *bumpless*: `I3` moves the gate from the current source to the remote one with no step in either command.
3. **No lateral manoeuvre in progress.** In **B3** the grant waits while the vehicle is crossing into the refuge. It is given before the lateral move starts

or after it ends, never halfway across.

==On the grant, `I2` and `I3` switch the mode to `REMOTE` and the current behaviour ends. From then on, **B5** is the same as when it starts from standstill:

- a lost link drops to the in-lane stop, or to **B4**;
- release at standstill leads to `SAFE_STOP`;
- re-engaging the ADS needs the HPC to be re-qualified.==

Taking over from **B1** deselects an HPC that was still driving. It does not come back on its own.

It fits inside the hold window. On a highway, **B2** enters at up to 60 km/h after its first descent. Slowing from 60 to 20 km/h at 2 m/s^2 takes 5.6 s, and the 5 s of situational awareness run during that descent, not after it. So a remote driver who accepts at once can take the vehicle about 6 s after **B2** begins, inside the 8–15 s `T_hold`. On a city road the descent from 30 km/h takes 1.4 s, and the 5 s of awareness set the time instead. If the RCC is slower, `T_hold` runs out, **B3** begins, and the take-over can still happen there.

Failure during a take-over from motion	**Island response**
The RCC accepts but never matches the command	No grant. The current behaviour continues and its timers expire normally
The link drops before the grant	The request is withdrawn. Nothing else changes: the island never gave up authority
The link drops after the grant	As for any **B5**: in-lane stop, or **B4** without the camera
The HPC requests take-over while the island sees it healthy	A capability drop, so `N2` enters **B1**; the request goes to the RCC
The camera is lost before the grant	**B4**; the request is withdrawn

B6 — joystick control at the vehicle

The device. A handheld joystick **pendant on a cable**, plugged into a CAN FD connector on the vehicle. It has proportional steering and speed controls, a forward / neutral / reverse selector, horn and hazards, and two safety controls:

- a **three-position enabling switch**, the dead-man: commands count only while it is held in the middle position;
- a latching **emergency-stop** button.

The cable is what keeps the operator at the vehicle, and it is the only link `N23` supports in this design. **A short-range radio link is deferred to the next design**, with IEC 62745 (cableless control of machinery) as its reference: that design must add pairing, a stop on loss of the radio link, and the radio band, which differs by region.

Connecting and entry.

- The pendant is plugged in at standstill and a key switch on the vehicle is turned: physical presence, twice over.
- The port has one connector, so only one pendant can be connected. Unplugging it ends control, and stops the vehicle.
- B6 starts when the pendant's enabling switch is held and `N3` grants `LOCAL`.
- If a B5 session is active, the local request **pre-empts** it. If the vehicle is moving under B5, the island first stops it, then grants `LOCAL` at standstill.

==**What the operator commands.** A steering setpoint and a speed setpoint, at 50 Hz on the SI-E2E/B profile, which fits CAN FD:

- speed limited to `v_local` = **5 km/h** forward and **3 km/h** in reverse;
- acceleration limited to 0.5 m/s^2, and the steering rate limited.==

B6 reverses on the operator's eyes alone, on both branches. Any island sensors still valid keep `I19` and `I20` armed, but **B6** requires none of them: it runs at rung 0.

==**When the island stops.** The island stops at once, at the emergency rate through `I6`, then `I5` secures:

- the enabling switch is released, or squeezed through to its third position;
- the emergency-stop button is pressed;
- the cable link is silent for `T_local` = **200 ms**, which is what a pulled or broken cable looks like.==

From 5 km/h that stop is about 0.85 m including all latencies; from 3 km/h in reverse, about 0.45 m. The emergency-stop button should also be **wired directly to the actuator gateway's stop input** — through the same cable, which makes it simple to do — in the same way as the §8.5 FSI fault line, so that the stop does not depend on island software. That is a Part-2 hardware item.

Leaving B6. The operator releases control at standstill, and the island goes to `SAFE_STOP`. **B6** never hands the drive back to the ADS. A vehicle recovered by hand is checked by a person before it is automated again.

Where the new nodes run

Node	**Partition (§9.1)**	**Why**
`N21` command and liveness path	**P1**, 50 Hz	It feeds a command source and supervises a link whose loss triggers a stop
`N21` status uplink	**P4**	Telemetry; it may be shed first
`N22` streamer	**P3s**, a new band just below P3	The heaviest load and the least critical. It must never take time from P0–P2, and during **B5** the P3 nominal controller is idle anyway. On the X5H it runs on the application cluster (§9.5)
`N23` local control	**P1**, 50 Hz	Same reasoning as `N21`'s command path

Failure modes and what the island does

Failure	Detected by	Island response
RCC link drops	N21: `T_link`	LinkLost: in-lane stop, or **B4** without the camera
Link slows down, or video freezes at the RCC	N21: command age (frame-stamp echo)	Degraded: `v_teleop` cut; LinkLost above the hard bound
Replayed, forged or corrupted command	N21: the RCC profile's checks (next design)	Command dropped, security event to `N8`; repeated drops are a link loss
Remote driver lets go or is incapacitated	dead-man flag	Speed setpoint to zero, brake held
Driver steers toward an obstacle	`I19`, `I20`, `N10`	Braked or vetoed whatever the command says
Camera lost during **B5**	`N17`	**B4**: the driver is no longer shown a validated picture
Two RCC sessions request the same vehicle	N21: one session at a time	The second request is refused
Local operator arrives during **B5**	`N23`	**B6** pre-empts: stop first, then `LOCAL`
Joystick cable pulled or broken	N23: `T_local`	Immediate stop, `I6` then `I5`
Joystick enabling switch released, or emergency stop pressed	`N23`, plus the hardwired stop	Immediate stop
Actuators not tracking, in either behaviour	`N9`	**B4**

These rows join the Part-4 fault-injection matrix of §12.

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7 New nodes to add on the HPC

The HPC side needs nodes whose only job is to make the interface narrow, typed and rate-limited (R5). Without them, "connect Autoware to the island" means publishing the full ROS graph over TSN.

#	Node	Purpose	Rate
H1	`si_bridge_hpc`	In-house E2E egress/ingress counterpart of N1 (§8.4). Wraps each safety payload with Data ID, sequence counter, CRC, length and gPTP timestamp; enforces the publication rate; is the *only* HPC node permitted to write to the island link.	per-interface
H2	`hpc_capability_reporter`	Condenses `diagnostic_graph_aggregator` / `hazard_status_converter` / `pipeline_latency_monitor` output into a **fixed-size capability vector** (perception OK, localization OK, planning OK, per-pipeline latency margin, confidence floor) plus a self-test summary. Replaces free-form diagnostics on the link.	10 Hz
H3	`hpc_corridor_publisher`	Emits a compact drivable-corridor strip (left/right boundary polylines, fixed point count) around the current trajectory, derived from the Lanelet2 map. This is the map, reduced to what the island can hold — enabling N5 and N7 without shipping the map.	10 Hz

#	Node	Purpose	Rate
H4	`hpc_odd_reporter`	ODD and geofence validity, weather/visibility class, unmapped-area and roadworks flags — as a small enumerated status, not raw sensor data. Also carries the HPC's **declared active operating domain** and its bounds — speed ceiling, required sensing range, assumed friction — plus an `odd_switch_state` enum (`STEADY` / `SWITCHING` / `EXHAUSTED`). This is the HPC's half of the B1 exchange: the island demands a switch, and this is the node that answers within `T_ack`. The bounds are what the island verifies against its own envelope, so they must be the domain's *declared* limits, not the vehicle's current state.	10 Hz
H5	`hpc_trajectory_conditioner`	Resamples and bounds the planned trajectory to the island's fixed-capacity message (**250 points**, as already enforced in the nano-ros port), guarantees monotonic timestamps, fills per-point velocity/acceleration, and rejects malformed trajectories *before* they are transmitted.	10 Hz
H6	`hpc_island_state_client`	Subscribes to island state and MRM state; republishes into `/system/fail_safe/mrm_state` and the AD API so that operator UI, logging and `command_mode_decider` see the island's verdict. Closes the loop for graded reaction steps 1–2.	10 Hz
H7	`hpc_heartbeat`	Deterministic, minimal heartbeat with a rolling self-test result. Kept separate from H2 so that it can be given the highest RT priority and the shortest code path.	50–100 Hz
H8	`hpc_refuge_publisher`	Streams the rolling, map-validated and law-checked **pull-over refuge** defined in §6.3, plus the explicit "no refuge available" declaration when it cannot honour the coverage guarantee. This is the node that makes a *lawful* island pull-over possible at all — the island's own detection (`N18`) can find clear space without it, but only as an `EMERGENCY_ONLY` stop.	2 Hz
H9	`hpc_shadow_control_monitor` *(optional)*	Runs the same controller configuration on the HPC and compares against the island's echoed command; flags divergence. **Field-monitoring and validation only — no safety authority.**	50 Hz

8 The interface contract

Deliberately narrow: 11 signals in — ten required and one optional — and 9 out. Every row is end-to-end protected (§8.4) and rate-declared.

8.1 HPC → Island (TSN/Ethernet, DDS or SOME/IP binding under nano-ros)

Signal	Topic	Type	Rate	Age limit	Protection
Trajectory (≤250 pts)	`/planning/scenario_planning/trajectory`	`autoware_planning_msgs/Trajectory`	10 Hz	300 ms	SI-E2E/A
Drivable corridor *(preferred source)*	`/si/in/corridor`	*new* `SafetyCorridor`	10 Hz	300 ms	SI-E2E/A
Kinematic state (reference)	`/localization/kinematic_state`	`nav_msgs/Odometry`	50 Hz	60 ms	SI-E2E/A

Signal	**Topic**	**Type**	**Rate**	**Age limit**	**Protection**
Acceleration (reference)	`/localization/acceleration`	`geometry_msgs/AccelWithCovarianceStamped`	50 Hz	60 ms	SI-E2E/A
Operation mode state	`/api/operation_mode/state`	`autoware_adapi_v1_msgs/OperationModeState`	10 Hz + on change	300 ms	SI-E2E/A
Capability vector	`/si/in/capability`	*new* `HpcCapability` (including `remote_takeover_requested` and its reason)	10 Hz	300 ms	SI-E2E/B
ODD status + declared active domain	`/si/in/odd_status`	*new* `OddStatus` (geofence validity, weather class, roadworks flags, **`active_odd` with its speed / sensing-range / friction bounds**, `odd_switch_state`)	10 Hz	300 ms	SI-E2E/B
Heartbeat + self-test	`/si/in/heartbeat`	*new* `HpcHeartbeat`	100 Hz	**20 ms**	SI-E2E/B
Pull-over refuge	`/si/in/refuge`	*new* `PullOverRefuge`	2 Hz	2 s	SI-E2E/A
Auxiliary commands	`/control/command/{gear,turn_indicators,hazard_lights}_cmd`	`autoware_vehicle_msgs/*`	10 Hz	300 ms	SI-E2E/A
Object list *(optional — §3.5)*	`/perception/object_recognition/objects`	`autoware_perception_msgs/PredictedObjects`, truncated to a bounded digest at `N20`	10 Hz	300 ms	SI-E2E/B

Under D7, **every input on this table is now degradable.** Each has an island-local substitute:

HPC input	**Island substitute when it is stale or absent**
Kinematic state, acceleration	`N4` dead reckoning — the island cross-checks the HPC estimate against it and, on divergence or loss, uses N4 alone
Drivable corridor	`N14` camera lane boundaries — and nothing below them: with no camera corridor the island enters B4
Pull-over refuge	The last refuge whose `validity_distance_m` still covers the stopping distance; then `N18`'s **island-detected** `EMERGENCY_ONLY` clearance, searched for at most `T_refuge`; then in-lane stop
Trajectory	The buffered last-valid trajectory, then `N7`'s generated MRM profile
Capability vector, ODD status	`N2` liveness verdict and `N17` island sensor health. After HPC loss, the road class of the last valid `active_odd` is latched for B2's `v_odd`; with none valid, the most restrictive class. A silent `active_odd` is not a satisfied one: silence past `T_ack` is exactly the B1 timeout, and escalates by §3.2's takeover rule rather than being substituted for
Object list *(optional)*	`N16`'s own perceived obstacles; then nothing at all. It is the one row whose substitute is *absence* — §3.5 is written so that losing it costs information, not capability

That is the mechanism that lets I9–I13, I19 and I20 keep running after HPC loss. **No row on this table is a hard dependency of the safety core** — and the last row is not a dependency of anything at all.

8.2 Island → HPC

Signal	Topic	Type	Rate
Island safety state	`/si/out/state`	*new* `IslandState` (mode — including `LANE_KEEP_HOLD` for B2 and `EMERGENCY_STOP` for B4 — fault code, authority holder, corridor source in use *or the blackout flag when there is none*, lateral deviation, remaining hold budget `T_hold` , the latched ODD class and its `v_odd`, the current `v_hold` and the `v_entry` the ramp is aimed at)	50 Hz
Safety world state	`/si/out/safety_world`	*new* `SafetyWorldState` (nearest in-path obstacle range and range-rate, TTC, free-space distance, perceived corridor, per-sensor validity)	20 Hz
Island sensor health	`/si/out/sensor_health`	*new* `IslandSensorHealth` (per-sensor availability, blockage, degradation cause , and which D8 ranks are fitted — the island's sensor rung)	5 Hz
Island ODD verdict + switch demand	`/si/out/odd_verdict`	*new* `OddVerdict` (visibility range, precipitation class, friction estimate, ambient light, derived `v_max`, ODD margin , **`odd_satisfiable`**, the failed ODD dimension, the remaining `T_odd` budget, and the accepted domain once there is one)	2–5 Hz
Fallback availability	`/si/out/fallback_availability`	*new* `FallbackAvailability` (**`pull_over_available`**, reason, distance to next refuge , and **`refuge_source`** — `HPC_NOMINATED` or `ISLAND_DETECTED` — so that the HPC and the operator can tell a lawful refuge from a clearance)	2 Hz
MRM state	`/system/fail_safe/mrm_state`	`autoware_adapi_v1_msgs/MrmState`	10 Hz + on change
Constraint / veto vector	`/si/out/constraints`	*new* `SafetyEnvelopeCommand` (accel limit, curvature limit, speed limit, mode: ALLOW / CONSTRAIN / MRM)	50 Hz
Executed control command (echo)	`/control/command/control_cmd`	`autoware_control_msgs/Control`	50 Hz
Vehicle status	`/vehicle/status/{velocity,steering,gear,control_mode}_status`	`autoware_vehicle_msgs/*`	50–100 Hz

`SafetyEnvelopeCommand` is the mechanism for graded-reaction steps 1 and 2: the island constrains the HPC rather than seizing control, and only escalates when the HPC fails to comply within a declared number of cycles.

`SafetyWorldState` is **advisory in this direction**. The HPC may use it to cross-validate its own perception — a radar track the island sees and the HPC does not is a strong SOTIF signal — and `H9` is the natural consumer. It must never become an input the HPC's nominal pipeline depends on, or the independence argument runs backwards.

8.3 Island → actuator gateway

CAN FD (or deterministic Ethernet) at **100 Hz**, protection profile **SI-E2E/B**, carrying steering, acceleration / braking, gear, hazard lamps and the MRM flag, with actuator feedback returned on the same bus for N9. This path shares nothing with the HPC link (R6, and RFC design principle 3).

8.4 In-house safety-message protection

D6 originally named AUTOSAR E2E. The WG has since scoped this design as **Autoware in-house**, so AUTOSAR is out. The *obligation* does not go with it: it comes from the communication threat analysis in EN 50159 and ISO 26262-6, and it is unchanged by the choice of implementation.

What every safety-relevant message must still let the receiver detect:

Threat	Detected by
Corruption	CRC over header and payload
Unintended repetition	Sequence counter
Incorrect sequence, reordering	Sequence counter
Loss	Sequence counter gap
Unacceptable delay, staleness	gPTP timestamp against a per-signal age limit
Insertion, masquerade	Data ID bound to the interface
Addressing error, wrong interface	Data ID
Truncation, malformed payload	Length field

Two profiles are enough for the HPC link and the vehicle bus — the RCC link, an open network, will need a third; it is deferred with the RCC communication (§8.6):

	SI-E2E/A — Ethernet, variable length	**SI-E2E/B — compact status and CAN FD**
Used for	Trajectory, corridor, refuge, odometry, acceleration, mode, auxiliary commands	Heartbeat, capability vector, ODD status, actuator commands and feedback
Data ID	16-bit	8-bit
Sequence counter	8-bit	4-bit
Length	16-bit	implicit (fixed)
Timestamp	64-bit gPTP	32-bit ms
CRC	CRC-32 over header + payload	CRC-8
Overhead	17 bytes	6 bytes

`N1` verifies on ingress, `H1` applies on egress, and **a verification failure is a safety event raised to `N8`, never a middleware exception**. That rule is what actually matters, and it is independent of which profile carries it.

nano-ros already lists an E2E safety protocol with CRC and an EN 50159 mapping in its evaluation checklist. **The first task in Part 3 is to determine whether that covers SI-E2E/A and /B outright**, in which case the island inherits it rather than implementing it — the preferable outcome, since it comes with the middleware's own verification evidence.

8.5 On-SoC FSI → Island

SPI plus a discrete `SOC_ERROR` GPIO carrying hardware faults, watchdog expiry and failover triggers, bypassing Linux and the TSN switch entirely. Consumed by N8 as a highest-priority fault source. When the HPC is an Orin/Thor-class part, this is the sub-millisecond detection path; the TSN heartbeat (20 ms age limit) is the vehicle-level-independent one.

8.6 The RCC link and the local control port

Two interfaces cross the island boundary that are not the HPC link: the cellular session to the RCC, which ends at `N21`, and the local control port, which ends at `N23`. Both carry commands into the gate, so both carry the same end-to-end protection as §8.4. The RCC link needs more than that.

The RCC communication is deferred to the next design. The two RCC tables below are what B5 requires of it — the signals, their rates and their age limits — and the profile after them is the proposal that design starts from. The transport, the session, the keys and the video channel are that design's to choose. The local port is designed here, **wired only**.

RCC → island (into `N21`).

Signal	**Topic**	**Type**	**Rate**	**Age limit**	**Protection**
Session control	`/si/rcc/session`	*new* `RccSessionControl` (request / confirm vehicle / release, operator ID, vehicle ID echo)	on event + 1 Hz	1 s	SI-E2E/R
Remote drive command	`/si/rcc/drive_cmd`	*new* `RemoteDriveCommand` (steering angle, signed speed setpoint, gear, dead-man, indicators / hazards / horn, **frame stamp**)	50 Hz	100 ms per command; command age $\leq$ 1.0 s	SI-E2E/R
RCC heartbeat	`/si/rcc/heartbeat`	*new* `RccHeartbeat` (echo of the island's last ping, for the round-trip time)	10 Hz	`T_link` = 300 ms	SI-E2E/R

Island → RCC (out of `N21`, with `N22`'s stream).

Signal	**Topic**	**Type**	**Rate**	**Protection**
Video	media channel	H.264 / H.265, a capture stamp on every frame; ≈ 2 Mbit/s (Least-Cost) to $\leq$ 6 Mbit/s (Maximum-Safety)	10–30 Hz	SRTP; the frame stamp is checked at the RCC and echoed back
Tele-operation telemetry	`/si/rcc/telemetry`	*new* `TeleopTelemetry` (radar tracks, ultrasonic clearances, `N4` ego state, granted `v_teleop`, guards armed)	20 Hz	SI-E2E/R
Link state	`/si/rcc/link_state`	*new* `RccLinkState` (round-trip time, command age, stream age, `v_teleop` and why it was cut)	10 Hz	SI-E2E/R

Signal	**Topic**	**Type**	**Rate**	**Protection**
Island status, take-over and assistance requests	`/si/rcc/status`	*new* `IslandStatusSummary` (island mode, fault summary, MRM outcome, refuge source, GNSS position, **remote take-over request** in **B1**–**B3**, with its reason, the current behaviour and its remaining budget, and an assistance request after **B3** or **B4**)	1 Hz + on change	SI-E2E/R

Local port ↔ `N23` (CAN FD, by cable).

Signal	**Direction**	**Content**	**Rate**	**Age limit**	**Protection**
Local drive command	device → island	steering setpoint, speed setpoint, F / N / R, enabling-switch position, emergency-stop state, horn / hazards	50 Hz	`T_local` = 200 ms	SI-E2E/B
Device status	device → island	pendant ID, connection state, enabling-switch self-test	5 Hz	1 s	SI-E2E/B
Island feedback	island → device	mode granted, speed, fault, `v_local`	5 Hz	—	SI-E2E/B

Why the RCC link will need a third profile — a proposal for the next design. The HPC link and the local port run on closed networks. A cellular link is an **open** network in the sense of EN 50159: anyone who can reach it can try to inject a message, and a CRC is not a defence against an attacker who can compute one. So the RCC link carries **SI-E2E/R**, which splits the job in two:

	SI-E2E/R — RCC link
Safety part — the same threats as §8.4	Data ID 16-bit · sequence counter **32-bit**, with a 64-message replay window · length 16-bit · CRC-32
Age	The **frame-stamp echo**. It is measured on the island's own clock, so no shared time base is needed
Session	Session ID 32-bit, bound to one authenticated operator and one vehicle
Security part — masquerade and tampering	A message authentication code from the secure channel: TLS 1.3 or DTLS 1.3, mutually authenticated. The keys are held in the island processor's hardware security module
Overhead	≈ 27 bytes, plus the channel's

A message that fails the security part is a **security** event; one that fails the safety part is a **safety** event. `N21` raises both to `N8`, and repeated failures of either kind count as a link loss. Commands and telemetry travel over the authenticated session. Video travels on its own SRTP media channel, because it is the bulk of the traffic and its integrity matters less than its age, which the frame-stamp echo checks anyway.

9 Execution partitions, timing, and middleware realisation

9.1 Execution partitions

Thirty-odd nodes do not mean thirty-odd threads. Following the declarative RT-configuration approach already demonstrated (contract facts → priority bands, `chain_aware` mapper), the island is organised as five executors with fixed priority bands:

Partition	Nodes	Period	Priority band
P0 — Safety core	I1, I2, I3, I5, I6, I7, I16, N1, N2, N3, N4, N8	10 ms (100 Hz)	1 — highest
P1 — Supervision + fallback control	I8, I13, N5, N7, N9, N10, N12, N18 · N21 command path, N23 *(§6.6)*	20 ms (50 Hz)	2
P2 — Safety sensing	I19, I20, N6, N13, N14, N15, N16, N17, N19, N20 *(optional)*	20–50 ms (20–50 Hz)	3
P3 — Nominal control	I9, I10, I11, I12, I14, I15	20–30 ms (33–50 Hz)	4
P3s — Tele-operation streaming	N22 *(§6.6)*	33–100 ms (10–30 Hz)	4b — below P3; idle outside B5
P4 — Non-real-time	N11, telemetry egress · N21 status uplink	100–1000 ms	5 — lowest

The priority order encodes the degradation strategy, and it is deliberate that **the QM nominal controller sits below ASIL-B sensing.** Under overload the island sheds the expensive MPC first: the cheap ASIL-B fallback controller in P1 takes over within one cycle, which is exactly what the two-tier design in §4.2 exists to make possible. Shedding sensing instead would silently downgrade an obstacle-aware MRM to a blind one — the worse outcome. P4 is shed before anything else.

P0 must complete without P2 or P3. This is the property the measured failover already exhibits — control cadence held at a 9.96 ms mean period through a full Linux outage — and it is the property the timing analysis in Part 2 must *prove* rather than observe. The corollary from §2.2 is that P0 reads P2's output only through an age-checked snapshot, and never blocks on it.

Sizing note. P2 is the new load introduced by D7, and it is the partition most likely to force a larger MCU than the current S32Z-class target. Camera processing — lane, road-edge and lateral free-space extraction — is now the dominant and unavoidable part of it, since the camera is the one mandatory sensor; 2D-lidar occupancy adds to it only where a lidar is fitted. Both are bounded, but not free. If open decision 8 resolves toward a certified smart camera that emits lane and object lists directly, N14 collapses to a bus parser and P2's worst case drops sharply — an argument for that option beyond the certification one.

9.2 Preliminary fault-reaction budget

To be reconciled with the FTTI in Part 2.

Stage	Budget
Detection — heartbeat/deadline miss (N2)	≤ 30 ms (3 missed 100 Hz heartbeats)
Detection — FSI hardware fault line	≤ 1 ms
Decision — N8 → N3 → I2/I3	≤ 20 ms (2 cycles)
Command — I1 → I16 → CAN FD	≤ 20 ms
Island reaction total	≤ **70 ms**
Actuator response	vehicle-specific

The 500 ms watchdog used in the current evaluation island is a coarse outer bound; the 30 ms heartbeat detection above is the target for the production interface, with the 500 ms watchdog retained as the independent backstop.

9.3 Middleware realisation on nano-ros

nano-ros is a `no_std` **Rust** ROS 2 client library for bare-metal and RTOS targets, with zenoh-pico, XRCE-DDS and Cyclone DDS as selectable RMW backends. Four of its properties do real work for this allocation, and one is a live design decision.

nano-ros property	What it buys the allocation
`no_std` Rust; `alloc`/`std` are opt-in features	Satisfies D5's "memory safe" requirement **at the language level**, not by coding convention.
148 Kani bounded-model-checking harnesses + **83 Verus** deductive proofs over CDR serialization, scheduling and protocol correctness	Reusable verification evidence for the *middleware element* of the ISO 26262 argument — normally the hardest part of a ROS-based safety case to construct.
Cyclone DDS backend **wire-compatible with stock ROS 2 RTPS**	The HPC↔island link needs no bridge and no type adaptation. Already demonstrated against unmodified Humble Autoware.
Declarative RMW / domain / topology selection (`system.toml` + launch XML → generated entry)	RMW and topology are **configuration, not code**. The decision below can be revisited without touching node source.
Verified targets: Zephyr, FreeRTOS, NuttX, ThreadX, bare-metal Cortex-M3 / ESP32-C3	Covers the S32Z-class production target and the FVP/QEMU evaluation path already in use.
Pub/sub, **services**, actions and parameters all complete	`mrm_handler` → MRM operators uses `tier4_system_msgs/srv/OperateMrm`. The service path exists — but see risk 1 below.

The allocation-policy decision

Per-RMW allocation behaviour differs, and it lands directly on **R4**:

RMW backend	Allocation behaviour	Fit for the T0 partition
XRCE-DDS	Allocates **only at entity setup**	Best — fully static after init
Zenoh / zenoh-pico	Requires an allocator; a bump allocator suffices on bare metal	Acceptable with a pre-sized bump arena
Cyclone DDS	Allocates **per message**	Admissible only with a pre-sized pool and bounded queue depth

The existing port uses Cyclone DDS for wire compatibility, and bounds that per-message allocation with `KEEP_LAST` depth 1 on every controller input plus fixed-capacity sequences (the 250-point trajectory bound). nano-ros ships `scripts/rmw-alloc-sites.py`, which reports **every allocation site** — that makes the bound *auditable rather than asserted*, and its report should become a standing review artefact for the T0 partition.

Recommendation: keep Cyclone DDS on the `N1`/`H1` boundary, where wire compatibility with stock Autoware is the entire point, and treat the intra-island transport for P0 as open decision 7 (§11).

Integration risks carried from the middleware

1. **The service path is not yet exercised in the ASI port.** `mrm_handler` ↔ MRM operators (`OperateMrm`) and the rejected-command reporting path both use services. Complete in nano-ros, listed as remaining validation work in the port. **Make this a phase-1 exit criterion.**
2. **Some Cyclone DDS embedded action paths are still in progress.** No node in this allocation uses ROS actions — keep it that way for the island, and prefer services or state topics if a future node is tempted.
3. **Execution-model exceptions.** Bare-metal deferred dispatch and Zephyr's component shape are flagged as exceptions in the nano-ros scheduling wiring matrix. The P0/P1/P2 priority bands in §9.1 must be checked against that matrix for the chosen target *before* Part 2's timing analysis.
4. **E2E.** nano-ros lists an E2E safety protocol (E2E CRC, EN 50159 mapping) in its evaluation checklist. With AUTOSAR now out of scope (§8.4), the question is simply whether that built-in protocol satisfies `SI-E2E/A` and `/B` directly. If it does, `N1`/`H1` become thin adapters rather than protocol implementations, and the island inherits the middleware's own verification evidence — **the outcome to aim for.**

9.4 Minimal deployable configurations

The full allocation is 18 migrated Autoware nodes plus 19 new ones. That is the *target*. This section answers a narrower question: **what is the smallest island that can pull the vehicle over after the HPC dies?**

Minimal sensor set

The set is a ladder, not a single answer, because it depends on the declared ODD. It has a floor — rung 0 — which is not a deployable configuration but the state the island degrades *into* when its sensors are lost. Listing it makes the point that B4 exists precisely so that the ladder cannot run out from under the design.

Rung	Sensors	Sufficient for
0 — the floor	chassis signals only — wheel speed, steering report, ESC yaw rate and acceleration; **no exteroceptive sensor**, or none the design may rely on: an island whose camera is lost is here, whatever else it still carries	B4 and nothing else: a blind emergency stop in the running lane. Not a configuration anyone deploys — it is what rungs A–E degrade to, and the reason B4 carries no precondition
A — irreducible	**forward camera** · chassis signals	B2 at the city-road `v_odd` (~30 km/h); B3 at or below ~30 km/h, the camera finding both the lane lines and the pull-over space, with a long indicator dwell and a very low lateral rate standing in for rear-corner radar; fair weather. **This is what the NTU golf-cart prototype can start with.**
B — design point	+ forward radar · rear-corner radar on the pull-over side	A credible L4 ODD: the highway `v_odd` (~60 km/h) for B2, B3 above ~30 km/h, weather robustness, closing speed on traffic ahead, awareness of the lateral space the manoeuvre crosses — and the radar-versus-camera visibility test that makes B1 possible
C — near-field	+ ultrasonic array	Near-field clearance for the final metres of B3: the stop may end tight to the kerb rather than within the shoulder; standstill and creep release; `I20` collision_detector runs (`N15` → `PointCloud2`)
D — full	+ island IMU / GNSS	A dynamics channel independent of the chassis bus; the friction dimension of §6.4; GNSS corroboration of the ODD class `v_odd` is set from
E — enriched	+ 2D lidar — *optional*	A second free-space channel and physical-edge confirmation for the camera's refuge; radar-versus-lidar visibility as a second instance. **Adds confidence, never capability**: nothing in B1–B6 requires it

Reasoning behind each, in the order D8 ranks them:

Sensor	Verdict	Why
Forward camera	**Mandatory, and the one that cannot be dropped**	Every behaviour above B4 is a question about the *lane*: B2 must hold it, B3 must find the space beside it and cross its outer line, and §6.4 must judge whether it can still be seen. The camera is the only island sensor that sees lane lines at all, and with classical road-edge and free-space extraction it also sees the pull-over space. Its loss is, by definition, B4.
Chassis signals	**Always present — not an island sensor**	Without `N4` there is no ego state after HPC loss and nothing to track a path against. Wheel speed, steering report and the ESC yaw rate and acceleration are on every vehicle's bus; they are what rung 0 stands on.
Rear-corner radar, pull-over side	**Mandatory above ~30 km/h**	The manoeuvre crosses lateral traffic space a forward camera cannot see. Cheap, weather-robust, CAN FD — the same part production blind-spot systems use. Below ~30 km/h it can be traded for a long indicator dwell and a very low lateral rate.
Forward radar	**Ranked second; mandatory for the highway class and once weather is in the ODD**	The only forward channel that survives fog, heavy rain and darkness — exactly the conditions in which the MRM is most likely to be needed. It supplies the range that lifts `v_max` to the highway `v_odd`, the point cloud `I19` runs on when no lidar is fitted, and half of the §6.4 visibility test.
Ultrasonic array	**Ranked third**	Required if the manoeuvre must terminate tight to a kerb, or if standstill-release safety is claimed. Droppable if "stop within the shoulder" is the accepted terminal condition — which is what rungs A and B accept. Cheap, short-range and blind beyond ~5 m, it buys a terminal condition, not speed: it is the only island sensor that sees the kerb, a bollard or a pedestrian in the last metre beside the vehicle, where the camera's view and radar's resolution both run out.

Sensor	Verdict	Why
Island IMU / GNSS	**Ranked fourth — recommended**	Gives `N4` a channel that does not share the chassis bus, and gives §6.4 a friction estimate it can claim margin on. GNSS corroborates the road class behind `v_odd`, downward only. Droppable: below rung D the chassis ESC signals stand in, with less independence and less margin.
2D lidar	**Optional**	Measures free space and the physical edge directly, and would confirm the camera's refuge well. But a behaviour written against an optional sensor is not deployable on the islands that lack it, so the lidar may only *confirm or disqualify* — never qualify a refuge or supply a corridor.

> **The ranking settles the sensor set, and it inverts the previous one.** Earlier revisions made the 2D lidar irreducible and the camera conditional. D8 reverses that, and the behaviours follow: the lane B2 holds and the space B3 pulls into are both found by the camera, the visibility test of §6.4 is radar-versus-camera, and B4 is defined as the loss of the camera. Every lower rung adds speed, or ODD width — rung B the highway class, rung C a tight-to-kerb end, rung D friction margin — and rung E adds only confidence.

Minimal node set — 21 mandatory

Group	Nodes	Count
Authority and state	`I1` gate · `I2` decider · `I3` switcher · `I7` mrm_handler · `N2` hpc_supervisor · `N3` state_machine · `N8` fault_manager	7
Ingress, ego state, actuation	`N1` gateway · `N4` vehicle_state_estimator · `I16` vehicle interface	3
MRM behaviours	`N18` pull_over_manager · `I6` mrm_emergency_stop_operator *(abort)* · `I5` stop_mode_operator *(terminal stop + auto-park)*	3
Fallback control	`N7` mrm_planner *(bounded Trajectory)* · `I13` pure_pursuit	2
Sensing	`N14` vision_guard · `N6` radar_guard · `N16` safety_world_model · `N17` sensor_health · `I19` autonomous_emergency_braking	5
Execution assurance	`N9` actuator_supervisor	1

Of those 21, the subset that B4 alone requires is small and worth naming, because it is the part of the island that must keep working when everything else has stopped being trustworthy: `N4` ego state · `N3` state machine · `N8` fault manager · `N17` sensor health (to *declare* the blackout) · `I7` mrm_handler · `I6` emergency-stop operator · `I1` gate · `I16` vehicle interface · `I5` to secure at standstill — **nine nodes, none of which consume an exteroceptive topic.** That is the property to hold on to: the B4 path shares no input with the sensing chain whose failure triggers it.

Recommended additions (+3): `I9` + `I12` + `I11`. The MRM needs a longitudinal law. Reusing Autoware's PID longitudinal controller — which means keeping the follower node and its plugin base — is preferable to writing one, for the same provenance reason the whole design rests on. **Note that `I10`**

`mpc_lateral_controller` is *not* in this set: the pull-over MRM does not need the MPC. That is the two-tier argument of §4.2 arriving at its conclusion.

So: **21 mandatory, 24 as recommended, against 37 in the full design.**

The §3.5 object option adds **one node and zero mandatory ones**: `N20` sits outside all three counts, and the nine-node B4 path above is unchanged by it. That is the arithmetic test the option was designed to pass.

What is deliberately excluded, and the risk accepted

A minimal set is only useful if the omissions are argued rather than overlooked.

Excluded	Why it is safe to omit *for MRM execution*	Risk accepted
`I10` mpc_lateral_controller	Nominal comfort controller. The MRM tracks a bounded path with pure pursuit.	Ride quality during the manoeuvre. Irrelevant.
`I8` control_validator, `N10` envelope_monitor	`N7` applies hard lateral-acceleration, steering-rate and jerk bounds at generation time.	Loss of the **guardian** role — the island can still fall back, but can no longer veto a *healthy* HPC. Restores in phase 2.
`N5` corridor_monitor	Pull-over deliberately leaves the lane; lane-departure checking must be suppressed during the manoeuvre in any case.	No lateral containment check during nominal driving.
`N12` time_quality_monitor	After HPC loss the gPTP master may be gone. The island runs on its local clock and age checks become local.	Cross-domain freshness claims are unavailable. Acceptable, since by then there is no cross-domain traffic.
`N11` event_recorder	Not needed to *execute*.	**Graded-reaction step 4 (audit) is unmet.** Required before any pilot, not before a bench demo.
`I14` shift_decider	`I5`'s `enable_auto_parking` covers the gear-P transition.	none
`I15` raw_vehicle_cmd_converter	Only needed for vehicles with pedal/-torque interfaces. **Not needed on the NTU golf cart**, whose interface is steering angle + velocity.	Vehicle-specific; re-add per platform.
`N13` scan_guard, `I20` collision_detector, `N15` proximity_guard	Rungs A and B have no ultrasonic (D8 ranks it third, below the design point) and no 2D lidar (optional). `I19` runs on `N6`'s radar point cloud instead.	No near-field guard for the final metres and standstill release , and no second free-space channel behind the camera's refuge. Pair with the "stop within the shoulder" terminal condition.
— *(no exclusion)*	B4 is unaffected by every row above: it consumes no exteroceptive topic.	none — which is the point of declaring it. Each omission narrows what the island can *do*, never what it can *stop doing*.
`N20` object_digest	Optional by construction (§3.5). It enriches B1–B4 and gates none of them.	The island loses the HPC self-consistency veto and the conservative object term. **No capability is lost** — which is the test the option was designed to pass.
`N19` odd_monitor	Not required to execute an MRM.	The §6.4 feature is simply absent.

9.5 Two design branches: Least-Cost and Maximum-Safety

D10 carries the design in two branches. They share everything in §§3–8: the six behaviours, the node set, the interface contract and the state machine. What differs is the hardware fitted to the island: which D8 sensors, and which processor. Because a behaviour may *require* only the camera, both branches deliver all six behaviours. What changes from branch to branch is the speed, the ODD and the evidence each behaviour can claim. This section states those adjustments. Where a behaviour's row in §3.1 names a sensor that a branch does not carry, the branch's adjustment below takes precedence.

Hardware of each branch

	Least-Cost	**Maximum-Safety**
Intent	The cheapest island that still delivers all six behaviours, using low-cost sensors only	As much sensing as the island's processor can carry
Forward camera (rank 1)	1 — low-resolution mono, on 100BASE-T1 automotive Ethernet	1 — on GMSL / CSI-2, at a higher resolution than Least-Cost
Radar (rank 2)	**none**	**4** — two at the front corners, with overlapping forward fields, and one at each rear corner
Ultrasonic (rank 3)	5	5
IMU / GNSS (rank 4)	1	1
2D lidar (rank 5, optional)	**1**	none
Processor	One of NXP **S32Z2**, Infineon **AURIX TC4x** (TriBoard), ST **Stellar** (evaluation board)	Renesas **R-Car X5H**
Sensor rung (§9.4)	Off the ladder: rungs A, C, D and E **without B**. No radar, so neither the highway class nor rear-corner awareness is available	Rung D, with corner radar on **both** sides rather than only the pull-over side
Island ODD it can claim	City-road class, about 30 km/h fallback speed, **fair weather**	Highway class (about 60 km/h fallback speed) and city-road class, **weather included**

Two placement proposals apply to both branches. Both are open decision 21(c).

- **The five ultrasonic sensors:** three across the front bumper, and two on the pull-over (kerb) side, one forward and one rearward of the wheelbase. The front three guard creep and the end of B3; the side pair watches the lateral strip that the vehicle moves into.
- **"Two radars at the front"** is read as two corner units, not one long-range unit plus one corner unit. Two corner units give two independent forward range channels, which §6.4's visibility test can cross-check, together with cut-in coverage. If the WG means one long-range forward unit, the highway case gains range and loses the second instance (open decision 23(a)).

Why Least-Cost is a city-road branch

Without radar, Least-Cost's forward range comes from the camera and the 2D lidar. A low-cost single-plane scanner reaches a few tens of metres on dark targets, less in rain. The table below sets that against the distance the island needs in order to stop for an obstacle. The figures assume 0.5 s from detection to brake, including the §9.2 budget, and 4 m/s^2 of obstacle braking.

Speed	**Distance to stop for an obstacle**	**Low-cost 2D lidar (≈ 20–30 m)**	**Front radar (≥ 70 m)**
30 km/h	≈ 13 m	**sufficient**	sufficient
60 km/h	≈ 43 m	**insufficient**	sufficient
100 km/h	≈ 110 m	insufficient	sufficient for a long-range unit

So the Least-Cost island can stand behind a fallback speed of about 30 km/h and no more. That caps the **vehicle's** ODD, not only the island's. B2 must reach `v_odd` inside its hold window. From 50 km/h to 30 km/h at 2 m/s^2 takes 2.8 s, which fits. From 100 km/h to 30 km/h takes 9.7 s and 176 m, which overruns an 8 s window before the HPC has had any time to recover. **A Least-Cost vehicle should therefore have an ODD limited to roads of about 50 km/h or less**: shuttles, campuses, city roads, and the NTU golf cart.

Weather sets the same limit from a second direction. The camera and the 2D lidar both lose range in fog, heavy rain and spray, and they lose it together. The island can therefore *detect* that visibility has failed, but it cannot *measure* how far it can still see, so it cannot back a reduced-visibility domain. The Least-Cost island ODD excludes fog and heavy rain (open decision 22).

Behaviours on each branch

	Least-Cost	**Maximum-Safety**
B1 — **ODD switch**	**Narrowed.** The island can verify only a target domain inside its own envelope: city-road class, fair weather. Visibility is estimated by the camera contrast test plus a **lidar-versus-camera** divergence test in place of radar-versus-camera. Because those two sensors fail together, the test can show that visibility has been lost but cannot confirm that a degraded domain is still satisfiable. If the HPC declares a domain the island cannot verify, such as fog at reduced speed, that declaration fails verification, `T_odd` runs out, and the successor is chosen as in §3.2. Friction is verified on the island IMU (rung D)	**Full.** Two **radar-versus-camera** instances, one per front radar, so that a single radar fault cannot mask a loss of visibility. Highway-class and reduced-visibility domains can be verified, because radar still measures range in fog. Friction is verified on the island IMU
B2 — **lane hold**	`v_odd` is **city-road class only** (about 30 km/h); there is no highway class to fall to (see above). The corridor comes from the camera; the lidar may narrow it but never create it. AEB (`I19`) runs on `N13`'s lidar `PointCloud2`. With no rear-facing sensor, the descent runs on the fixed fallback profile (≤ 2 m/s^2) with hazard lamps on at entry. `T_hold` is 8–15 s; the 2.8 s descent leaves most of the window to the HPC	Highway `v_odd` (about 60 km/h) is reachable, because the front radars supply the range. The **rear radars watch closing traffic during the descent**, the most exposed part of the hold (100 to 60 km/h is 5.6 s and 124 m). It is proposed that they may choose the descent rate within a declared band but never push the arrival at `v_odd` past its deadline, and an obstacle ahead always overrides (open decision 23(c)). The front-corner radars add cut-ins to `N16` as hazards. AEB runs on `N6`

	Least-Cost	**Maximum-Safety**
B3 — **pull over**	**Always at or below about 30 km/h**, which is rung A's regime (`v_entry` is already at or below `v_odd`). The camera finds the refuge. With no rear-corner radar, the manoeuvre relies on a long indicator dwell and a very low lateral rate, and the **side ultrasonic pair** watches the strip being entered: it may disqualify the space and abort the manoeuvre, never approve one. The lidar confirms the physical edge. The ultrasonics allow a tight-to-kerb end. The vehicle pulls over to the **instrumented side only**	Above 30 km/h, up to the highway `v_entry`. **Rear-corner radar on both sides**, so a refuge `H8` nominates on either side can be used; an island-detected refuge stays on the kerb side (open decision 23(b)). The front-corner radars check that the space ahead is free (for example, no vehicle already stopped on the shoulder). The ultrasonics allow a tight-to-kerb end on the kerb side
B4 — **emergency stop**	**Unchanged**: the trigger is the loss of the camera. A lidar that is still working keeps `I19` armed and may steepen the ramp for a target in path, and nothing else. The ultrasonics guard the final metre	**Unchanged.** Radars that are still working keep `I19` armed and may steepen the ramp. **The rear radars do not soften B4**: a surviving sensor may only make B4 more conservative (§3.5), and braking less hard because of a vehicle behind is not more conservative
B5 — **tele-operation**	The stream is the camera (required), plus the lidar scan and ultrasonic clearances; there are no radar tracks. **Forward driving only**: nothing on the island looks backwards, so reversing under B5 is not permitted. The video is encoded **in the camera**, as a second stream that `N22` forwards together with the island's metadata, because none of the three microcontrollers is offered with a hardware video encoder. `v_teleop` is lower, because the picture is low resolution	The stream is the camera plus **all four radars' tracks**, giving the remote driver all-round awareness, plus ultrasonic clearances. **Reversing at creep speed is permitted** on the rear radars, with the last metre unguarded within the radars' minimum range. The video is encoded on the X5H's application cluster, physically separate from the safety cluster
B6 — **joystick**	Unchanged (rung 0). The front ultrasonics guard forward creep; reversing relies on the operator's eyes alone	Unchanged (rung 0). The radars and ultrasonics stay armed as guards in both directions

Node set on each branch

Node	**Least-Cost**	**Maximum-Safety**
`N6` radar_guard	**absent**	present, four radars (front pair, rear pair)
`N13` scan_guard	**present**: the AEB channel, and a confirming channel for the corridor and the refuge	absent
`N14` vision_guard	Ethernet camera, low resolution	GMSL / CSI-2 camera, higher resolution
`N15` proximity_guard, `I20`	present	present
`N19` odd_monitor	camera contrast + lidar-versus-camera; fair weather only	camera contrast + two radar-versus-camera instances
`I19` AEB input	`N13` lidar	`N6` radar
`N22` teleop_streamer	forwards the camera's own encoded stream	encodes on the application cluster
The §9.4 minimum of 21	21, with `N13` in place of `N6`	21 as listed

The nine-node B4 path is the same on both branches. It consumes no exteroceptive topic, so the choice of sensors cannot affect it.

Processing load and where it runs

The load is dominated by one node. The table is an order-of-magnitude estimate for sizing, assuming about 80 operations per pixel for classical lane, road-edge and free-space extraction (inverse perspective mapping, gradient, line fitting, column scan). Part 2 replaces these figures with measurements.

Workload	Least-Cost	Maximum-Safety
`N14` camera, classical	640 ? 400 mono, 15 Hz → ≈ **0.3 Gop/s**	≈ 1.3 MP, 20 Hz → ≈ **2 Gop/s**
`N13` lidar, ~1080 points at 40 Hz	≈ 5 Mop/s	—
`N6` radar, 4 ? 64 tracks at 20 Hz	—	< 10 Mop/s
`N15`, `N16`, `N17`, `N19`	tens of Mop/s	tens of Mop/s
Camera ingest	raw ≈ 31 Mbit/s over 100BASE-T1	serial link into the SoC's camera input
B5 video encoding	in the camera	hardware codec on the SoC (to be confirmed for X5H)

Least-Cost: what the three microcontrollers can carry. On every candidate, P0 and P1 run on lockstep cores. The question is where `N14`'s pixel work runs.

Candidate	Published resources	Where `N14` runs	Verdict
NXP S32Z2	8 ? Cortex-R52 at up to 1 GHz with Neon, configurable as lockstep or split clusters; a 25 GFLOPS DSP/ML processor; 19 MB on-chip SRAM; LPDDR4	Pixel stages on the DSP or on Neon; line fitting on an R52 core. Uses well under one core-equivalent	**Recommended.** The most headroom, and already the S32Z-class target nano-ros is verified for (§9.3)
Infineon AURIX TC4x	Up to 6 ? TriCore 1.8 at 500 MHz with lockstep; on TC4Dx, a PPU (an ARC EV71 vector DSP); 10 MB SRAM	On the **PPU**. Without it, 0.3 Gop/s takes most of one scalar core	Acceptable **with the PPU**. The safety integrity of the PPU path still has to be argued
ST Stellar (SR6P)	Cortex-R52+ cores at 400–500 MHz, depending on the part; no vector accelerator in the published material	On an R52+ core, near saturation	The tightest. Pair it with a **certified smart camera** (open decision 8(b)), so that `N14` becomes a bus parser

A MIPI CSI-2 receiver was not found in the published material for any of the three. A GMSL camera would therefore need a bridge, so the Least-Cost branch specifies an automotive Ethernet camera instead (open decision 21(b)). The frame buffers, about 1 MB for raw, rectified and gradient images, fit in on-chip SRAM on all three.

Maximum-Safety: what the X5H changes. The X5H's published resources are 6 ? Cortex-R52 in dual lockstep (over 60k DMIPS, ASIL D), 32 ? Cortex-A720AE (over 1,000k DMIPS), an NPU of up to 400 TOPS, and a GPU. For this island, processing capacity stops being the limit, and the two-tier split of §4.2 maps directly onto the hardware:

X5H cluster	Hosts
Cortex-R52, lockstep (ASIL D)	P0, P1 and P2: the whole safety chain, including `N14` at the higher resolution (≈ 2 Gop/s, within reach of two of the six cores with Neon), `N6` for four radars, and `N15`–`N17`, `N19`, `I19` and `I20`
Cortex-A720AE (application)	P3, the QM nominal controller (MPC), bounded by the ASIL-D gate `I1` on the R52 cluster, which is exactly §4.2's two tiers; P4 and `N11`; `N21` and `N22`, which §6.5 already wanted in a separate low partition
NPU	Unused by the branch definition. A learned detector would be admissible only under §3.5's conservative-only rule (it may add a hazard, never create a corridor or a validity flag), and §6.2's selection principle (processable without a neural network) still governs the safety chain

With capacity no longer binding, the limits on Maximum-Safety are elsewhere:

- **The safety cluster's share.** Everything that decides runs on the six R52 cores.
- **I/O.** Four radars need four CAN FD channels.
- **Independence.** The island must not use the same SoC family as the HPC, because a shared silicon or software defect is exactly the common cause that R7 exists to avoid. Freedom from interference between the two clusters has to be argued in Part 4 (open decision 23(d)).
- **Optional additions.** If headroom is to be spent, the first candidate is camera resolution and frame rate, which extends lane-detection range and so raises `v_corridor`. The second is the optional 2D lidar (rung E), which adds confidence but no capability.

> ==**What separates the two branches is a single sensor class.** Every Least-Cost restriction above (city class only, fair weather only, B3 at or below 30 km/h on one side, no reversing under B5) follows from having no radar, not from the microcontroller. One forward radar is the cheapest addition that would change the most: it restores the highway range, weather robustness and §6.4's radar-versus-camera test. That makes it the first upgrade to price if Least-Cost proves too narrow (open decision 22).

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10 Diagrams

10.1 Node allocation across the two domains

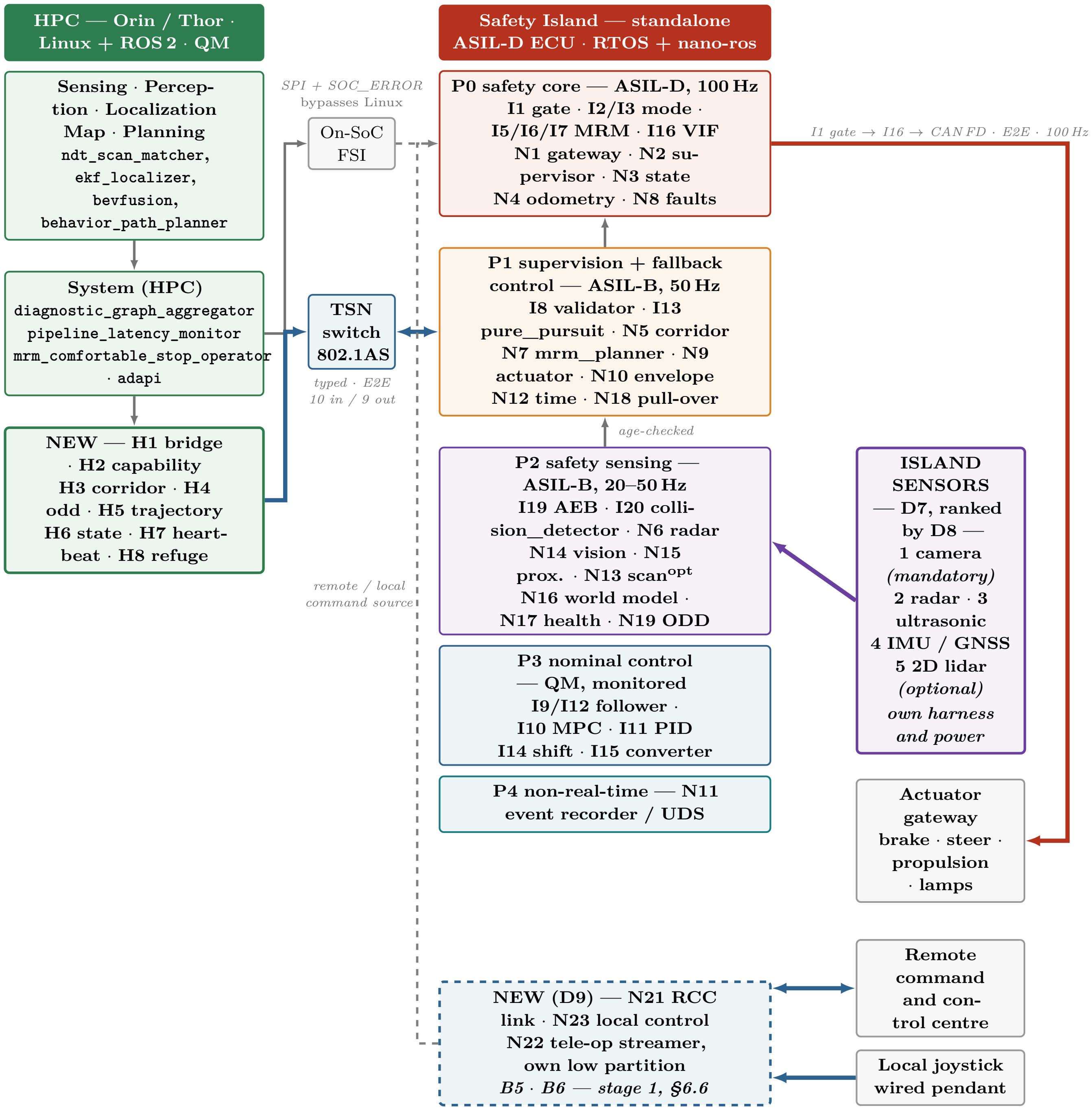

HPC — Orin / Thor · Linux + ROS 2 · QM
Sensing · Perception · Localization Map · Planning ndt_scan_matcher, ekf_localizer, bevfusion, behavior_path_planner
System (HPC) diagnostic_graph_aggregator pipeline_latency_monitor mrm_comfortable_stop_operator · adapi
NEW — H1 bridge · H2 capability H3 corridor · H4 odd · H5 trajectory H6 state · H7 heartbeat · H8 refuge
SPI + SOC_ERROR bypasses Linux
On-SoC FSI
TSN switch 802.1AS
typed · E2E 10 in / 9 out
remote / local command source
Safety Island — standalone ASIL-D ECU · RTOS + nano-ros
P0 safety core — ASIL-D, 100 Hz I1 gate · I2/I3 mode · I5/I6/I7 MRM · I16 VIF N1 gateway · N2 supervisor · N3 state N4 odometry · N8 faults
I1 gate → I16 → CAN FD · E2E · 100 Hz
P1 supervision + fallback control — ASIL-B, 50 Hz I8 validator · I13 pure_pursuit · N5 corridor N7 mrm_planner · N9 actuator · N10 envelope N12 time · N18 pull-over
age-checked
P2 safety sensing — ASIL-B, 20–50 Hz I19 AEB · I20 collision_detector · N6 radar N14 vision · N15 prox. · N13 scan^opt N16 world model · N17 health · N19 ODD
ISLAND SENSORS — D7, ranked by D8 — 1 camera (mandatory) 2 radar · 3 ultrasonic 4 IMU / GNSS 5 2D lidar (optional) own harness and power
P3 nominal control — QM, monitored I9/I12 follower · I10 MPC · I11 PID I14 shift · I15 converter
P4 non-real-time — N11 event recorder / UDS
Actuator gateway brake · steer · propulsion · lamps
NEW (D9) — N21 RCC link · N23 local control N22 tele-op streamer, own low partition B5 · B6 — stage 1, §6.6
Remote command and control centre
Local joystick wired pendant

Figure 15. *Node allocation across the two domains. Two sensor sets, two harnesses, two power rails.*

Purple is the sensing chain added by D7. Two sensor sets, two harnesses, two power rails: the vehicle is perceived twice, at very different resolutions, by two domains that share no failure mode. The only paths that survive HPC loss are those originating inside the island boundary — and after D7 that now includes a view of the road ahead.

The dashed blue box is D9's communication module and local control port (§6.5). It is drawn inside the island boundary on purpose: the remote driver of B5 and the operator of B6 reach the gate without passing through the HPC, so both survive its loss. It is dashed because the WG has not yet accepted their stage-1 design (§6.6).

10.2 The island sensing chain

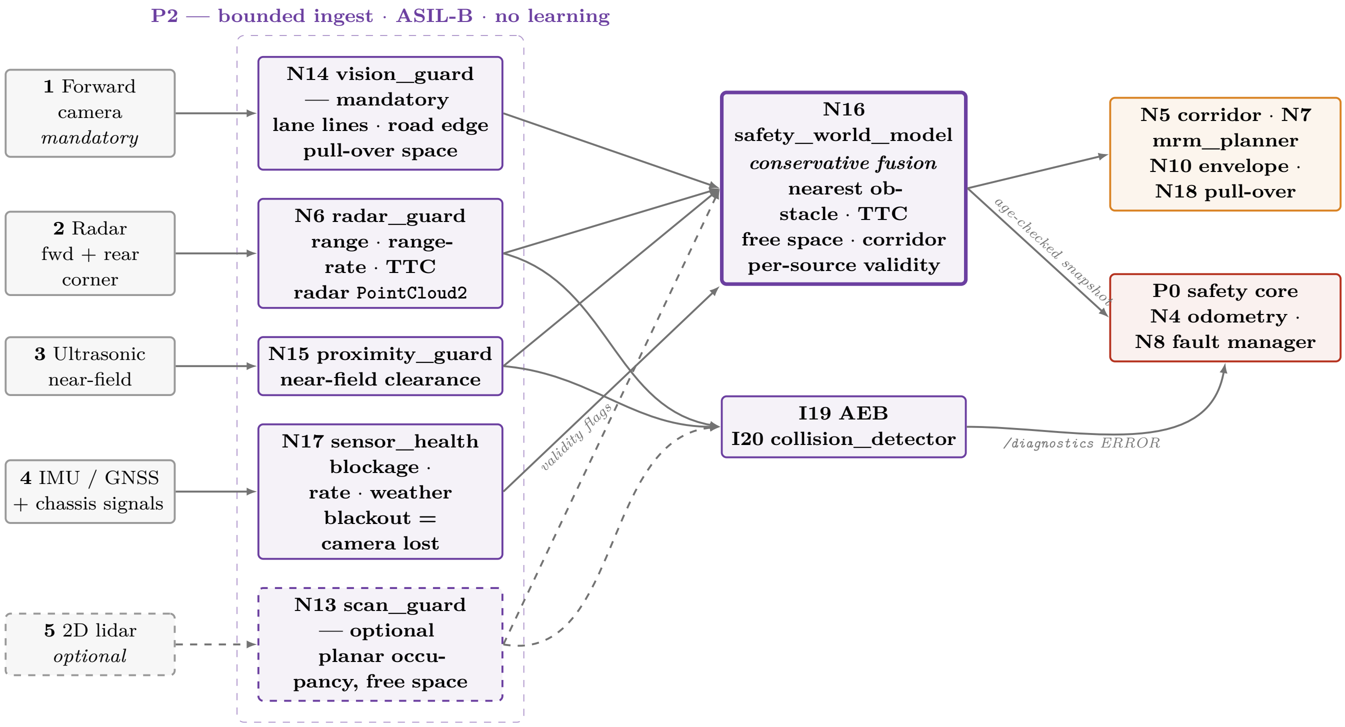

P2 — bounded ingest · ASIL-B · no learning
1 Forward camera mandatory
2 Radar fwd + rear corner
3 Ultrasonic near-field
4 IMU / GNSS + chassis signals
5 2D lidar optional
N14 vision_guard — mandatory lane lines · road edge pull-over space
N6 radar_guard range · range-rate · TTC radar PointCloud2
N15 proximity_guard near-field clearance
N17 sensor_health blockage · rate · weather blackout = camera lost
N13 scan_guard — optional planar occupancy, free space
N16 safety_world_model conservative fusion nearest obstacle · TTC free space · corridor per-source validity
I19 AEB I20 collision_detector
N5 corridor · N7 mrm_planner N10 envelope · N18 pull-over
P0 safety core N4 odometry · N8 fault manager
validity flags
age-checked snapshot
/diagnostics ERROR

Figure 16. *The island sensing chain. Camera mandatory, radar second, ultrasonic third, IMU / GNSS fourth; the dashed 2D lidar is optional. Bounded ingest, conservative fusion, an age-checked snapshot.*

Solid boxes and edges are the ranked sensors; dashed ones exist only where D8's optional 2D lidar is fitted. The ultrasonic array, ranked third, is solid but arrives at rung C: it adds `N15`'s near-field clearance and lets `autoware_collision_detector` run on its `PointCloud2`. Read the solid part alone and it is still complete: the camera feeds the corridor and the lateral free space, radar feeds the in-path range and — as a `PointCloud2` from `N6` — `autoware_autonomous_emergency_braking`, unmodified. Where a 2D lidar is fitted, `N13` adds a bounded `PointCloud2` to the same migrated nodes, *and* feeds `N16` for the fused world state that the supervision tier consumes — as a confirming source only. The safety core reads `N16` only through an age-checked snapshot, so a stall anywhere in this diagram degrades the MRM to blind deceleration rather than blocking it.

The two blue dash-outlined boxes are §3.5's option, drawn that way because they may simply not be there. Note what is **absent**: there is no edge from the digest to `N17`, and its box says so. The digest is excluded by construction from the blackout determination, so a live HPC publishing objects can never hide the fact that the island itself has gone blind. Note also which consumers it *does* reach — `N16`, where it can only add a hazard, and `N10`, where it powers the self-consistency veto. It reaches no node on the P0 path at all.

10.3 Failover sequence — HPC loss during autonomous driving

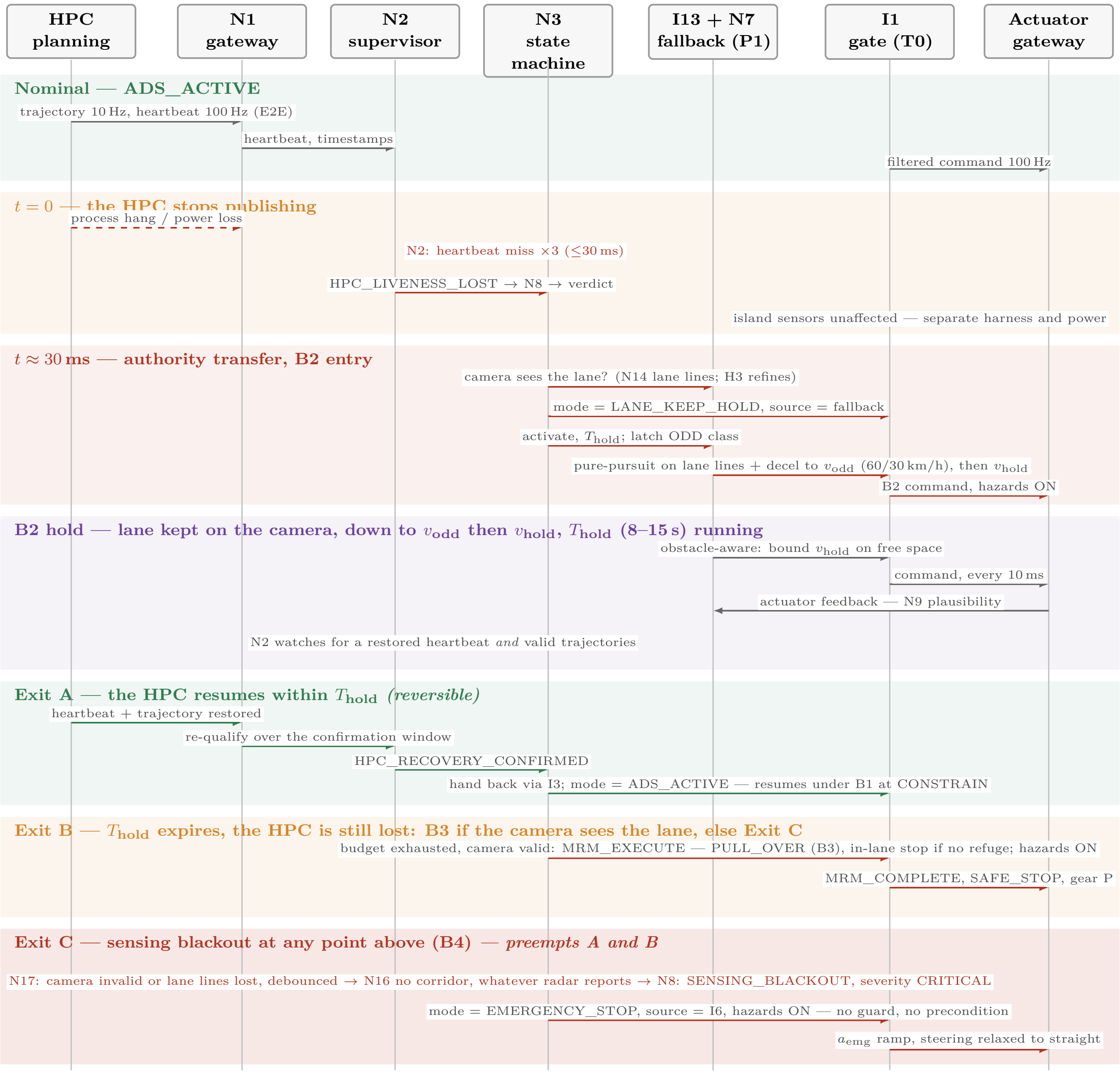

HPC planning
N1 gateway
N2 supervisor
N3 state machine
I13 + N7 fallback (P1)
I1 gate (T0)
Actuator gateway
Nominal — ADS_ACTIVE
trajectory 10 Hz, heartbeat 100 Hz (E2E)
heartbeat, timestamps
filtered command 100 Hz
t = 0 — the HPC stops publishing
process hang / power loss
N2: heartbeat miss ×3 (≤30 ms)
HPC_LIVENESS_LOST → N8 → verdict
island sensors unaffected — separate harness and power
t ≈ 30 ms — authority transfer, B2 entry
camera sees the lane? (N14 lane lines; H3 refines)
mode = LANE_KEEP_HOLD, source = fallback
activate, T_{hold}; latch ODD class
pure-pursuit on lane lines + decel to v_{odd} (60/30 km/h), then v_{hold}
B2 command, hazards ON
B2 hold — lane kept on the camera, down to v_{odd} then v_{hold}, T_{hold} (8–15 s) running
obstacle-aware: bound v_{hold} on free space
command, every 10 ms
actuator feedback — N9 plausibility
N2 watches for a restored heartbeat *and* valid trajectories
Exit A — the HPC resumes within T_{hold} *(reversible)*
heartbeat + trajectory restored
re-qualify over the confirmation window
HPC_RECOVERY_CONFIRMED
hand back via I3; mode = ADS_ACTIVE — resumes under B1 at CONSTRAIN
Exit B — T_{hold} expires, the HPC is still lost: B3 if the camera sees the lane, else Exit C
budget exhausted, camera valid: MRM_EXECUTE — PULL_OVER (B3), in-lane stop if no refuge; hazards ON
MRM_COMPLETE, SAFE_STOP, gear P
Exit C — sensing blackout at any point above (B4) — *preempts A and B*
N17: camera invalid or lane lines lost, debounced → N16 no corridor, whatever radar reports → N8: SENSING_BLACKOUT, severity CRITICAL
mode = EMERGENCY_STOP, source = I6, hazards ON — no guard, no precondition
a_{emg} ramp, steering relaxed to straight

Figure 17. *Failover sequence — HPC loss during autonomous driving. Time runs downwards; Exit C can preempt the other two.*

Exit C is drawn last but is not last in precedence: it can preempt Exits A and B at any moment, including mid-hand-back. Note what is *absent* from it — no `N16` query, no corridor lookup, no `N18` refuge check. B4 is the only path in this diagram that completes without consulting a sensing node, which is what allows it to run when the sensing nodes are the thing that failed.

The critical span is detection to authority transfer: three missed heartbeats at 100 Hz, then two decision cycles. **Nothing in the shaded transfer band reads a topic the HPC produces** — the ODD class is latched from the last `active_odd` the island had already verified and buffered, not requested from an HPC that is gone, and a missing one selects the most restrictive class — and the `alt` branch on world-state validity is the §2.2 staleness rule made explicit. Island perception improves the outcome; its absence never blocks the transfer.

The exits are the point of the diagram. HPC loss does not commit the vehicle to a terminal manoeuvre: it commits it to **B2**, a bounded lane-keeping hold, from which Exit A returns authority to a re-qualified HPC and Exit B forks — to B3 when the camera still sees the lane, to Exit C (B4) when it does not; there is no third way out of the hold. Within the window, the first seconds go to bringing the vehicle down to `v_odd` (§3.1); the rest is the HPC's chance to return. Note what Exit A does *not* do — it does not accept the first heartbeat that reappears, and it does not return to `ALLOW`. Note also that the corridor check sits at B2's entry, not inside it: an island that cannot see the lane never enters the hold at all — it goes to Exit C, which is B4, directly.

10.4 Command-source arbitration inside the island

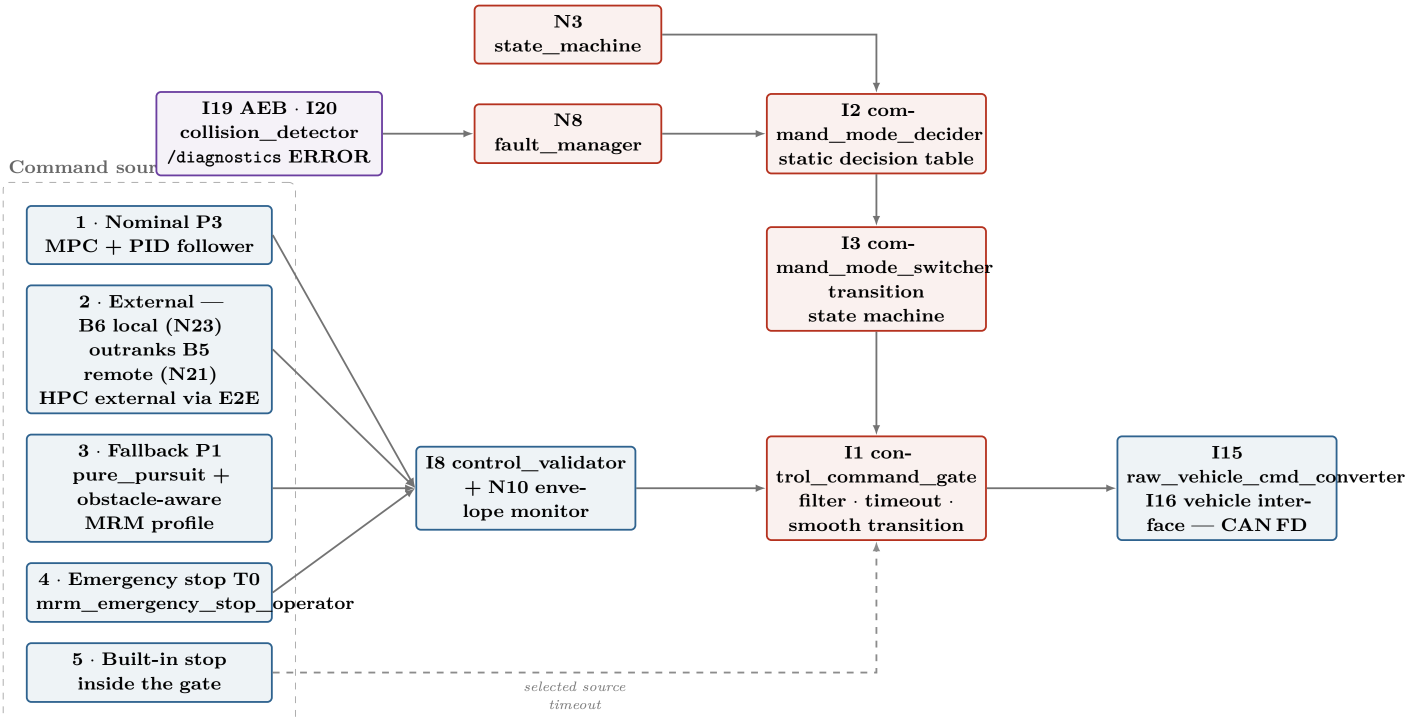
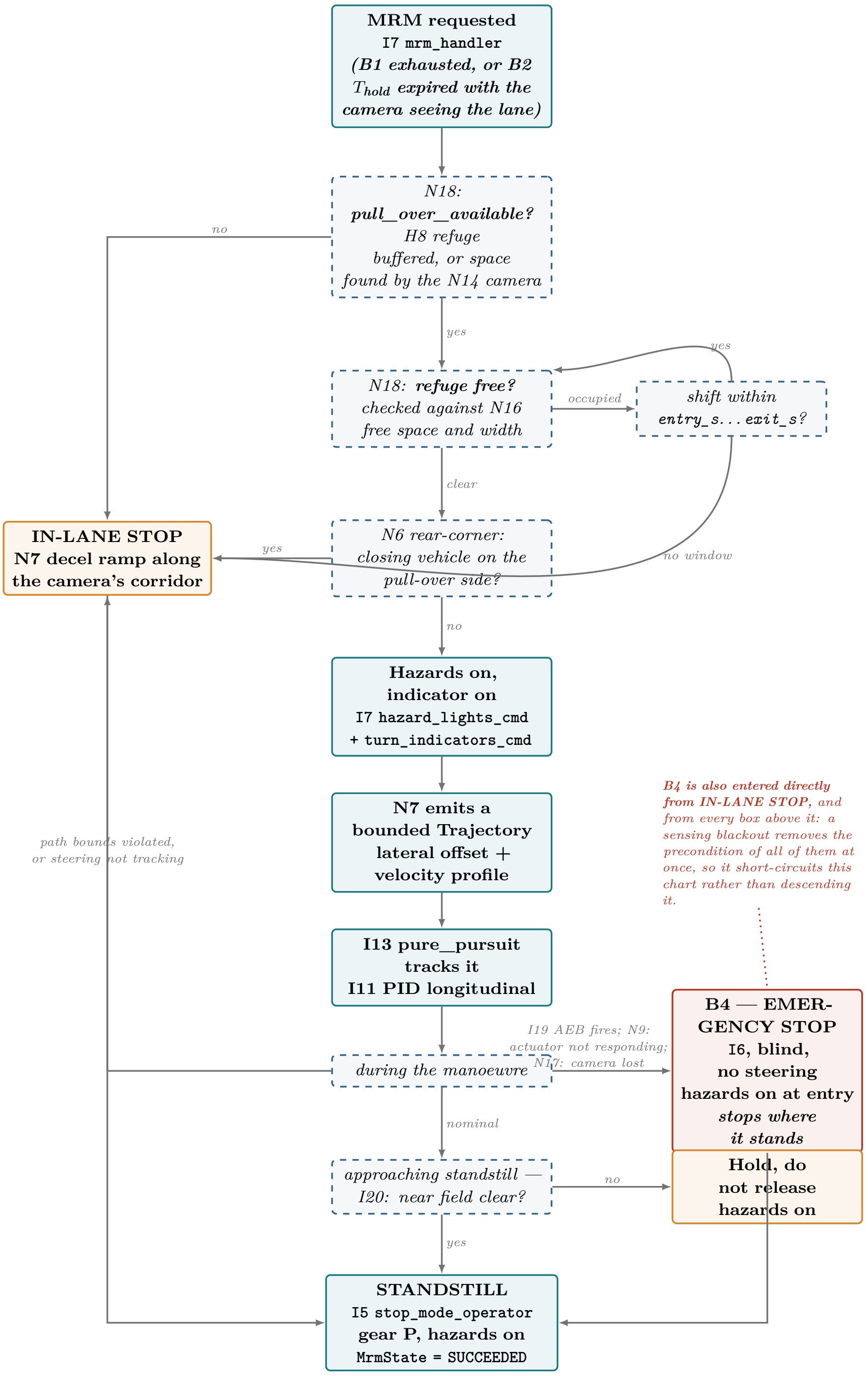

N3
state_machine
I19 AEB · I20
collision_detector
/diagnostics ERROR
N8
fault_manager
I2 command_mode_decider
static decision table
Command sour
1 · Nominal P3
MPC + PID follower
2 · External —
B6 local (N23)
outranks B5
remote (N21)
HPC external via E2E
I3 command_mode_switcher
transition
state machine
3 · Fallback P1
pure_pursuit +
obstacle-aware
MRM profile
I8 control_validator
+ N10 envelope monitor
I1 control_command_gate
filter · timeout ·
smooth transition
I15
raw_vehicle_cmd_converter
I16 vehicle interface — CAN FD
4 · Emergency stop T0
mrm_emergency_stop_operator
5 · Built-in stop
inside the gate
selected source
timeout

Figure 18. *Command-source arbitration inside the island. Rank order is fixed at build time.*

Rank order is fixed at build time. Every source except the gate's own built-in stop passes through validation; the built-in stop exists precisely because it must work when validation's inputs are gone. `I19` and `I20` do **not** appear as command sources — they are fault sources. Island perception escalates through the fault manager and the state machine, never straight to the actuators.

10.5 Pull-over decision and abort ladder


MRM requested
I7 mrm_handler
(B1 exhausted, or B2 T_{hold} expired with the camera seeing the lane)
N18: pull_over_available? H8 refuge buffered, or space found by the N14 camera
no
yes
N18: refuge free? checked against N16 free space and width
occupied
shift within entry_s. . . exit_s?
yes
no window
clear
N6 rear-corner: closing vehicle on the pull-over side?
yes
IN-LANE STOP
N7 decel ramp along the camera's corridor
no
Hazards on, indicator on
I7 hazard_lights_cmd + turn_indicators_cmd
N7 emits a bounded Trajectory lateral offset + velocity profile
I13 pure_pursuit tracks it
I11 PID longitudinal
path bounds violated, or steering not tracking
during the manoeuvre
I19 AEB fires; N9: actuator not responding; N17: camera lost
B4 — EMERGENCY STOP
I6, blind, no steering
hazards on at entry
stops where it stands
nominal
approaching standstill — I20: near field clear?
no
Hold, do not release hazards on
yes
STANDSTILL
I5 stop_mode_operator
gear P, hazards on
MrmState = SUCCEEDED

B4 is also entered directly from IN-LANE STOP, and from every box above it: a sensing blackout removes the precondition of all of them at once, so it short-circuits this chart rather than descending it.

Figure 19. *Pull-over decision and abort ladder. Every branch off the happy path lands on a simpler manoeuvre — the last of them is B4.*

Every branch out of the happy path lands on a *simpler* manoeuvre. Pull-over degrades to in-lane stop, in-lane stop degrades to B4 emergency stop, emergency stop degrades to the gate's built-in stop. The manoeuvre never escalates in complexity under fault — which is what makes attempting the harder one acceptable. Note the two edges into B4 that do not pass through the pull-over logic at all: a sensing blackout removes the *precondition* of every box above it at once, so it short-circuits the ladder rather than descending it.

The `SEARCH` box is where the island produces a stopping place of its own rather than verifying one, and its three outcomes are not interchangeable. **Found** rejoins the ordinary path at `VERIFY` — an island-detected clearance is verified free exactly as a nominated refuge is, because provenance changes what may be *claimed* about a refuge, never what must be *checked*. **Positively none** descends one rung to the in-lane stop, which still has a corridor. **Inconclusive** is the only one that reaches B4, and it does so for the same reason the blackout edge does: a search that cannot conclude is a statement about the island's own capability.

10.6 ODD degradation ladder

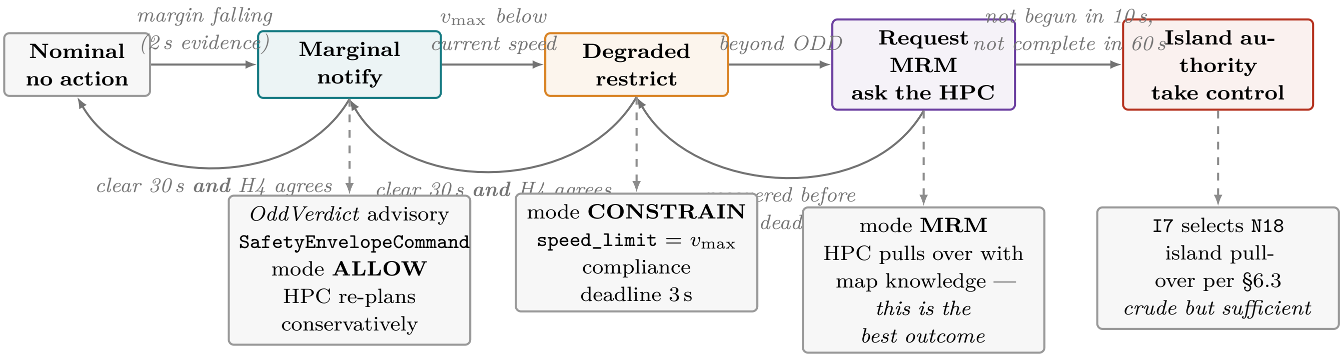

Nominal
no action
margin falling
(2 s evidence)
Marginal
notify
$v_{\max}$ below
current speed
Degraded
restrict
beyond ODD
Request
MRM
ask the HPC
not begun in 10 s,
not complete in 60 s
Island authority
take control
clear 30 s and H4 agrees
clear 30 s and H4 agrees
ered before
dead
OddVerdict advisory
SafetyEnvelopeCommand
mode ALLOW
HPC re-plans
conservatively
mode CONSTRAIN
speed_limit = $v_{\max}$
compliance
deadline 3 s
mode MRM
HPC pulls over with
map knowledge —
this is the
best outcome
I7 selects N18
island pull-over per §6.3
crude but sufficient

Figure 20. *The ODD degradation ladder. Every rung is reversible until authority is taken.*

Time runs in seconds here, not milliseconds. The escalation is **deliberately reluctant**: the island gives the HPC four chances to handle its own ODD exit, because a planned pull-over with map knowledge beats the island's geometric one every time. Only non-compliance — which the island detects from vehicle status alone, needing no HPC cooperation — moves authority.

`SwitchDemanded` is the rung that carries the drive forward rather than down. Every other edge in this diagram either holds the vehicle where it is or moves it closer to a stop; the edge from `SwitchDemanded` back to `Nominal` is the one that returns the vehicle to normal L4 operation — in a narrower domain than it left. Note that it returns to `Nominal` and not to `Marginal`: once a domain has been declared, verified and held, the vehicle is no longer operating outside its envelope. The envelope changed.

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11 Open decisions for the WG

These change the allocation and should be settled before Part 2.

1. **Target or legacy command architecture (§4.3).** The recommendation is the target set (`control_command_gate` + `command_mode_decider/switcher` + `stop_mode_operator`). This must be pinned to a specific Autoware release, because the existing ASI port targets Humble-era interfaces.
2. **Two-tier controller (§4.2).** Does the WG accept a QM-rated MPC

bounded by an ASIL-D gate, or must the entire control module reach ASIL-B/D? The answer determines whether `autoware_pure_pursuit` (I13) is in scope for phase 1.

3. **Island sensor set (D7, §6.2).** Settled in principle, and now ranked by D8: camera mandatory, radar second, ultrasonic third, IMU / GNSS fourth, 2D lidar optional. What remains is the rung each deployment commits to (§9.4) and the mounting — in particular the camera's field of view, which must cover the road edge on the pull-over side as well as the lane ahead, because B3 now finds its space with it. Each sensor added costs P2 budget and a safety manual.

4. **Corridor abstraction (H3/N5).** With N14 available the map corridor is no longer the only source, so this is now a question of *preference order* rather than necessity. The proposal is now split by who drives: while the HPC drives, map corridor first, cross-checked against the camera; once the island drives, the camera is required and the map corridor only refines it. The heading tube is withdrawn as a corridor source — its loss-of-lane case is B4. Confirm that ordering, and the age at which each source expires.

5. **Message set ownership.** `SafetyCorridor`, `HpcCapability`, `OddStatus`, `HpcHeartbeat`, `IslandState`, `SafetyEnvelopeCommand`, `SafetyWorldState`, `IslandSensorHealth`, `PullOverRefuge`, `OddVerdict`, `FallbackAvailability` are new. Proposal: a single `autoware_safety_island_msgs` package, contributed upstream so both domains generate from the same `.msg` sources — as the nano-ros port already does for wire compatibility.

6. **In-house protection profiles (§8.4).** AUTOSAR is out of scope by WG decision, so `SI-E2E/A` and `SI-E2E/B` are proposed in its place. Confirm the field widths and the CRC polynomials — and, first, whether nano-ros's built-in E2E already covers them, in which case the island inherits the middleware's verification evidence instead of writing its own.

7. **Intra-island RMW backend (§9.3).** Cyclone DDS on the HPC boundary is settled by wire compatibility. For the T0 partition, XRCE-DDS gives static-after-init allocation, zenoh-pico gives a bounded bump arena, Cyclone DDS needs a pre-sized pool. Whether the island runs one backend or two is the open question, and it is a `system.toml` decision rather than a code decision.

8. **Camera processing on the island (N14).** Two routes. *(a)* Classical CV — inverse perspective mapping plus line fitting — which is bounded, auditable, and cheap to certify, but fragile in poor lane marking and adverse light. *(b)* A certified automotive smart camera that emits lane and object lists over CAN/Ethernet, moving the perception burden and its ASIL evidence to the supplier. **(b) is recommended**: it collapses N14 to a parser, removes learning-based code from the island entirely, and buys supplier safety evidence. It costs BOM and vendor lock-in. If the WG prefers (a), the island's ODD must be narrowed accordingly. D8 raises the stakes of this decision: the camera is now the one mandatory sensor, and B3 asks of it more than lane lines — the road edge and the drivable space beside

the lane. Classical CV does the first well and the second poorly, so (a) would confine the island-detected refuge to roads with a marked edge line or a high-contrast kerb. Whichever route, the camera's output must include road-edge and lateral free-space fields, not lanes alone.

9. **Do island sensors also feed the HPC?** The design says no — one-way, sensors to island only, with the island forwarding an advisory `SafetyWorldState` upward. Confirm, because the alternative (a tap into the HPC's sensing) is tempting for cross-validation and would quietly create the reverse dependency the independence argument forbids.
10. **`MrmState` has no pull-over value (§6.3).** `autoware_adapi_v1_msgs/MrmState.behavior` carries only `NONE`, `EMERGENCY_STOP` and `COMFORTABLE_STOP`, yet `mrm_handler` already defines a `pull_over_manager` slot. Proposal: contribute a `PULL_OVER` value upstream rather than overloading `COMFORTABLE_STOP`, which would make the AD API and the operator UI report the wrong manoeuvre.
11. **Can the HPC honour the refuge coverage guarantee (§6.3)?** It must guarantee that the buffered refuge set always covers the current worst-case stopping distance, and declare explicitly when it cannot. This is a real constraint on the planner, and it should be confirmed with the planning maintainers before the interface is fixed.
12. **ODD verdict arbitration (§6.4).** The proposal is that the more restrictive of the HPC's and the island's verdict binds, and that disagreement is recorded as a SOTIF event. Confirm — the alternative, HPC-wins, removes the only mechanism that catches a confidently-wrong HPC.
13. **Island ODD declaration (§6.4).** The MRM's claimed operating domain — speed ceiling, road geometry, weather — is narrower than the vehicle's ODD and must be stated as a first-class requirement, not discovered during validation. B2's **`v_odd` table** is now part of it: the road classes and the safe speed for each — **~60 km/h highway, ~30 km/h city road** are the starting proposal — and which class applies when `active_odd` names a domain the table does not list (proposed: the most restrictive). The table is per sensor rung, since rung A cannot support the highway value.
14. **B2's hold window and the hand-back criteria (§3.1, §3.2).** Three numbers and one policy question, and they are the most consequential open items in the behaviour set. *(a)* `T_hold` — how long the island may hold the lane awaiting HPC recovery. Too short and every transient HPC hiccup becomes a pull-over on a live carriageway; too long and the vehicle loiters under a fallback controller with a narrow ODD. The range is now **8–15 s** (it was 2–5 s), because the window must contain the descent to `v_odd` — 5.6 s from 100 to 60 km/h at 2 m/s^2 — and still leave the HPC time to return (§3.1). What remains open is the value per road class within that range, and the fallback deceleration ($\leq$ 2 m/s^2 proposed) used to reach `v_odd`. *(b)* `v_hold` — the constrained speed held during B2, from the §6.4 speed law evaluated against the *island's* ODD, not the vehicle's, and capped by `v_odd` (decision 13) Now a **ramp** rather than a constant (§3.1): the open numbers are the `v_exit(t)` decay shape — linear in time, linear in remaining `T_`

`hold`, or a constant-deceleration profile — and `v_entry`, the speed the ramp must land on at `T_hold`. A ramp that is too aggressive throws away the hold's whole purpose by stopping the vehicle before the HPC has had its window; one that is too gentle leaves B3 to begin with a deceleration it should have inherited. *(c)* The **recovery confirmation window** — how long a returned HPC must publish valid, in-envelope trajectories before `N2` will qualify it. It must be long enough that a crashed-and-restarted HPC cannot re-acquire authority on its first plausible message. *(d)* The policy question: **may a recovered HPC be trusted at all within the same drive cycle?** The design says yes, under confirmation and re-entering B1 at `CONSTRAIN`, because the alternative — treating every transient loss as terminal — makes the vehicle brittle in exactly the scenarios the island exists to survive. The conservative alternative (hold, then always pull over) is defensible and cheaper to argue for in the safety case. This is a WG decision, not a design one.

15. **B4's blackout declaration and stop rate (§3.1, §6.2).** B4 is new in version 20260903 and its parameters are unset. Three numbers and one policy question. *(a)* The **blackout debounce** — how long the camera must be invalid, or the lane lines lost, before `N17` declares it. Since D8 this is a single-sensor debounce, and it is also the bridging budget over which `H3`'s corridor may carry a camera dropout. Too short and a tunnel mouth, a spray burst from an overtaking truck, or a low-sun frame triggers an emergency stop on a live carriageway; too long and the vehicle travels blind. The starting proposal is **3 consecutive `N16` cycles** (≈ 60–150 ms), bounded additionally by a hard **distance-travelled-blind** cap, because at speed the time budget and the distance budget are not the same constraint. *(b)* `a_emg` — the emergency deceleration rate. It is **not** AEB's rate: AEB brakes to avoid a detected obstacle, whereas B4 has no obstacle to stop short of, so following-vehicle risk rather than stopping distance is the dominant term. *(c)* The **steering-relaxation rate** — how fast the held steering angle is allowed to return toward straight while the vehicle decelerates. Zero means the vehicle continues its current arc; too fast is a lateral disturbance of its own. *(d)* The policy question: **must B4 also alert?** Hazard lamps are in the design already. Whether a blind stop additionally raises a remote-assistance request, a V2X hazard message, or the horn is an operational-concept decision the WG owns, and it interacts with whether the deployment assumes a remote operator at all.

16. **The optional HPC object list (§3.5, §6.2, §8.1).** New in version 20260903, and the first question is whether to take it at all. *(a)* **Adopt, or not.** For: the self-consistency veto is the cheapest guardian function in the whole design — it needs no island sensor, only `N10`/`I8` and the digest, and it catches the failure mode island perception is worst at catching, namely a *healthy* HPC planning through an object it has itself just published. Against: it is the first HPC-produced topic the island consumes for anything other than supervising the HPC, and every such input is one more argument to make in the safety case. **Recommendation: adopt at phase 2, behind the monotone-conservatism rule**, which is what makes the argument

short. *(b)* **The digest bounds** — maximum object count `K_obj`, prediction horizon `H_obj`, and the field set. These fix the worst-case message size, and therefore the R4/R5 argument. Starting proposal: the **16 highest-threat objects**, **3 s** of predicted path, and pose, extent, velocity and class only. *(c)* **The carry-over budget** — how long, and how far, the last valid digest survives HPC silence. Two bounds rather than one, for the reason decision 15 gives. Starting proposal: **500 ms or 10 m, whichever comes first.** *(d)* **The E2E profile.** SI-E2E/B is proposed on the ground that corruption can only cause *nuisance* braking, never a missed hazard. If the WG judges nuisance braking at speed to be a hazard in its own right, the row moves to SI-E2E/A. This is the sharpest of the four questions, and it is the one that tests whether the monotone-conservatism argument is believed.

17. **The ODD switch exchange (B1, §6.4).** B1 is now a negotiation rather than a constraint, and four things about it are unset. *(a)* `T_ack` and `T_odd` — **2 s** to declare a new active domain, **10 s** to be measurably in it, are starting proposals only. `T_odd` is additionally bounded from above by `t_margin` − `t_pullover` (§3.1), so what the WG is really fixing is a *ceiling* and a default, not a single number. *(b)* **Who owns the ODD catalogue.** The design says the HPC does, and that the island never names a target domain — it only rejects answers. That keeps planning off the island (R4, R5) but means the island cannot tell *no satisfiable domain exists* from *the HPC failed to find one.* Both route to B3, so nothing unsafe follows; what is lost is diagnostic resolution in the field data. *(c)* **What `active_odd`'s bounds must contain** for the island's acceptance test to be meaningful. Speed ceiling, required sensing range and assumed friction are proposed because the island measures all three. Any bound the island cannot measure is one it cannot verify, and belongs in the message only as recorded evidence. *(d)* The policy question: **may the vehicle switch domains more than once in a drive cycle?** The design does not forbid it, and repeated narrowing is the natural behaviour in worsening weather. The conservative alternative — one switch, then B3 — is easier to argue and materially less useful.

18. **The island-detected refuge and `T_refuge` (B3, §6.3).** The island now carries a second refuge source, which is the first time it produces a stopping *place* rather than verifying one. *(a)* `T_refuge` — **5 s** is proposed, with a distance-travelled-searching cap alongside it for the reason decision 15 gives. It is bounded below by how long the sensors need to establish a clearance with confidence, and above by the speed the vehicle is still carrying while it searches. *(b)* **The clearance criteria** — minimum width, minimum length, maximum lateral displacement, and what surface evidence (if any) is required — all of them now measured by the camera, so each needs a stated camera accuracy at the range the search runs at. The island cannot classify a surface, so `surface_class = UNKNOWN` is proposed, which means the criteria must be geometric alone. *(c)* **Is an `EMERGENCY_ONLY` clearance acceptable at all?** This is the real question. The island cannot tell a hard shoulder from a cycle lane, a bus stop or a driveway, so admitting the source means accepting an unlawful stopping place in preference to a stop in a live running lane. The design says yes, second in priority and never

displacing a nominated refuge, on the ground that a stopped vehicle in a running lane is the worse outcome. A WG that disagrees should say so, and the consequence is that B3 degenerates to the in-lane stop whenever `H8` is silent. *(d)* **Inconclusive versus none.** The design distinguishes a search that *positively determines* no refuge exists — which abandons to the in-lane stop, corridor still in hand — from one that runs out of `T_refuge` *without a determination*, which goes to B4. The distinction is load-bearing (§3.1) and it must be implementable: `N18` has to be able to say which of the two it reached, and a WG that judges that undecidable in practice collapses both to B4 and should say so explicitly.

19. **B5 remote driving and the RCC link (D9, §6.6, §8.6).** The stage-1 design is proposed; these items remain. *(a)* **Ratify the take-over from a moving vehicle.** B5 may now be entered from B1–B3 on the ADS's request. The numbers to ratify are `T_sa` = 5 s of situational awareness, and a matched command within ?1.5° and ?2 km/h for 0.5 s. Also to decide: whether an HPC's own request should enter B1, as proposed. *(b)* **Ratify the stage-1 numbers:** `T_link` = 300 ms; a command-age bound of 600 ms soft and 1.0 s hard; `v_cap` = 20 km/h; a design loop time of 1.8 s, including 1.2 s of driver reaction; `a_tele` = 3 m/s^2. *(c)* **Re-engaging the ADS from `SAFE_STOP`.** Proposed: allowed, on the RCC's request, once the HPC has been re-qualified over B2's confirmation window and the ODD is satisfiable. Otherwise every remote recovery ends the drive cycle, including one that followed a transient fault. *(d)* **Deferred to the next design: the RCC communication.** The module and its redundancy (two SIMs, or two modems on two carriers); who owns the certificate authority for operator and vehicle keys; the protection profile (SI-E2E/R is the proposal); the video transport; and whether zenoh-pico's TLS transport is adequate. *(e)* **Stage 2: a path-based remote-assistance mode**, in which the driver sends a short path and the island's fallback tier (N7 → I13) drives it. It tolerates latency far better than direct driving, and needs a remote-path input on N7. *(f)* **The RCC's human-factors requirements**, and the operator-to-vehicle ratio: 1 : 1 while B5 is active is proposed.

20. **B6 joystick control (D9, §6.6).** The stage-1 design is proposed; these items remain. *(a)* **Entry from motion.** Proposed: never — a moving B5 is stopped before `LOCAL` is granted. A test-track mode that allows it belongs outside this design. *(b)* **Ratify the numbers:** `v_local` = 5 km/h forward and 3 km/h in reverse; 0.5 m/s^2; `T_local` = 200 ms. *(c)* **The device — wired only in this design.** The grade of the pendant's enabling switch and emergency stop, where the CAN FD connector sits, and the cable's length. **A short-range radio link is deferred to the next design**, with IEC 62745 as its reference. *(d)* **The hardwired emergency stop** from the controller's receiver to the actuator gateway, so that the stop does not depend on island software. A Part-2 hardware item.

21. **Least-Cost hardware (D10, §9.5).** *(a)* **Which microcontroller.** The recommendation is the **NXP S32Z2**. It has the most headroom for `N14`'s pixel work (Neon plus a DSP), and it is the S32Z-class target nano-ros is

already verified for. The **AURIX TC4x** is acceptable in a variant with the PPU, provided the safety integrity of the PPU path is argued. The **Stellar** is acceptable only with a certified smart camera (open decision 8(b)). *(b)* **The camera interface.** No CSI-2 receiver was found on any of the three, so Least-Cost specifies a 100BASE-T1 Ethernet camera: raw low-resolution mono at about 31 Mbit/s, plus an encoded second stream for B5. This should be confirmed against each reference manual. *(c)* **Ultrasonic placement, on both branches.** The proposal is three sensors across the front and two on the kerb side. The side pair is what lets Least-Cost's B3 watch the strip it moves into without a rear-corner radar.

22. **Least-Cost restricts the vehicle's ODD, not only the island's (§9.5).** Without radar, the island can back a fallback speed of about 30 km/h in fair weather and nothing more. B2's descent to `v_odd` fits inside the 8–15 s window only from about 50 km/h or less. The WG is asked to confirm that a Least-Cost vehicle is limited to an ODD of about 50 km/h or less that **excludes fog and heavy rain**, or else to price the alternative: a **single forward radar**, the cheapest addition that lifts the highway range, weather robustness and §6.4's radar-versus-camera test together.

23. **Maximum-Safety details (§9.5).** *(a)* **"Two front radars"** is read as two front-corner units, which give two independent visibility-test instances and cut-in coverage. One long-range unit plus one corner unit would trade the second instance for range. *(b)* **Pulling over to either side.** Rear-corner radar on both sides makes a refuge `H8` nominates on the far side usable. The proposal is that the island's own refuge search stays on the kerb side, where the ultrasonics are mounted. *(c)* **Using the rear radars during B2's descent.** The proposal is that they may choose the descent rate within a declared band, but may never push the arrival at `v_odd` past its deadline, and that an obstacle ahead always overrides. They are deliberately *not* allowed to soften B4. *(d)* **Independence of the X5H.** The island must not use the HPC's SoC family (R7, common cause), and freedom from interference between the X5H's application cluster and its lockstep R52 cluster must be argued in Part 4.

12 Next steps

Part	Content
2	Timing and FTTI: worst-case chain latency per interface, fault-reaction budget derivation, RT contract files (`*.contract.yaml` + platform files) for both domains, checked against the nano-ros scheduling wiring matrix for the chosen target.
3	Interaction specification: full message definitions for `autoware_safety_island_msgs`, per-signal E2E configuration, QoS table, and the complete island state machine with transition guards.
4	Safety case skeleton: HARA extract, safety goals, ASIL allocation and decomposition argument, independence analysis, and the fault-injection test matrix (HPC hang, link flood, stale trajectory, E2E corruption, actuator non-response, power loss, **simultaneous blindness of every island sensor — the B4 case** · loss of the RCC link mid-B5 and of the local link mid-B6 — and, if §3.5 is adopted, a corrupted or phantom-object digest and a digest that goes silent mid-manoeuvre).

Implementation sequencing follows the existing port. The current `autoware-safety-island` on nano-ros already covers I9–I12.

Phase	**Adds**	**Capability reached**
0 — done	I9–I12	Trajectory following on the island, closed loop against unmodified Autoware
1	P0: I1–I3, I5–I7, I16, N1–N4, N8	**Autonomous fallback.** Authority transfer on HPC loss; blind but bounded MRM
2	P1: I8, I13, N5, N7, N9, N10, N12	**Guardian role.** Command validation, safety envelope, veto, two-tier control
3	P2: I19, I20, N6, N13–N17 + the island sensor harness	**Obstacle-aware MRM** per D7. The island stops for what it can see.
4	N18 + H8	**Pull-over MRM** (§6.3). The island moves the vehicle out of the running lane.
5	N19	**ODD degradation** (§6.4). Independent visibility and friction verdict, graded restriction of a healthy HPC.

The minimal configuration in §9.4 cuts across this: **phases 1, 3 and 4 reduced to their mandatory nodes deliver a pull-over-capable island in 21 nodes**, deferring the phase-2 guardian role. That is the shortest credible path to a demonstrable L4 fallback, and it is the sequence recommended for the golf-cart prototype.

The optional §3.5 object digest (N20) can be added at **phase 2**, where it serves B1's self-consistency veto with no island sensing at all, or deferred to phase 3 with the rest of the sensing chain. It is on no phase's critical path, by construction.

Phases 1 and 2 are unchanged by D7 and remain the critical path — D7 improves the *quality* of the fallback, it does not create it. Phase 3 can proceed in parallel on the sensing side, since N13–N17 have no dependency on P0 beyond the world-state contract in §2.2.

Sources: `SafetyIsland_RefDesignWG.pdf` *(Reference Design WG, 2026-08-14);* `SaftyIsland_ReferenceDesign.md; This project researches the design of safety islan.md; AutowareSW.md; autoware_universe` *package inventory,* `main` *branch, retrieved 2026-08-27;* `autoware-safety-island` *(branch* `nano-ros`*, NEWSLab NTU); AWF progress update 2026-08-20, "Porting the Autoware Safety Island onto nano-ros";* `github.com/NEWSLabNTU/nano-ros`*; the nano-ros book,* `newslabntu.github.io/nano-ros-book`*.*

Reviewer comment log

Reviewer: ____________________ Date: ____________

Use this to collect the points you want on the agenda. *Class* is a hint for triage: **E** error of fact, **A** allocation disagreement, **S** safety-argument gap, **Q** question, **N** nit.

Page	**Line**	**Class**	**Comment**

Page	Line	Class	Comment

Overall assessment

1. Is the allocation defensible? *Does the right work run on the right side of the boundary?*

2. Is the safety argument sound? *Particularly the two-tier controller: a QM-rated MPC bounded by an ASIL-D gate.*

3. What is missing? *A failure mode, an interface, a deadline the design does not name.*

4. Which open decision worries you most, and why?

Please return marked-up copies before the session so that recurring points can be grouped into the agenda rather than discovered during it.